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Mitsubishi Industrial Robot

CRn-500 Series

“Personal Computer Support Software” Instruction Manual
(3A-01C-WINE/3A-02C-WINE)

MELFA
BFP-A8090-E

Revision History

Date of print	Specifications No.	Revision details
2000-4-28	BFP-A8090-*	First release
2000-7-10	BFP-A8090-A	• Change “file open/save as” (Program Editor) • Add “Program conversion” • Change “Program Management”
2001-9-3	BFP-A8090-B	• Add “Copying and pasting the position array variables” in Program Editing tool. • Change “Search” and “Replace” in Program Editing tool. • Change “Program monitor” • Add “backup/restore the Robot Origin Parameter” in Maintenance tool. • Change “Backup(Load/Save)” in Maintenance tool. • Add “Print of Parameters” in Maintenance tool.
2001-10-31	BFP-A8090-C	(Corresponds to the version G2) • Add “History” function to Program Editing tool. • Change “Position Variable Batch Edit Window” in Program Editing tool. • Add “Find In Files” function to Program Editing tool. • Change “Program Manager Setup Window” in Program Editing tool. • Add “Viewpoint change function using the mouse” to Robot Graphic Display. • Add “Reading, editing and writing robot origin data” to Maintenance tool. • Add “Instructions related to position data” to conversion targets of program conversion.
2002-3-18	BFP-A8090-D	(Corresponds to the version D1) • Corresponds to the Windows2000/XP • Add “Multiple Robot Controller Communication” function. • Change “Special-Purpose I/O Signal” to “Named I/O Signal”. • Accelerate JOG Operation of Simulation. • Add “OVRD setting function” to Simulation.
2002-12-20	BFP-A8090-E	(Corresponds to the version D2) • The floppy disks is abolished. • The description addition which encloses communication middleware “MelfaRXM.OCX” in the standard version, and the Chapter about a setup “6. Setup of the ommunication middleware “MelfaRXM.ocx”” is added.

INTRODUCTION

Thank you for purchasing Mitsubishi Electric Industrial Robot MELFA.

This document is the operation manual for the Personal Computer Support Software.

By fully utilizing the features of this software, you can perform the initial startup of the robot, and create, edit and control the robot programs.

To operate the robot in a safe manner, be sure to read this manual thoroughly in advance. Also, keep this manual in a location that provides an easy access whenever you need to refer to it.

Target Version of This Manual

This manual is for the Personal Computer Support Software of version D2 or later.

Personal Computer Support Software D2 edition corresponds to the version H7 of a robot controller.

Target Readers of This Manual

This manual is written on the assumption that the readers have sufficient knowledge of the basic operation of the personal computers, Windows 95/98/Me/NT 4.0/2000/XP as well as the robot controllers.

If you are unfamiliar with the basic operation of the personal computers, please read the user's manual of your personal computer.

Symbols Used in This Manual



DANGER

Indicates that incorrect handling is most likely to cause hazardous conditions, resulting in death or severe injury of the operator.



WARNING

Indicates a possibility that incorrect handling may cause hazardous conditions, resulting in death or severe injury of the operator.



CAUTION

Indicates that incorrect handling may cause hazardous conditions, resulting in injury of the operator, or only physical damage.

Please read this operation manual thoroughly and be sure to learn the correct operating procedures.

- No part of this manual may be reproduced by any means or in any form, without prior consent from Mitsubishi.
- The details of this manual are subject to change without notice.
- An effort has been made to make full descriptions in this manual. However, if any discrepancies or unclear points are found, please contact Mitsubishi.

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How to Ensure Stable Communication Between the Personal Computer Support Software and the Robot Controller

When communicating with the robot controller (hereinafter referred to as the R/C) using the Personal Computer Support Software (hereinafter referred to as the Software), depending on the personal computer model and settings the communication may become unstable in a batch backup or program upload/download where large amounts of data are transmitted. To ensure stable communication with the R/C, change the communication settings (communication protocol) of the R/C and the Software as shown below. If the communication settings of the R/C and the Software do not match, normal communication cannot be established. Be sure to change the settings on both the R/C and Software sides.

(1) The Robot Controller

As for the R/C communication settings, change the following parameters:

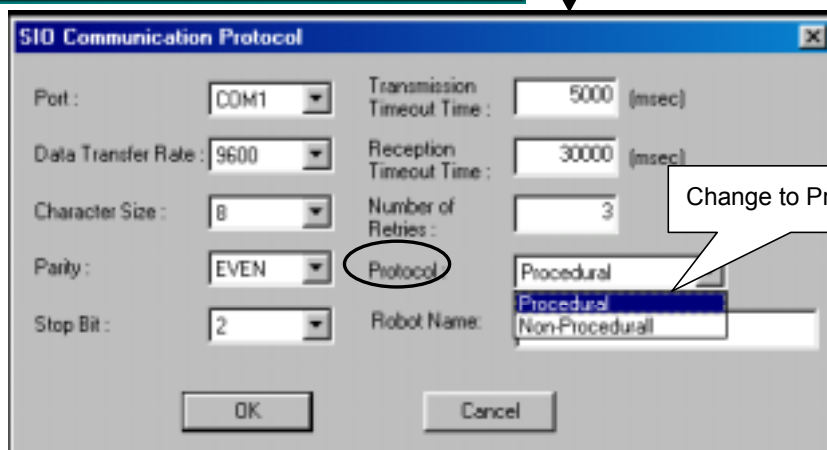
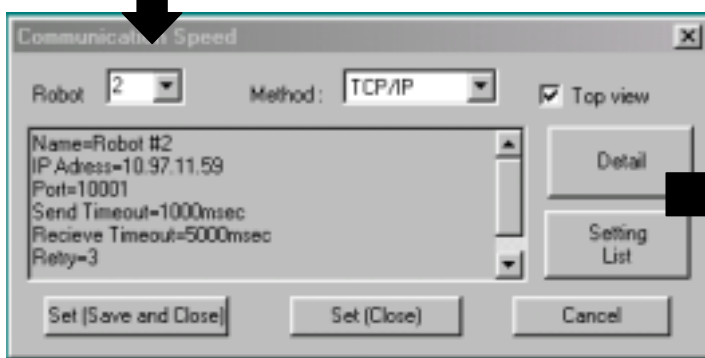
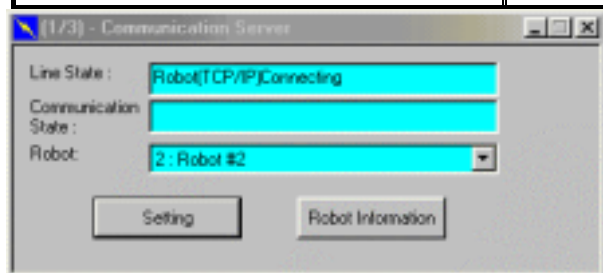
Parameter	Default	Change to
communication protocol (CPRC232)	0 (Non-Procedural)	1 (Procedural)

* To communicate with the personal computer via an extended RS-232C port by using an extension serial interface board, change the parameters of the extended RC-232C port.

(2) The Personal Computer Support Software

Change the communication settings of the Personal Computer Support Software through the "communication server."

Parameter	Default	Change to
Protocol	Non-Procedural	Procedural



* When communicating with the R/C using your custom software, reset the R/C protocol setting to "Non-Procedural" in advance.

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1. Before starting use

This chapter explains the precautions to be observed when using this personal computer software.

1.1. Using the instruction manual

The contents of this instruction manual are briefly explained in this section. Refer to the required section as necessary.

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The precautions before starting use are explained. Please read before starting use.

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The methods of starting up this personal computer software are explained.

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The functions of this personal computer software, and the basic Windows operations required for using this software are explained.

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This is a tutorial section for learning the series of operation methods when using this software for the first time.

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The operation methods of this software are explained in detail.

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The setup methods of the communication middleware(MelfaRXM.ocx) are explained.

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The functions of this software are shown as a tree format.

Windows 95/98/Me/NT/2000 and XP are products of Microsoft Corp.

1.2. Confirming the product

The configuration of the purchased product is shown below. Check that no parts are missing.

(1) The check of media

Standard version

- (i) "Personal Computer Support Software" CD-ROM (product) 1 disk
- (ii) "Personal Computer Support Software" Instruction Manual(this manual).....1 book

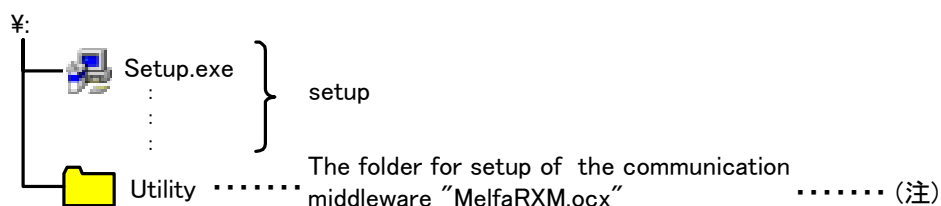
Mini version

- (i) "Personal Computer Support Software mini" CD-ROM (product) 1 disk
- (ii) "Personal Computer Support Software mini" Instruction Manual(this manual) 1 book

(2) The check of media contents

The following items are included on the CD-ROM.

Keep in mind that the contents included in CD-ROM differ with the [standard version] and the [mini version].



Caution) The [Utility] folder is included in only the standard version. Keep in mind that it does not go into the mini version.

(3) About the communication middleware 「MelfaRXM.ocx」

MelfaRXM.ocx is the ActiveX control that communicates to robot-controller. You can create the Windows Application of "MELFA ROBOT" by using this control.

Please refer to "6. Setup of the communication middleware "MelfaRXM.ocx" " about an setup of "MelfaRXM.ocx" .

In case of using only the function of " Personal Computer Support Software ", you don't need to install this software.

1.3. Items to be prepared by user

The items to be prepared by the user to use this personal computer software are explained in this section.

(1) Personal computer system

Prepare the following model.

* Personal computer that runs with Microsoft 95/98/Me/NT 4.0/2000/XP.

Details of working environment

Item	Min. required environment	Recommended environment
CPU	Pentium133MHz	Pentium III 450 MHz or more(*)
Main memory	32MB or more	128MB or more
Hard disk	100MB of open space or more	200MB of open space or more
Display	Must have graphic function that can display SVGA (800 × 600) or more, and must be capable of displaying 16 or more colors.	
Disk unit	CD-ROM drive	
Keyboard	PC/AT compatible keyboard	
Pointing device	Device that operates in Windows environment	
Communication port	Must have serial communication port that operates in Windows environment. (Min. 9600bps: 1 port)	

(*) When you calculate cycle time with this software, please use the personal computer of greater performance one than that.

(2) Software

Microsoft Windows 95, 98, Me, NT4.0, 2000 or XP.

2. Preparation before use

The methods of setting up this personal computer software are explained in this chapter. The setup work includes the following steps.

- i) Installing the personal computer software into the personal computer
 - Read "2.1 Installation methods".
- ii) Setting up the printer (Not necessary when not printing on paper)
 - Read "2.2 Setting up the printer".
- iii) Connecting the robot and personal computer
 - Read "2.3 Connecting to the robot".

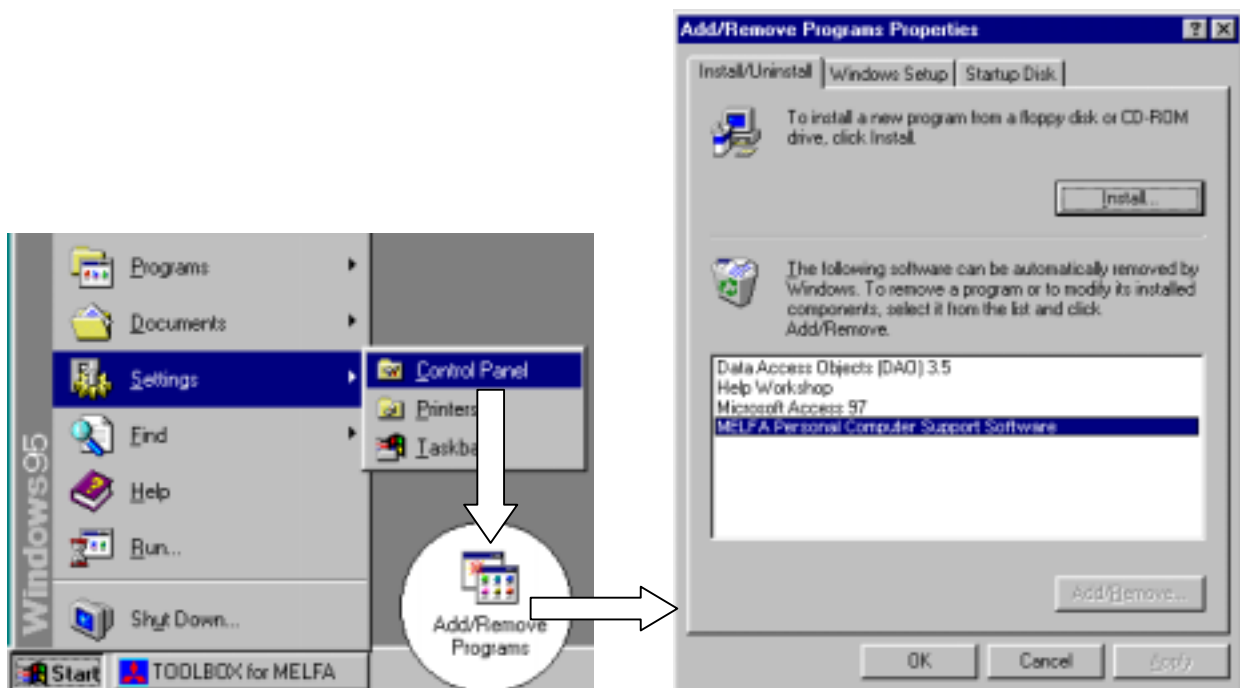
2.1. Installation methods

The methods for installing the personal computer software are explained in this section. The software can be installed from the CD-ROM or from the floppy disks.

2.1.1. Uninstalling the old version

If an older version of the "Personal Computer Support Software" is installed in the personal computer, always uninstall the older version before installing the newer version.

Select "MELFA Personal Computer Software" with [Start] → [Settings] → [Control Panel] → [Add/Remove Program], and then click on the Add/Remove Program button.

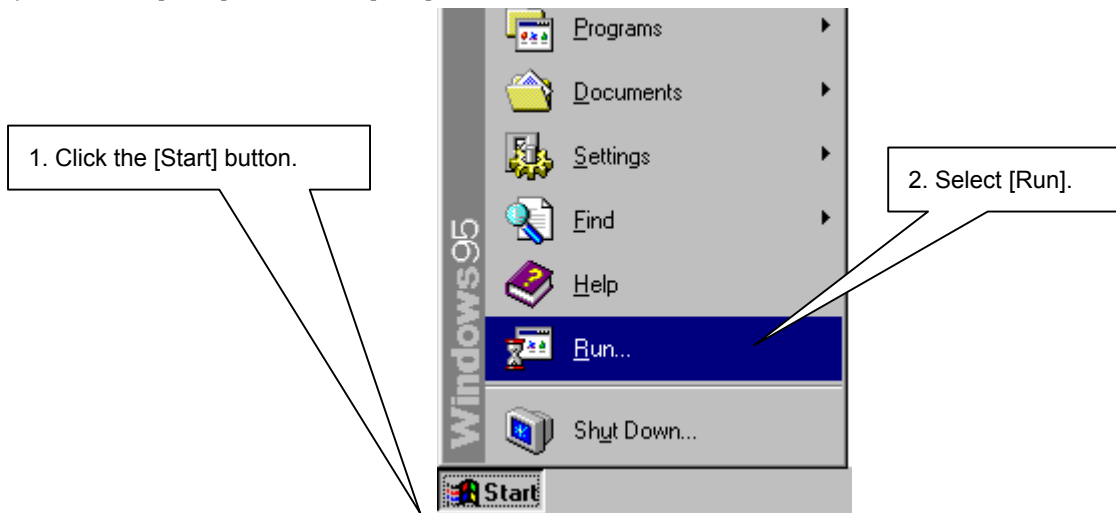


2. Preparation before use

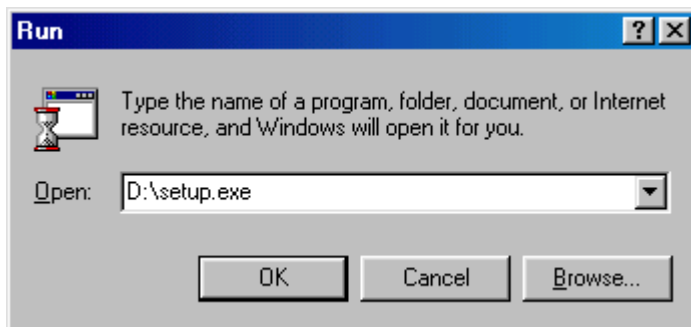
2.1.2. Installation

The method for installing from the CD-ROM is explained in this section.

- 1) Set "Personal Computer Support Software" CD-ROM in the personal computer's CD-ROM drive. The Setup screen will start up automatically. Follow the procedure from step 4) below. If the screen does not start up automatically, carry out the procedure from step 2).
- 2) Select the [Start] button and [Run].



- 3) Check the CD-ROM drive name. Input as shown below.
"Drive name":¥setup.exe "
(If the CD-ROM drive is "D:", this will be "D:¥setup.exe".)



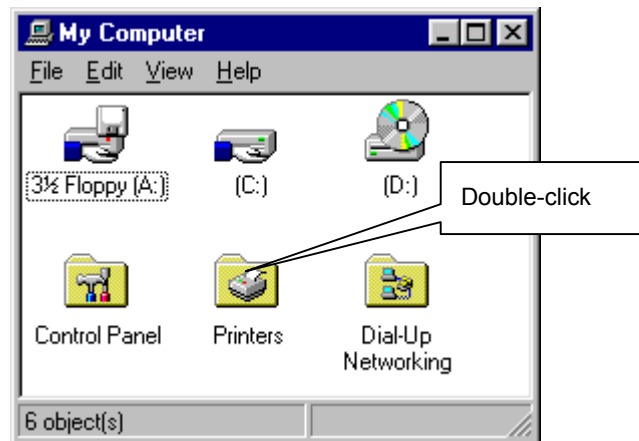
Click on the "OK" button.

- 4) The installer will start. Follow the instructions given on the screen. The basic flow is as shown below. Just click on the "OK" button when indicated on the screen.
 - i) The start of installation is announced.
 - ii) Select the install direction
 - iii) Start copying of installation file from CD-ROM
 - iv) Completion of setup is notified

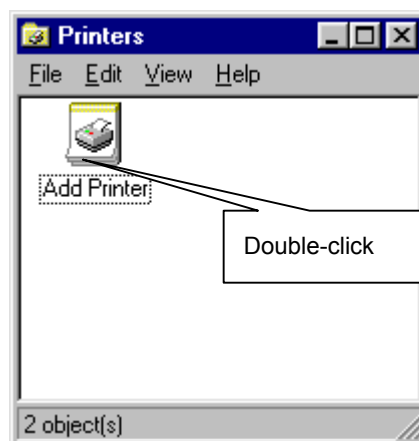
2.2. Setting up the printer

To print programs with this “Personal Computer Support Software”, the printer must be setup. The method for setting up the printer is explained in this section.

- 1) Double click on [My Computer] → [Printer].



- 2) Double click on [Add Printer].



- 3) Follow the instructions on the screen. The basic flow is as shown below. Refer to the instruction manual of the printer being used for details.
 - i) Select the type of connected printer
 - ii) Select the connected printer product name and maker
 - iii) Select the port used by the printer
 - iv) Set the printer name
 - v) Set whether to test the printing → Completed

2.3. Connecting to the robot

The method of connecting the controller body to the personal computer is explained in this section.

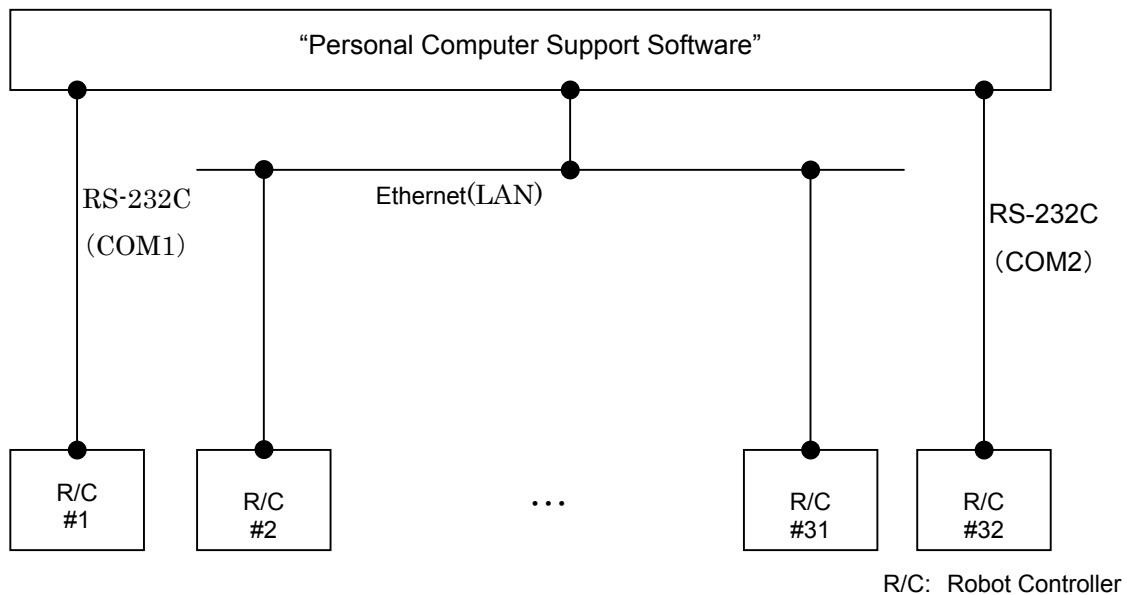
Either an RS-232-C connection or Ethernet can be used to connect the personal computer to the controller. The methods are explained in the following order in this section.

- 1) Using RS-232-C
- 2) Using Ethernet
- 3) Setting the communication server

* Usually, the personal computer is provided with RS-232-C as a standard. When using Ethernet, the "Ethernet interface" option is required on the robot controller and a network card is required on the personal computer.

Caution

Personal computer support software of D1 version or later can simultaneously connect with a maximum of 32 robot controllers. Please note that personal computer support software of C2 version or earlier cannot simultaneously connect with multiple robot controllers. Also, RS-232C and Ethernet can be used together.



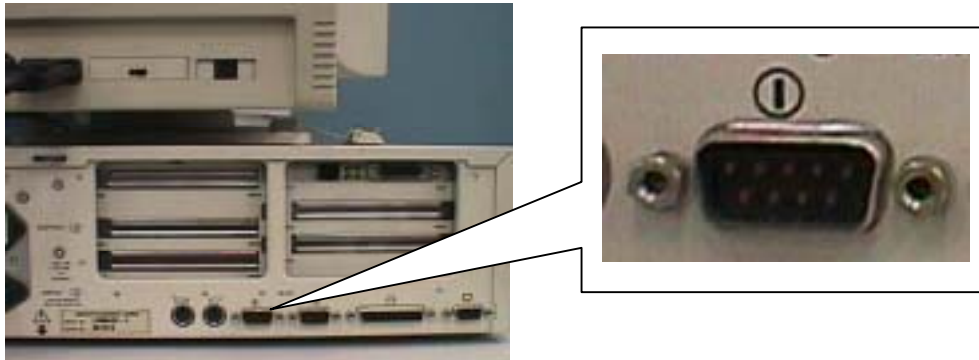
Example of connecting multiple robot controllers

* When using Ethernet, if communication setting is performed for unconnected robots, retry processing is performed frequently and, as a result, the execution speed slows down. Therefore, set the communication setting for unconnected or powered-off robots to "Not used."

* 32-robot connection is a theoretical value; when 32 robots are actually connected, it does not guarantee the same performance as when only one robot is connected. For example, if all of the 32 robots are monitored, the status update slows down compared to when only one robot is connected. Also, such setting may affect applications that use other networks in Windows 95/98/Me.

2.3.1. Using RS-232-C

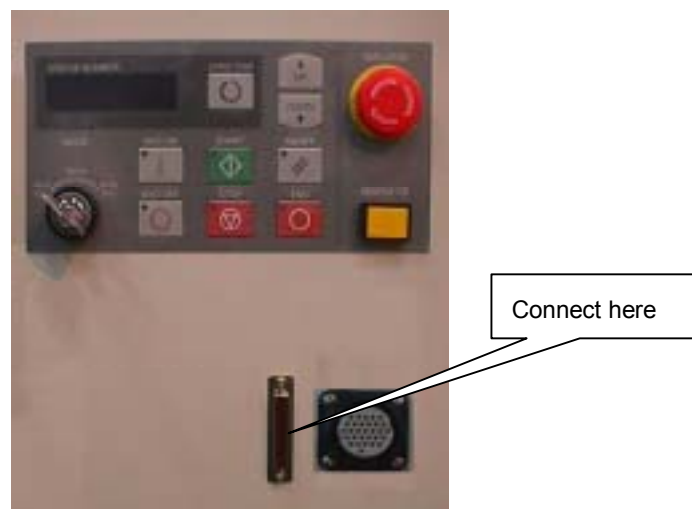
- 1) An RS-232-C cable is used for the connection between the personal computer and controller. (For a DOS/V personal computer, a 9-pin connector on the robot side and a 25-pin connector on the controller side is used.)
- 2) Connect the personal computer side of the cable to the standard RS-232-C connector (RS-232-C serial port 1).



(Personal computer's RS-232-C connector)

When connecting to a port other than the RS-232-C serial port 1, refer to section "2.3.3 Setting the communication server".

- 3) Connect the controller side to the RS-232-C connector on the front of the controller.



(Controller's RS-232-C connector)

2. Preparation before use

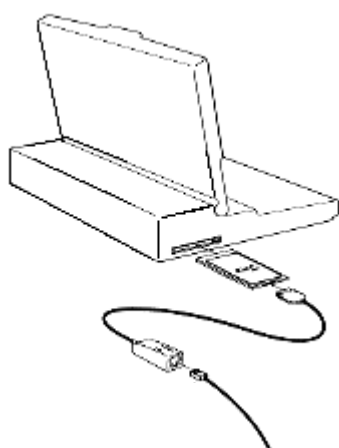
2.3.2. Using Ethernet

* To use Ethernet, the robot controller must be provided with the "Ethernet interface" option, and the personal computer must be provided with a network card. The network must also be set.

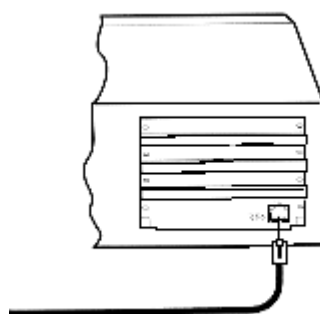
Connect the modular plug to the personal computer's network card connector.

Connect the personal computer to the network. Contact the network manager for details on setting the network address (IP address, Subnet Mask, default gateway, etc.) in the personal computer.

Connection example :



<Laptop computer>



<Desktop computer>

Refer to the "Ethernet Interface Option Instruction Manual" for details on connecting to the robot controller.

When using Ethernet, refer to section "2.3.3 "Personal computer support software" communication settings" and change the software's communication specifications.

2.3.3. Personal computer support software communication settings

Caution

Communication with the robot controller may be disabled if the communication specifications are changed. Take special care when making changes.

The personal computer support software communication settings are made with the "communication server".

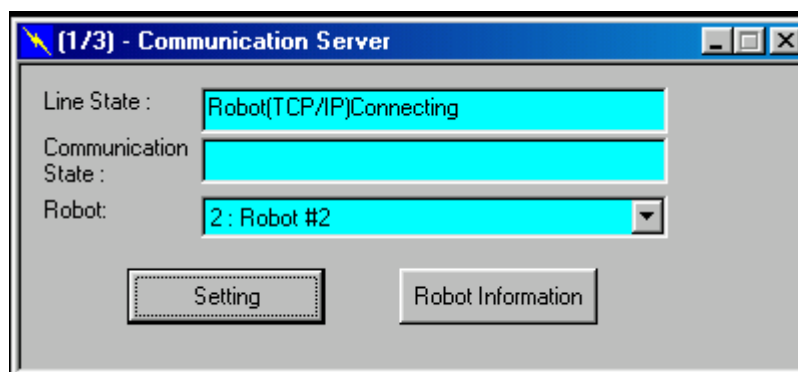
The "communication server" is set to the following default values when installed.

Item	Value
Communication method	RS-232C
Device No.	1
Port	COM1
Baud rate	9600
Data length	8
Parity	Even
Stop bit	2
Transmission timeout	5000 msec
Reception timeout	30000 msec
No. of retries	4 times
Protocol	Non-procedural
Robot Name	

When the communication specifications have been changed in the robot controller or when using the Ethernet interface option, change the communication specifications in the personal computer with the following procedure.

The communication server is automatically started when the "Personal Computer Support Software" is started up.

Note that the server is iconized when started up, so click on  on the task bar, and return to the original size.



[Title Bar] (AA/BB) indicates that AA is the number of robot controllers connected, and BB is the number of robot controllers for which communication settings have been made.

[Line State] The connection status of the communication line with the robot is displayed. The status color indicates the status of the robot controller that is currently being selected.

2. Preparation before use

Status	Content	Color
Connecting	Indicates that the connection with the robot has been established.	Blue
Connection wait	Indicates that a communication to verify connection is being made in the case of RS-232C connection. Indicates the wait status for communication port connection in the case of TCP/IP connection.	Green
Connection error	Displayed when the data reception enable signal cannot be detected because a cable has been disconnected or the robot has not been started in the case of RS-232C connection. Displayed when the communication port cannot be opened in the case of TCP/IP connection.	Red
Communication Setting error	Displayed when the communication port cannot be opened in the case of RS-232C connection. This is not displayed in the case of TCP/IP connection	Red
Waiting	Indicates the idling status displayed at the start of remote maintenance.	Green

[Communication State]..... The contents of communication with the robot controller are displayed.

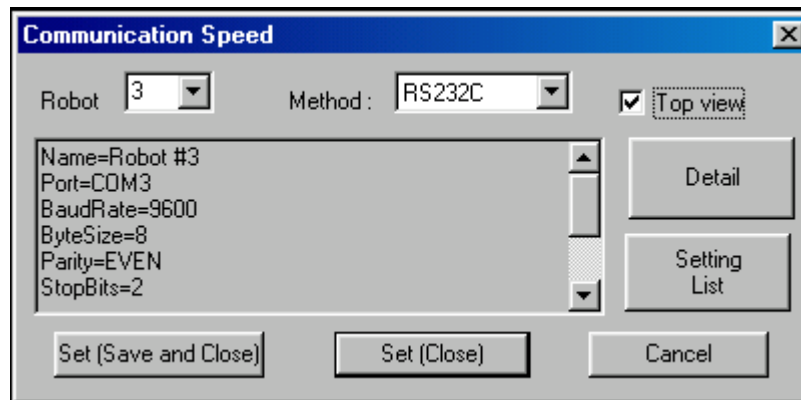
[Robot]..... Select the Robot Controller which wants to display Line State and Communication State. Only the Robot Controller by which a Communication setup was carried out is displayed.

[Setting] button..... Changes the communication specifications.

[Robot Information] button Information on the currently connected robot can be referenced.

Communication setting

The following type of window will open when the [Communication Setting] button is clicked on.



Robot..... Select the Robot Controller(1-32) for communication.

Communication method Select the method to be used for communication.

Either RS-232-C or TCP/IP can be selected.

Select "TCP/IP" when carrying out communication with Ethernet.

Set the robot controllers that are not connected to "Not used." Otherwise, the performance of the personal computer may be degraded. Robot controller 1 cannot be set to "Not used."

[Details] button A window for changing the communication specifications will open. The window that opens is different depending on whether RS-232-C or TCP/IP is selected for the communication method.

[Robot controller connection setting list] button The communication settings can be specified in a table format. This is convenient when performing communication settings for a large

number of robot controllers at one time.

[Set(Save and Close)] button The communication specifications are changed and the change results are saved. Even when the personal computer support software is started up next, communication will be carried out with the specifications set here.

[Set(Close)] button The communication specifications are changed. The change results are not saved, so when the personal computer support software is started up next, communication will be carried out with the previously set specifications.

[Cancel] button The communication setting is quit without changing the communication specifications.

[Top view] If this check box is set to ON, the dialog that indicates the communication status is displayed on top of the screen. If this check box is set to OFF, the dialog can be hidden.

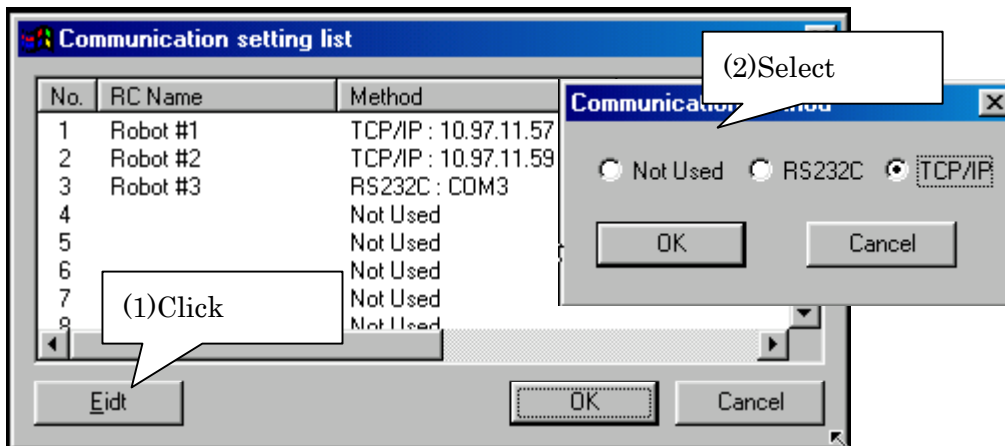
Caution

When completed setting the communication server, iconize the server by clicking on the  button.

The communication server will automatically quit when all of the personal computer support software applications are quit. Thus, do not quit the server with the  button. Communication with the robot controller will be disabled when the communication server is quit.

2. Preparation before use

Communication setting list



Select the item you want to set and click the [Edit] button. A dialog box for selecting the communication method appears. Select a communication method, and then click the [OK] button. A communication setting dialog box for the selected communication method appears.

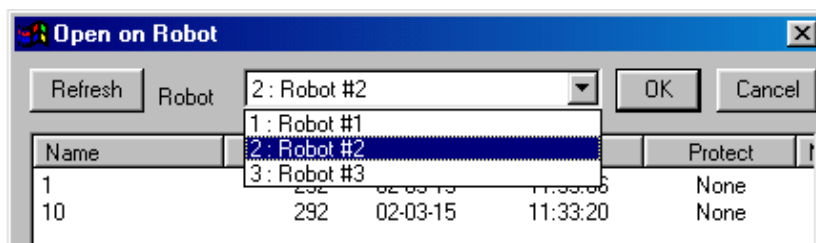
After editing is completed, click the [OK] button.

Robot controller switching operation using each tool

Once the communication settings of multiple robot controllers are made, you can switch the robot controller you want to operate by manipulating the combo box for selecting a robot controller that is provided in each tool.

The following figure shows the operation for “opening a program on the robot” in program editing.

Clicking the ▼ area of the combo box displays the list of the robot controllers for which communication settings have been made. Select a robot controller using the mouse.



Caution

- Do not perform duplicate communication settings for the same robot controller. Doing so will generate communication errors or problems such as missing program files in the controller. Example of incorrect use: Connect with the same robot controller using an RS-232C and a TCP/IP.
- Before changing communication settings, be sure to close all the tools except the communication server. Otherwise, mismatching will occur in the robot controller list.
- When connecting multiple robot controllers for use, execute the operations of the robot controllers only after verifying the target of operation.

Details setting (Only for RS-232-C)

The screenshot shows a dialog box titled "SIO Communication Protocol". It contains the following settings:

- Port : COM3
- Data Transfer Rate : 9600
- Character Size : 8
- Parity : EVEN
- Stop Bit : 2
- Transmission Timeout Time : 5000 (msec)
- Reception Timeout Time : 30000 (msec)
- Number of Retries : 3
- Protocol : Non-Procedural
- Robot Name : Robot #3

At the bottom, there are "OK" and "Cancel" buttons.

- Port used : Select from COM1, COM2, COM3 or COM4.
Selection of COM1-COM8 is possible for the C1 version or later.
Select the port to which the cable is connected.
- Baud rate : Select from 4800, 9600 or 19200.
- Character size : 7 or 8 can be selected, but 8 should be set.
- Parity : Select from NON (none), ODD or EVEN.
- Stop bit : Select from 1, 1.5 or 2.
- Transmission timeout : The timeout time during transmission can be set.
- Reception timeout : The timeout time during reception can be set.
- No. of retries : Set the No. of times to retry communication.
- Protocol used : Select from Non-Procedural or Procedural.
- Robot name : A nickname consisting of up to 32 half-size characters can be assigned to a robot controller (D1 version or later). It is not required, but the controller name assigned here will be displayed in places where you need to select a robot controller. So assigning nicknames is convenient in distinguishing them when multiple robot controllers are connected.

After changing to new settings, click on the [OK] button. The Communication Setting screen will reappear, so click on the [Set(Close)] button. (If the same settings are to be used the next time, click on the [Set(Save and Close)] button.)

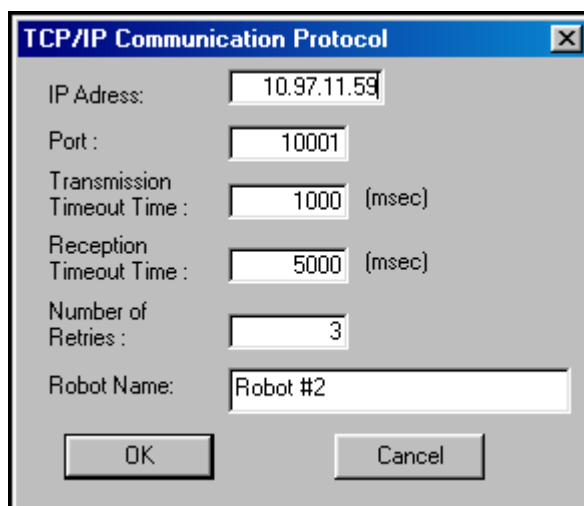
Set the following to perform a high-speed, stable communication:

- Baud rate :19200 bps
- Protocol used :Procedural

It is also necessary to change the communication settings of the robot controllers at this time.

2. Preparation before use

Details setting (for TCP/IP)



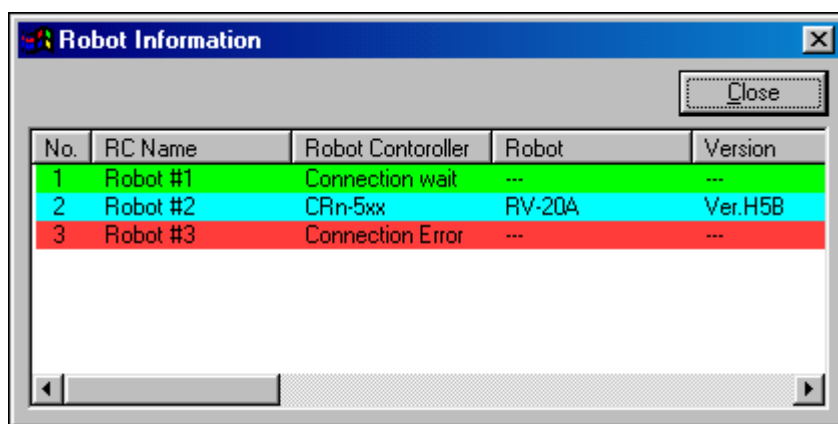
The dialog box titled "TCP/IP Communication Protocol" contains the following fields and buttons:

- IP Address: 10.97.11.59
- Port: 10001
- Transmission Timeout Time: 1000 (msec)
- Reception Timeout Time: 5000 (msec)
- Number of Retries: 3
- Robot Name: Robot #2
- Buttons: OK, Cancel

After setting the IP address assigned to the robot controller in [IP Address], click on the [OK] button. The Communication Setting screen will reappear, so click on the [Set(Close)] button. (If the same settings are to be used the next time, click on the [Set(Save and Close)] button.)

Robot Information

The information on the robot controllers for which communication settings have been made is displayed in a list format.



The dialog box titled "Robot Information" displays a table with the following data:

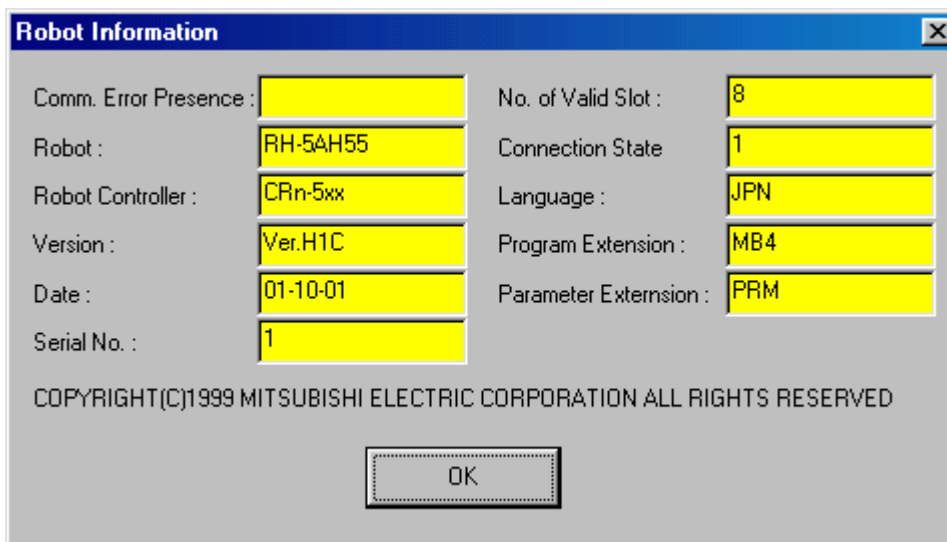
No.	RC Name	Robot Controller	Robot	Version
1	Robot #1	Connection wait	---	---
2	Robot #2	CRn-5xx	RV-20A	Ver.H5B
3	Robot #3	Connection Error	---	---

The table is scrollable, and a "Close" button is located in the top right corner.

If the connection has already been completed, the robot controller information is displayed after the controller model number.

If the connection has not been completed, the description of a communication error is displayed in the controller model number field.

Double-clicking the controller number in the list displays the information on the robot selected in the dialog box.



The image shows a 'Robot Information' dialog box with a blue title bar and a close button. It contains two columns of text labels and yellow input fields. The first column includes 'Comm. Error Presence', 'Robot', 'Robot Controller', 'Version', 'Date', and 'Serial No.'. The second column includes 'No. of Valid Slot', 'Connection State', 'Language', 'Program Extension', and 'Parameter Extension'. At the bottom, there is a copyright notice and an 'OK' button.

Comm. Error Presence :		No. of Valid Slot :	8
Robot :	RH-5AH55	Connection State	1
Robot Controller :	CRn-5xx	Language :	JPN
Version :	Ver.H1C	Program Extension :	MB4
Date :	01-10-01	Parameter Extension :	PRM
Serial No. :	1		

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OK

[OK] button : The Communication Server screen will reappear.

3. Basic functions and window operations

The functions provided with this personal computer software and the basic Windows operation methods are explained in this chapter.

3.1. Basic functions

The “Personal Computer Support Software” functions are explained in this section.

This “Personal Computer Support Software” has the following functions. Each function corresponds to the tools explained in Chapter 5 and following. A list of functions (Windows) is given in 6. Appendix (Function tree) for reference.

Function	Details	
Applicable model	• Personal computer that runs on Microsoft Windows 95/98/Me/NT4.0/2000/Xp.	
Program editing functions	Editing functions	* MELFA BASIC IV language compatible * Simultaneous display of multiple editing screens * Command input, command description * Editing of position data * File operation (Writing to controller, floppy disk and personal computer) * Search, Find in files and Replace function (With character, line No. or label) * Copy, cut, paste, insert (per character, line), undo (per command statement, position variable) * Line No. automatic generation, renumbering * Batch grammar check * Command template * Position variable batch editing * Position variable template * Print, print preview
	Management functions	* Program file management (List, copy, move, delete, content comparison, rename, protect)
	Debug functions	* Direct editing of programs in controller * Confirmation of program operation (step execution, direct execution) * Cycle time measurement
Simulation function	* Confirmation of robot program operation with off-line simulation using CG [Computer Graphics]. * Calculation of cycle time	
Monitor functions	* Robot movement monitor (robot operation state, stop signal, error monitor, program monitor (execution program, variables), general-purpose input/output signal (forced output possible), dedicated input/output signals, movement operation (movement range, current position, hand, etc.) * Operation monitor (operating time accumulation, production information, robot version) * Servo monitor (position, speed, current, load, power)	
Maintenance function	* Parameter setting * Batch backup	
Remote maintenance function	* Monitoring and maintenance of robot at remote location using telephone line. (A separate modem is required to use this.)	

Software shows the following :

STD : Personal Computer Support Software(standard version)

mini : Personal Computer Support Software mini

The simulation function becomes correspondence only of a standard version(STD).

3.2. Basic Windows operations

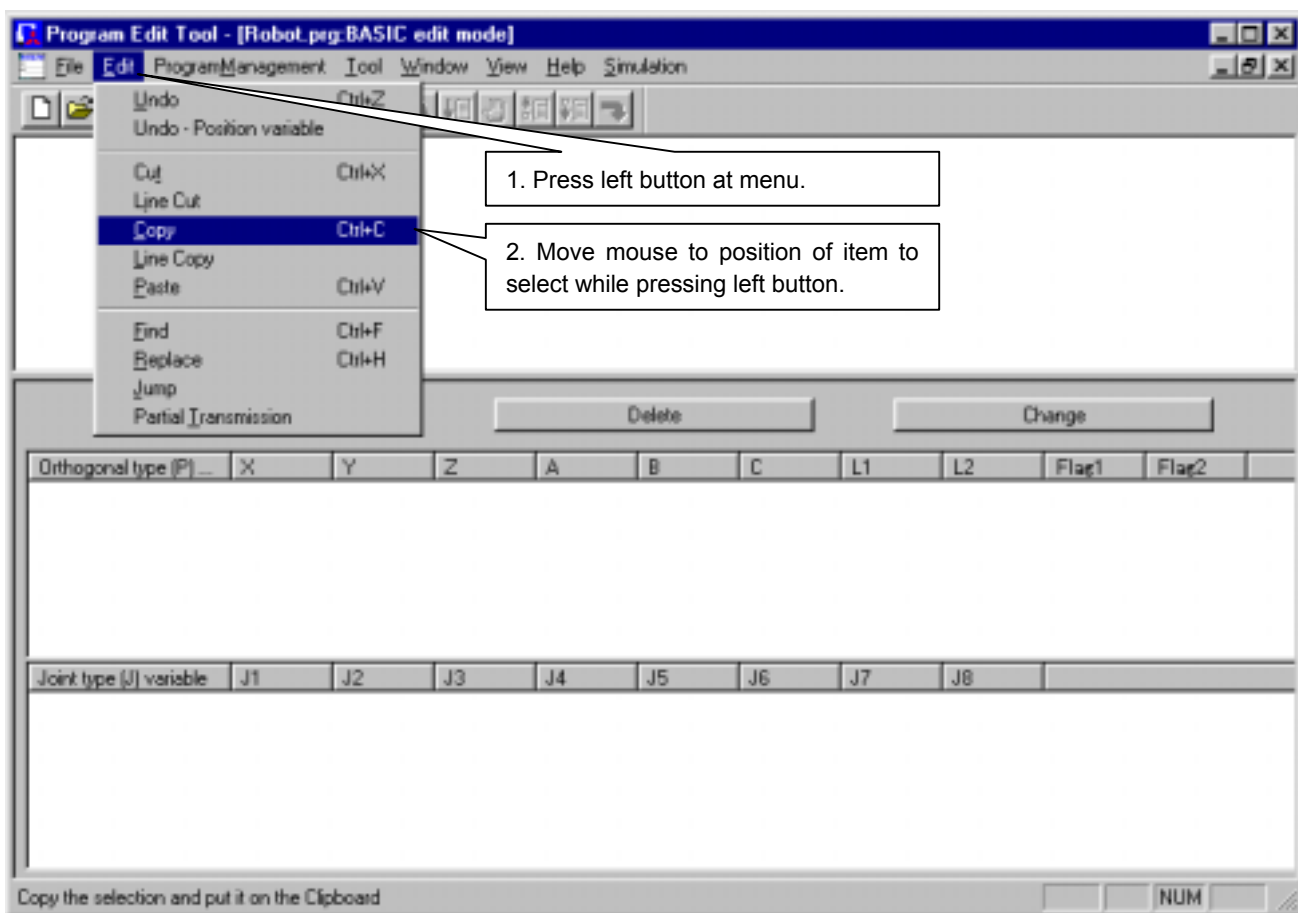
The basic mouse operations of Windows, required when a first-time user of Windows is using this personal computer software are explained in this section.

(1) Mouse operations

Selecting a menu

When selecting a menu, move the mouse cursor to the menu characters, and then press the left mouse button.

A list will appear from the selected menu, so while holding down the left mouse button, move the mouse cursor to the item to be selected, and then release the left button.



Click

This means to press the left mouse button once. This is used to press buttons, etc.

Double-click

This means to press the left mouse button twice quickly. This is used to select an item from the list.

4. Starting use

This chapter explains the series of software operations in a tutorial method for first-time users of this personal computer support software.

Here, a robot program will be created using the personal computer support software, and the program operation will be confirmed with simulation. Then, the program will be downloaded to the robot controller and the operation confirmed by operating the actual robot.

4.1. Starting

First, start by starting up the personal computer support software.

- (1) Connect the cable with the following procedure.

Connect the personal computer and controller with an RS-232-C cable.

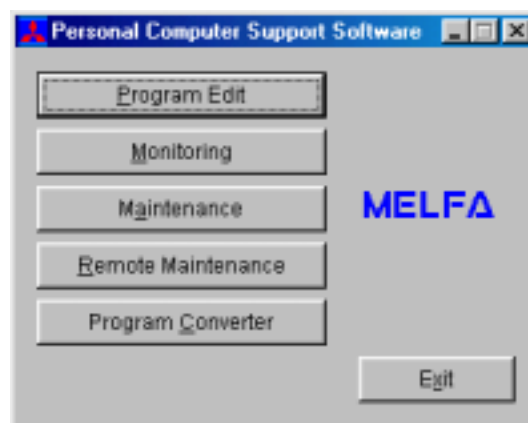
Connect the personal computer side to the standard RS-232-C connector (RS-232-C serial port).

Connect the controller side to the RS-232-C connector on the front of the controller.

- (2) Turn the controller power ON.

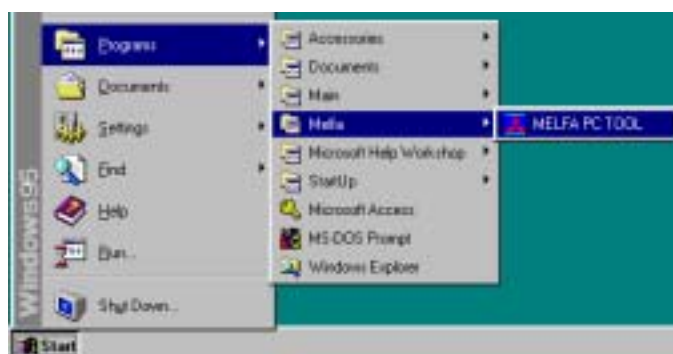
- (3) Start up the personal computer support software with the following procedure.

Turn the personal computer power ON. The following Main screen will start up automatically. If the screen does not start up, select the Windows [Start] button, and then [Program] → [MELFA] → [Personal Computer Software].



(Main screen)

(Note) When using Windows NT, double-click on the MelfaTool.exe icon in the folder (normally C:\Program Files\Melfa ENG) in which the personal computer tools are installed.



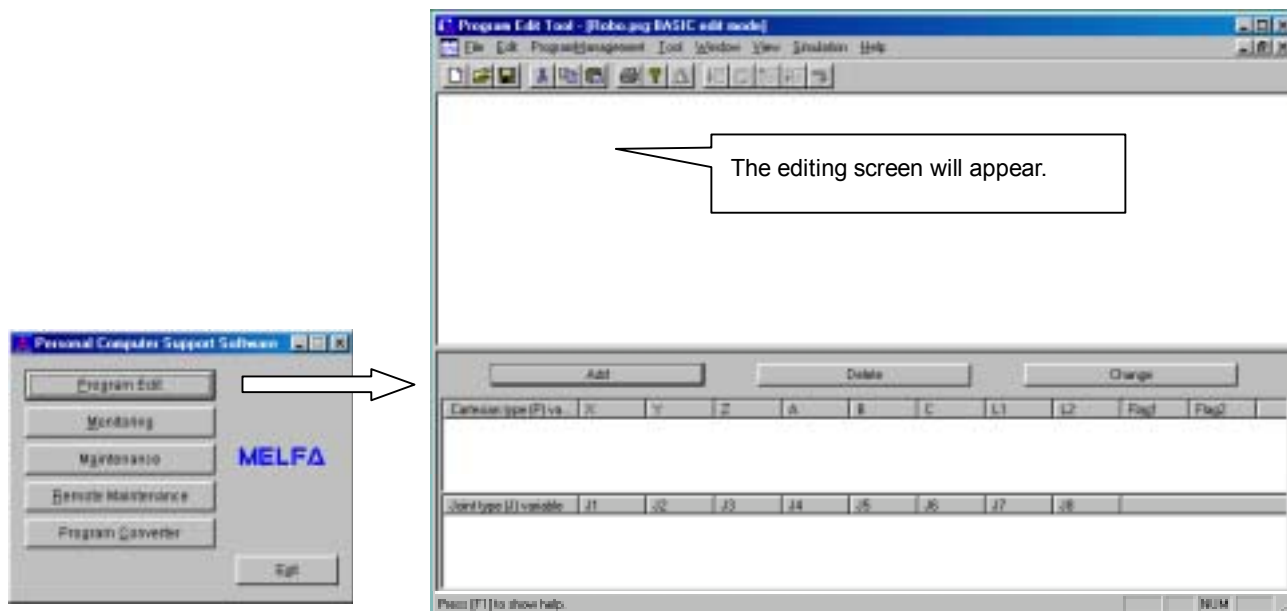
(Windows screen when Start button is selected).

4.2. Creating a program

Next, try editing an actual program.

(1) Starting the program editing tool

Click on the [Program Edit] button on the Main screen. The program editing tool, as shown below, will start up.

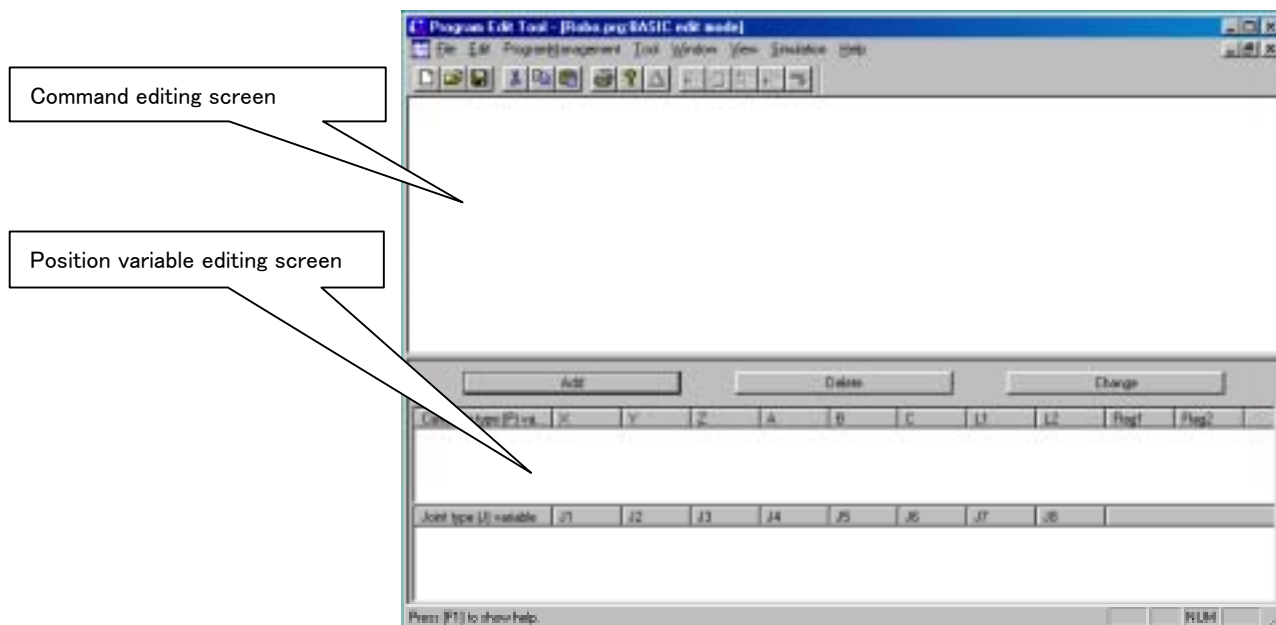


(Entire screen from Main screen to Program editing screen)

(2) Newly creating a program file

To create a new program file, first click on [New] under the [File] menu, or click on the  button.

The following type of editing screen will appear.



(Editing screen)

(3) Editing the program

Here, a simple program using the three positions P10, P11 and P12 will be created. The program contents will move the robot in order between the three points. The list is shown below.

Program name: P100

10 GETM 1	'Declare to move mechanism No. 1 robot
20 MOV P10	'Move to position P10
30 MOV P11	'Move to position P11
40 MOV P12	'Move to position P12
50 END	'End program

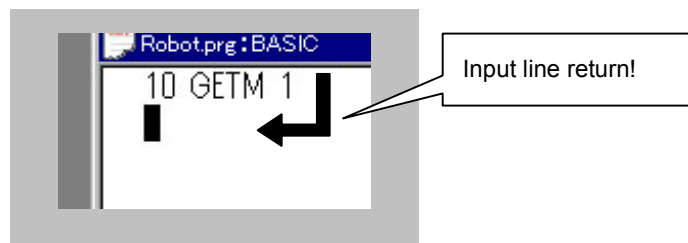
An example for inputting the first line is shown. The upper side of the editing window is the program command statement editing screen.

Input as shown below using the keyboard, and then press the Enter key.



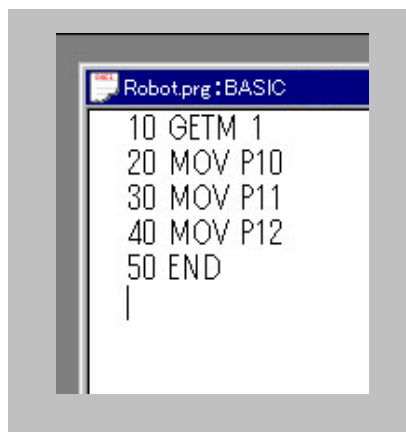
(1st line: "10 GETM 1")

The cursor will move to the next line. To change the contents of the program line, change the contents at the line, and then always press the Enter key at that line.



(Press Enter key)

Input the remaining program in the same manner.



(Editing window containing the following program.)

4.3. Teaching a position

(1) Editing the position variable

Next, edit the values of the position variables P10, P11 and P12 input in the program command statement to the following values. The values marked with an × in the table are not input here.

Robot.prg: BASIC

10 GETM 1
20 MOV P10
30 MOV P11
40 MOV P12
50 END
|

Caution: The following values are for the RV-20VA type robot.

Position variable name	X	Y	Z	A	B	C	L 1	L 2	Flag1	Flag2
P10	1060	-680	870	×	×	×	×	×	6	0
P11	900	300	1800	×	×	×	×	×	6	0
P12	933	0	1588	×	×	×	×	×	6	0

(Unit:mm)

(2) Add a position variable.

The lower side of the editing screen is the position variable editing screen.

AddDeleteChange

Cartesian type (P)...

X

Y

Z

A

B

C

L1

L2

Flag1

Flag2

List of Cartesian type variables

Joint type (J) variable

J1

J2

J3

J4

J5

J6

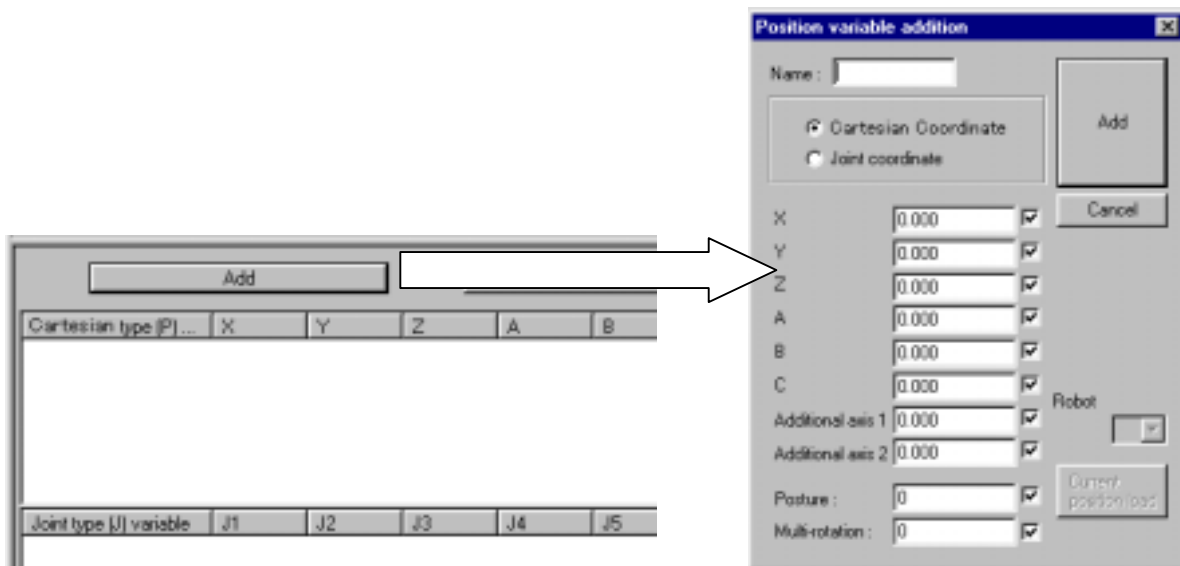
J7

J8

List of joint type variables

(Position variable editing screen)

Input the position variable P10 value as shown below.
When the [Add] button is clicked on, the Position variable addition screen will appear.



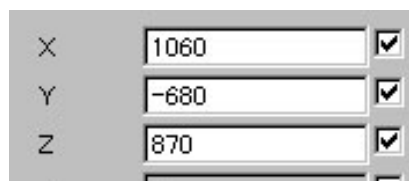
(Addition screen)

Input the variable name "P10" in the [Variable Name] area.



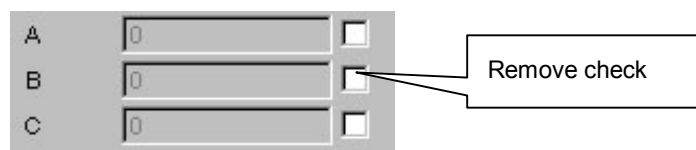
(Variable name area)

Input each value (X: 1060, Y: -680, Z: 870, Posture: 6, Multi-rotation: 0) in the [X] [Y] [Z] [Posture] and [Multi-Rotation] areas.



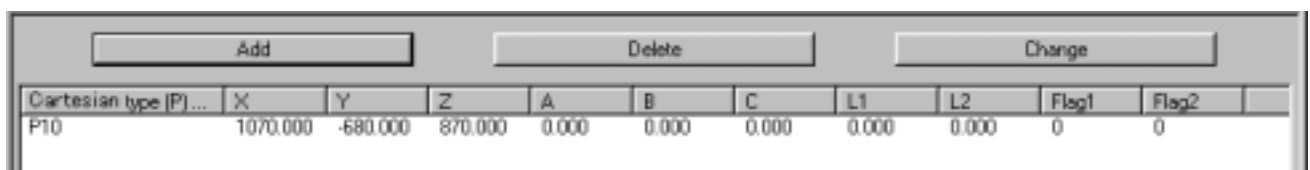
(Value input area)

In this case, the [Additional axis 1] [Additional axis 2] [A] [B] and [C] values are not defined, so remove the checks.



(Removing the check)

Check the described details, and click on the [Add] button. The position variables will be registered into the program, and the P10 position variable will appear in the position variable list as shown below.

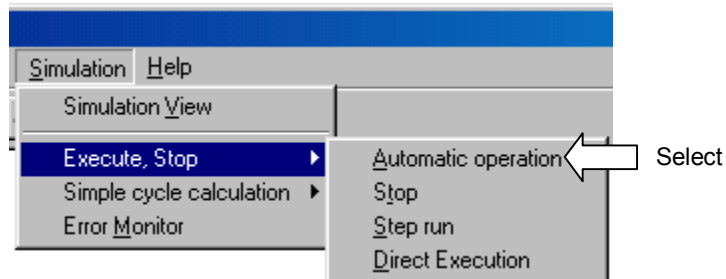


(State with P10 in list)

Add the remaining two position variables in the same manner.

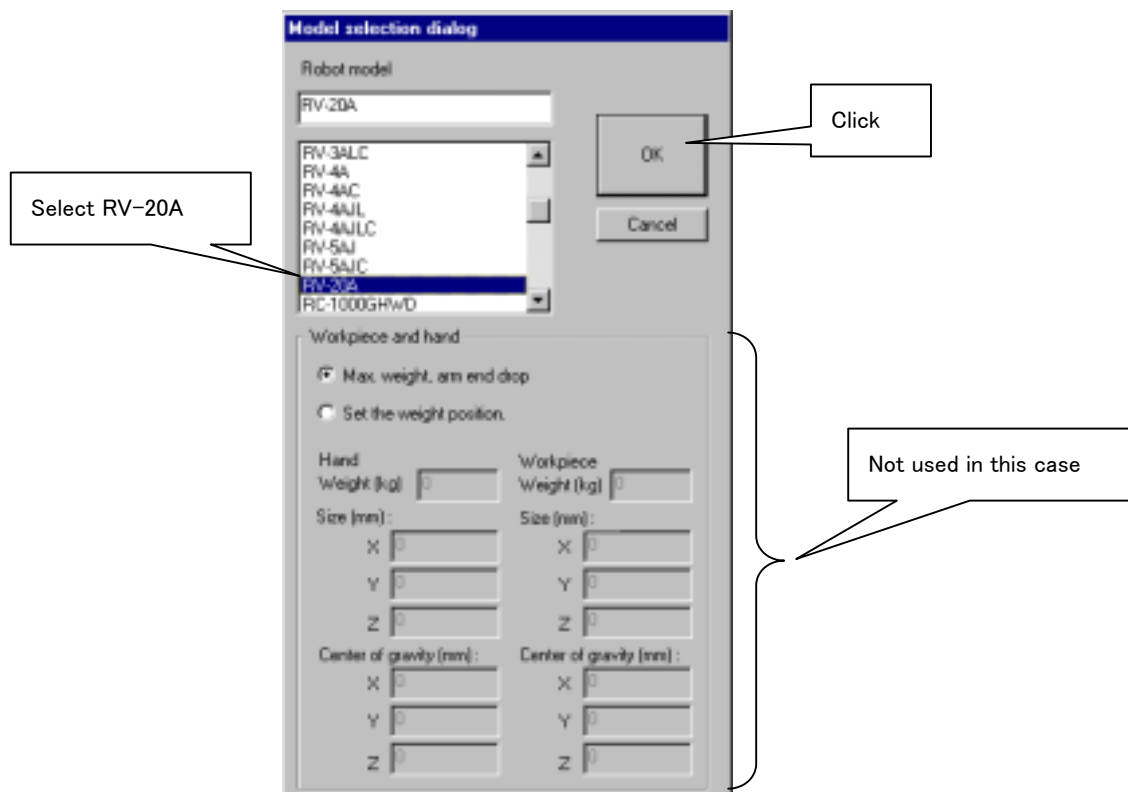
4.4. Confirming the operation with simulation

Next, try confirming the operation of the created robot program using simulation. Select [Execute, Stop] → [Automatic Operation] from the [Simulation] menu.



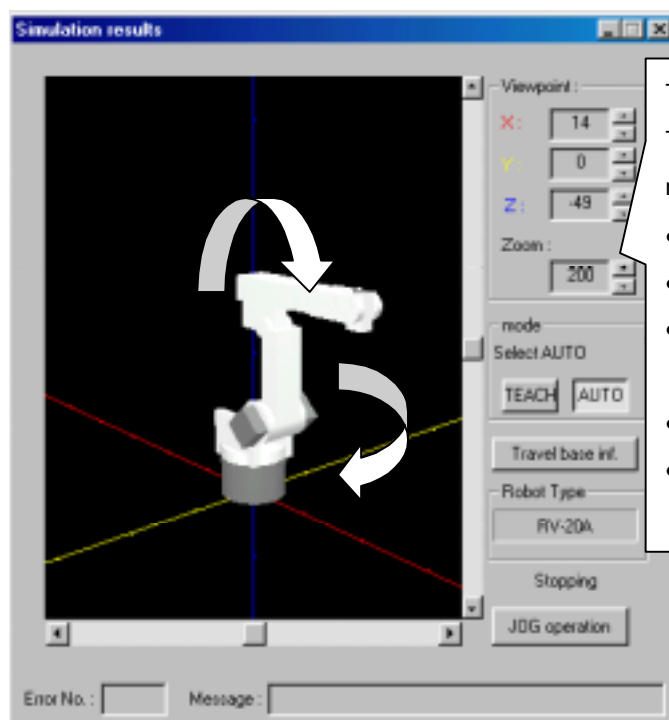
(Simulation menu)

The following type of robot type and hand/workpiece setting screen will appear. In this case, select "RV-20A" from the model list, and click on the [OK] button.



(Model and hand/workpiece setting screen)

The following type of simulation screen will appear, and simulation of the robot will start. Confirm that the robot moves as written in the program.



(Simulation screen)

Try changing the viewpoint.

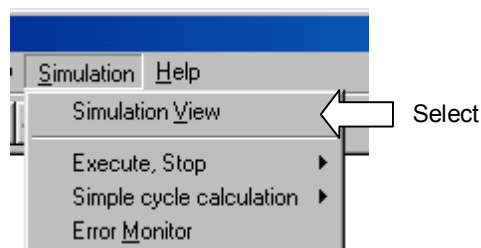
The viewpoint can also be changed by the following mouse operations:

- Left button (left and right) → Rotation of Z-axis center
- Left button (up and down) → Rotation of X-axis center
- Left + right buttons (left and right) →
Rotation of Y-axis center
- Right button (up, down, left and right) → Move
- [Shift] key + left button (up and down) →
Enlargement/reduction

Did the robot move correctly? If it moved correctly, try changing the position variable value and moving the robot. Refer to the Robot Language Instruction Manual enclosed with the controller for the definitions of the position variables.

(Caution: When not using the RV-20A type robot, change the program position variable values to match the model being used. Then check the movement with simulation, and then check the movement with the actual robot, as explained in the following section.)

After confirming the movement, select [Simulation] under the [Simulation View] menu, and close the Simulation screen.



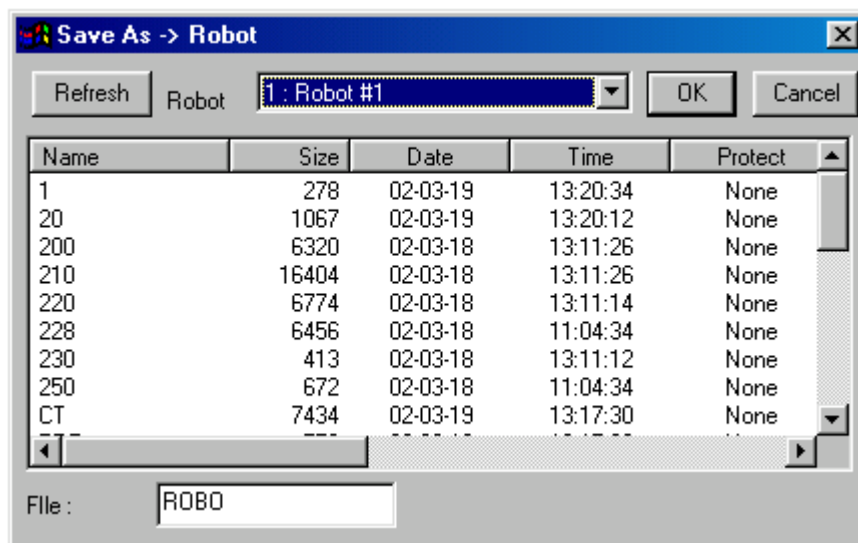
(Simulation menu)

4.5. Saving the program

After confirming the robot movement with simulation, save the program in the controller with the following procedure. Here, the program will be saved in the controller with the name "100".

Click on [Save As->Robot] under the [File] menu, or click on the  button. A screen for designating the program.

A window for designating program name appears when [Save As -> Robot] is selected.



(File dialog)

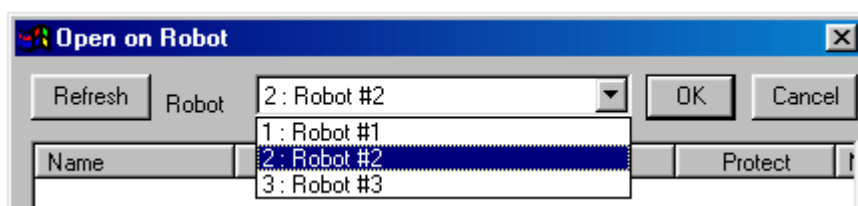
When  button is clicked, a dialog shown on the right appears. Select [ROBOT] and click on the [OK] button.

(Caution: When saving the program in the personal computer, click on the radio button next to "PC".)



Input "100" in the [File Name] area, and then click on the [OK] button.

To save the program in a robot controller other than robot controller 1, select the robot controller you want to save the program to.



(File name and OK button)

A dialog showing the save work progression will appear, and the edited program will be saved in the robot controller.

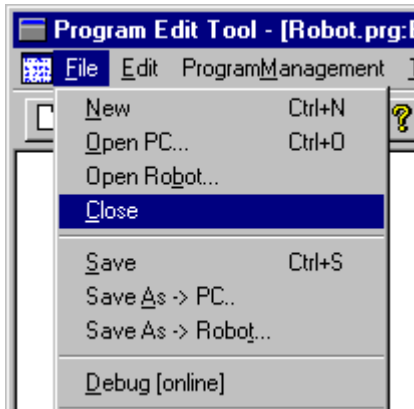


(Screen announcing save)

4. Starting use

Exiting the editing screen

Click on "Close" under the [File] menu, or click on the [x] button on the editing screen. The editing screen will close. This completes the program editing work.



("Close" in menu)



(Exit button)

4.6. Confirming the movement

Try starting the "P100" program with the robot controller's operating panel. Did the program run as anticipated?

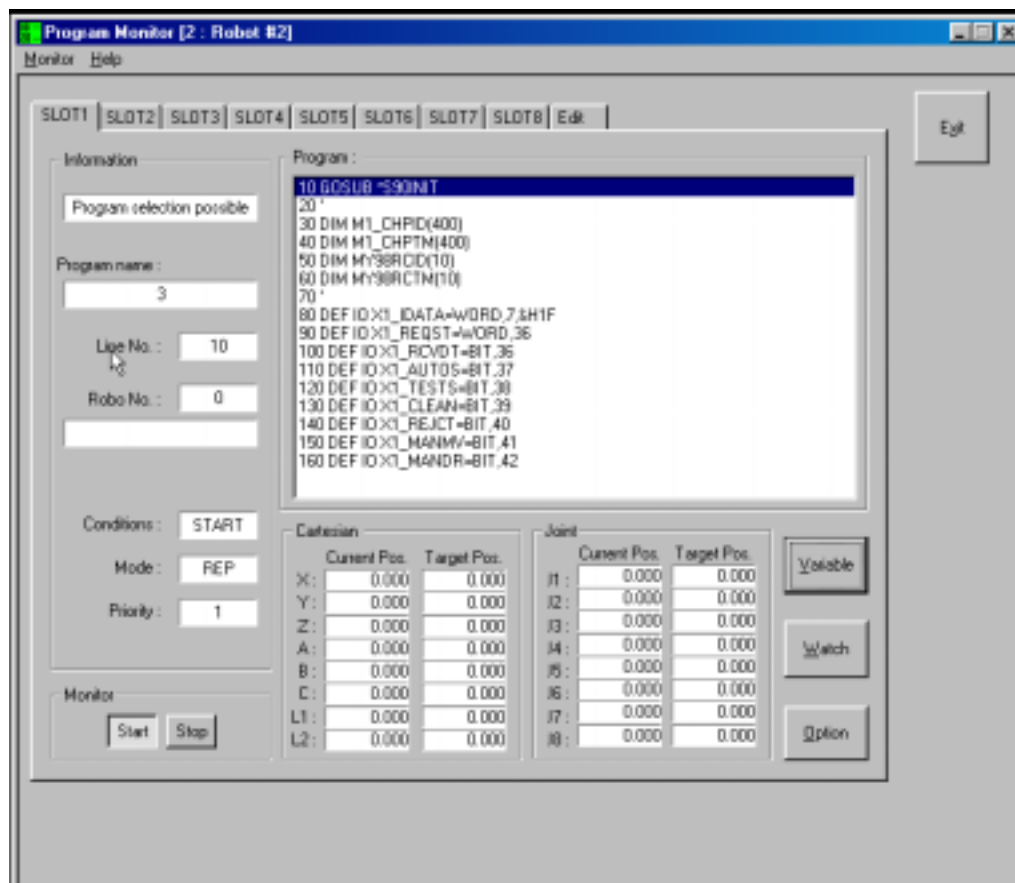
Try monitoring the robot movement

When the [Monitoring Tool] button on the Main screen is clicked on, the monitoring tool will start.

Click on [(4) Program Monitor] to start the program monitor.



Look at the program monitor and confirm the robot movement and program.

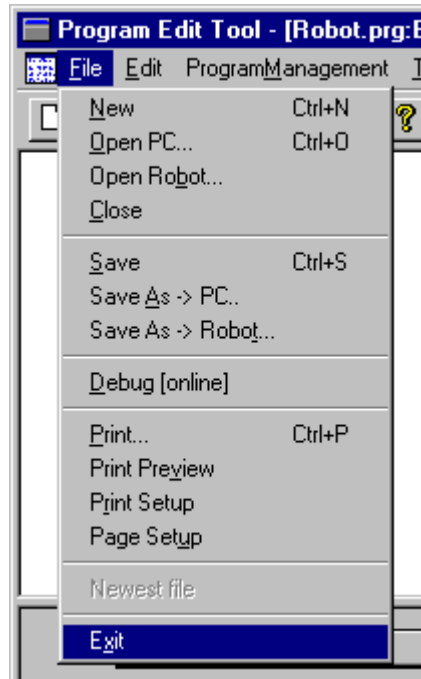


4.7. Exiting the operations

Did the robot move correct? Finally, exit the personal computer support software.

(1) Exiting the “Personal Computer Support Software”

Exit the program editing tool and monitoring tool. Each tool will exit when [Exit] is selected from under the [File] menu.



(Exiting the application)

(2) Exiting window

Exit windows by selecting [Shut Down...] -> [Shut down the computer?] -> [OK] under the [Start] button. Do not turn the personal computer off until a message indicating that it is okay to turn the personal computer power OFF appears on the screen. Depending on the personal computer, the power may turn OFF automatically.

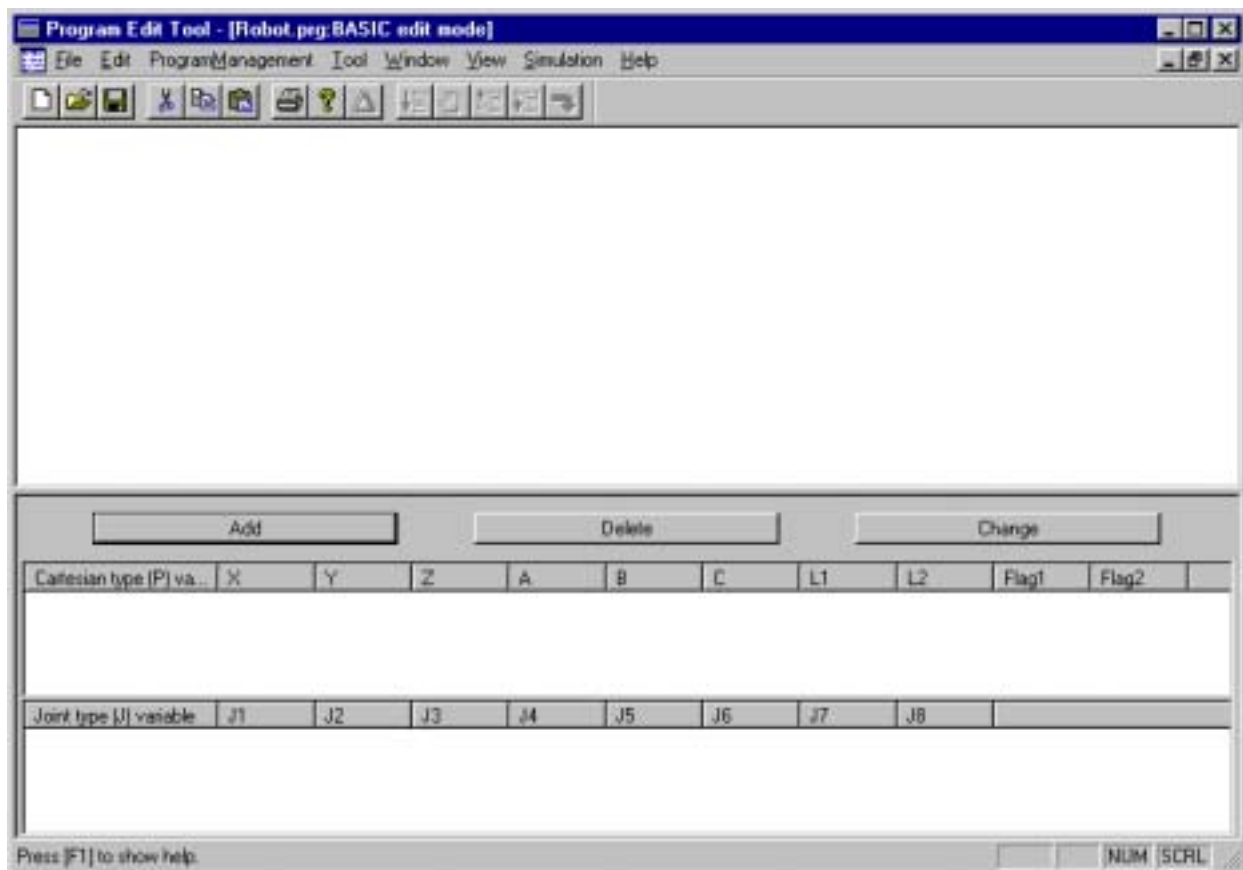
5. Explanation of each tool

The detailed operation methods of the personal computer support software are explained for each tool in this chapter.

5.1. Program editing tool

The methods of operating the program editing tool are explained in this section.

With the program editing tool, the robot program is created, the robot movement is debugged, and simulation (only for standard version) is carried out.



(Program editing tool)

5. Explanation of each tool

Setting the screen

The following settings can be customized with the program editing tool.

- (1). Font The font used with the program editing tool can be changed.
- (2). Variable display area setting The display ratio of the "orthogonal coordinate type variables" and the "joint coordinate type variables" on the Position Variable Edit screen can be changed.
- (3). Syntax check setting Whether to check the syntax, and whether to display messages if there are no syntax errors when saving the program can be set.
- (4) Programming Language The robot programming language can be changed. The language which can be switched is MELFA-BASIC IV and Movemaster-command. (Ver.B1 or more)
- (5) History The number of the document used recently displayed on a [File] menu is changed. (Ver.C2 or more)

[Setting method]

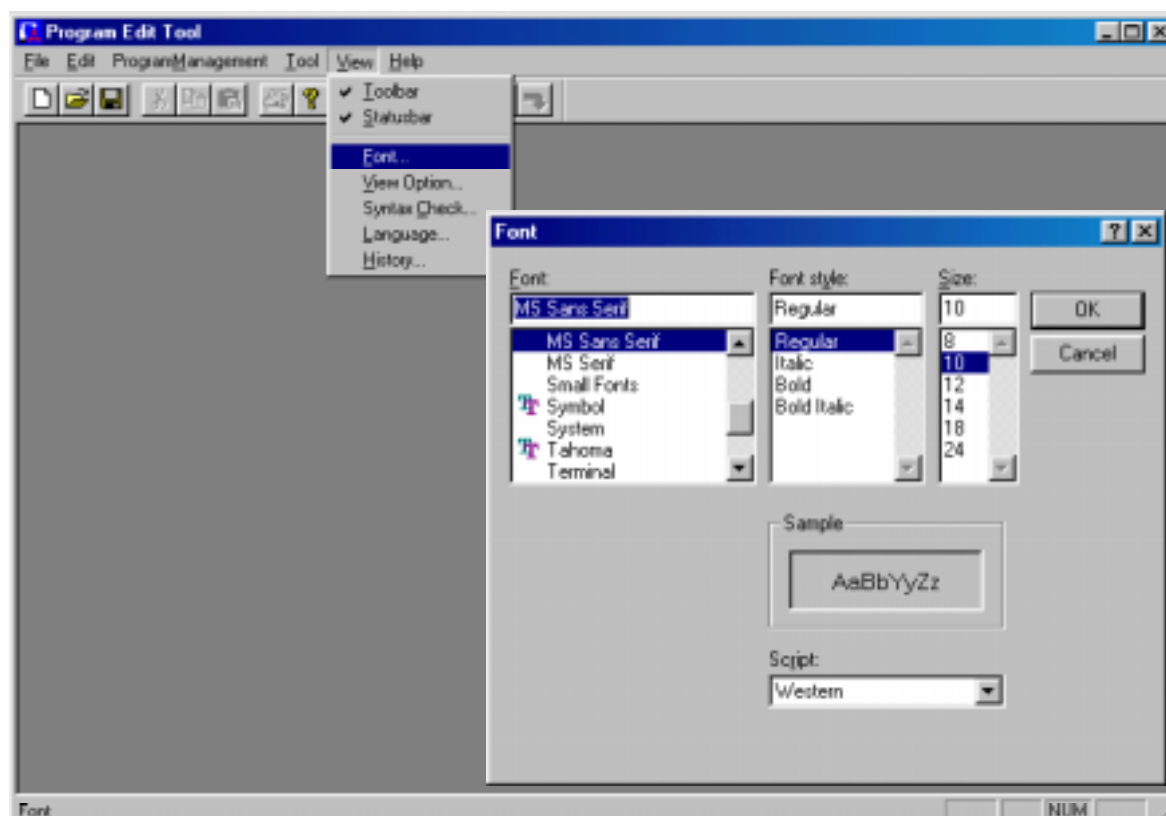
Click on the menu "File" - "Close", and close all of the program edit tool programs.

These settings can be made from the "View" menu.



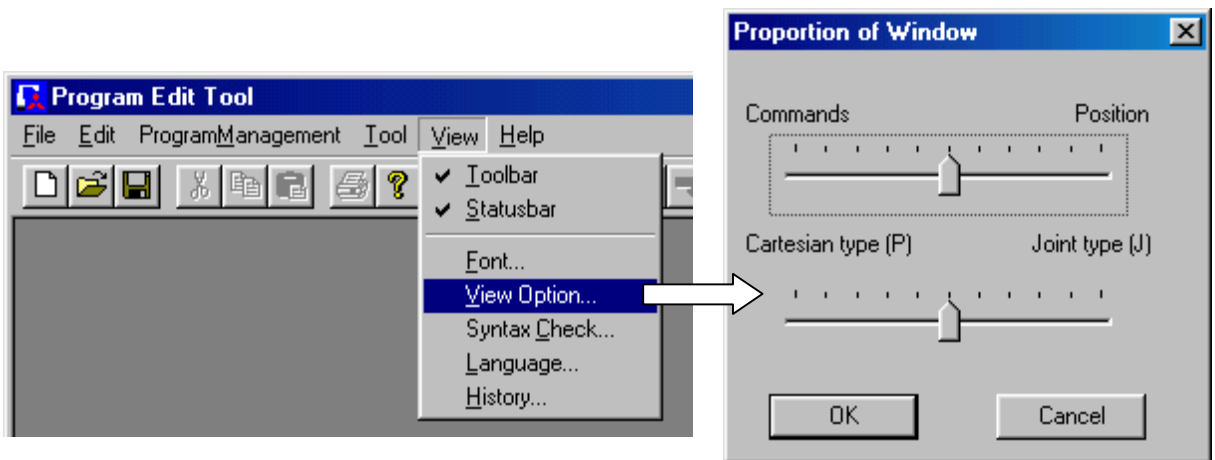
Font

The font used with the program editing tool can be changed. Click on the menu "View" - "Font".

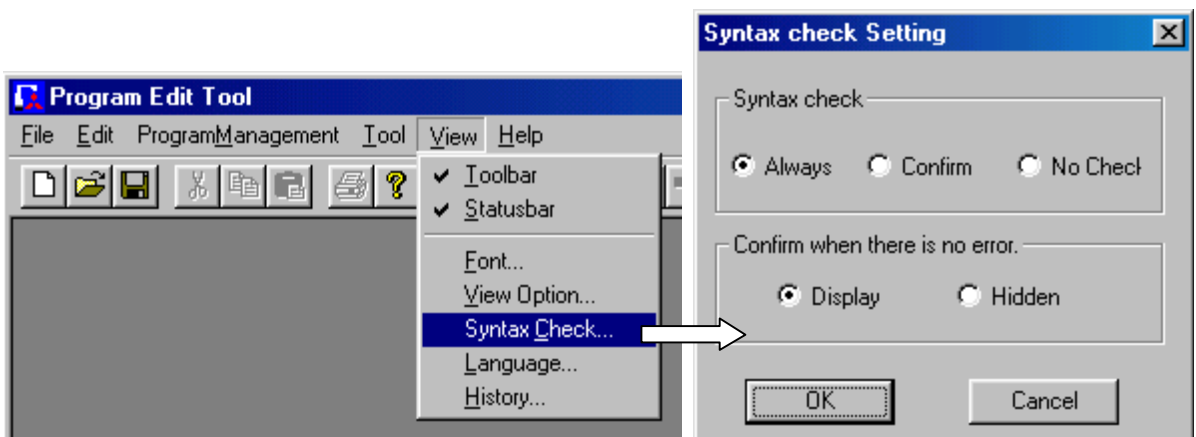


Variable display area setting

The display ratio of the "Cartesian type variables" and the "joint type variables" on the Position Variable Edit screen can be changed.

**Syntax check setting**

Whether to check the syntax, and whether to display messages if there are no syntax errors when saving the program can be set.



When each item is set, the following will occur.

Syntax check	Always	When the edited program is saved, the syntax will always be checked.
	Confirm	A message box confirming whether to check the syntax will appear when the edited program is saved.
	No Check	The syntax will not be checked when the edited program is saved.
Confirmation when there is no error.	Display	If there are no errors in the syntax check, the message "No syntax errors" will appear.
	Hidden	If there are no errors in the syntax check, no message will appear.

5. Explanation of each tool

Program Language

The robot programming language can be changed. The language which can be changed is as follows.

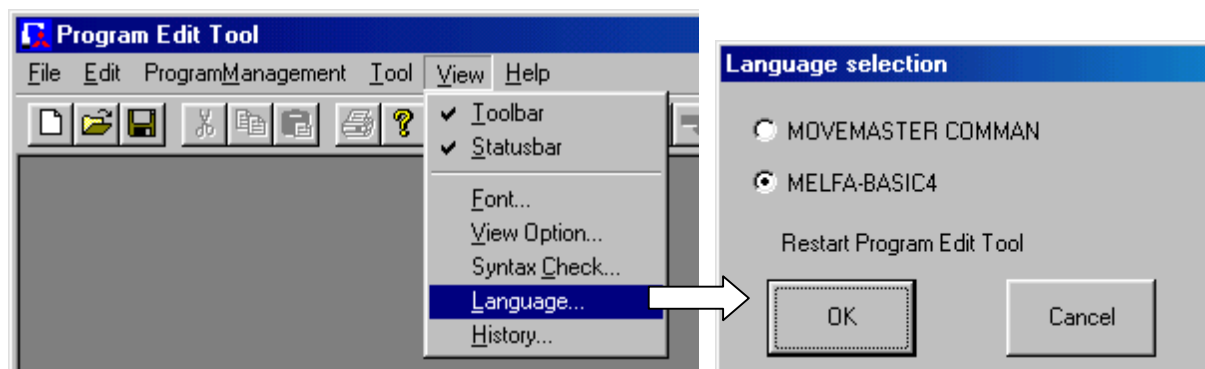
MELFA-BASIC IV

MOVEMASTER command

This change is possible by this software Ver.B1 or more.

However, MOVEMASTER command has the limitation in the robot model which can be used.

Check the "Standard Specifications", etc., to confirm that the model in use is compatible with the MOVEMASTER command.



If the language to be used has been changed in the Language Settings, end program editing once, and then restart.

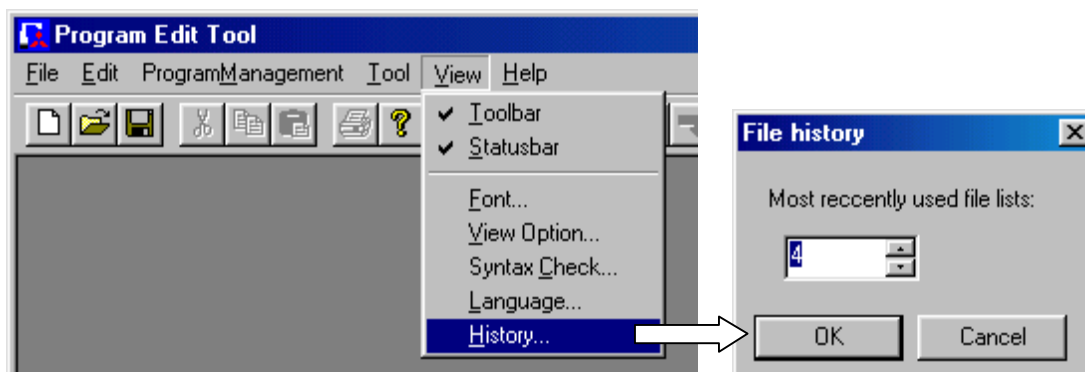
Caution) When MOVEMASTER command is used, the simulation function cannot be used.

History

The number of the document used recently displayed on a [File] menu is changed.

Please input the numerical values from 1 to 16.

This is possible by this software Ver.C2 or more.

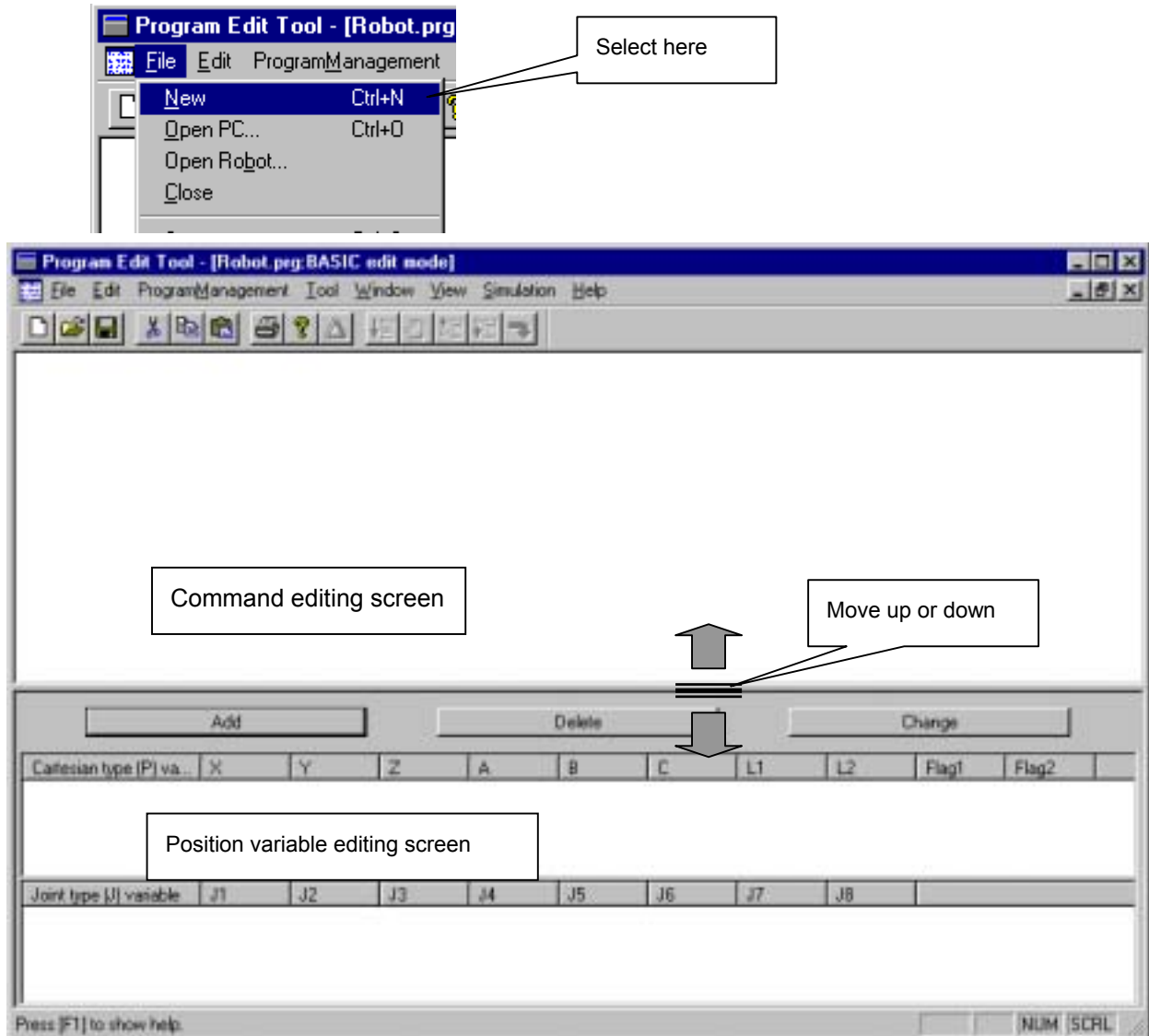


In order to confirm change, end program editing once, and then restart.

5.1.1. Editing screen

The methods of displaying and operating the editing screen are explained.

Select [New] under the [File] menu to display the program editing window.



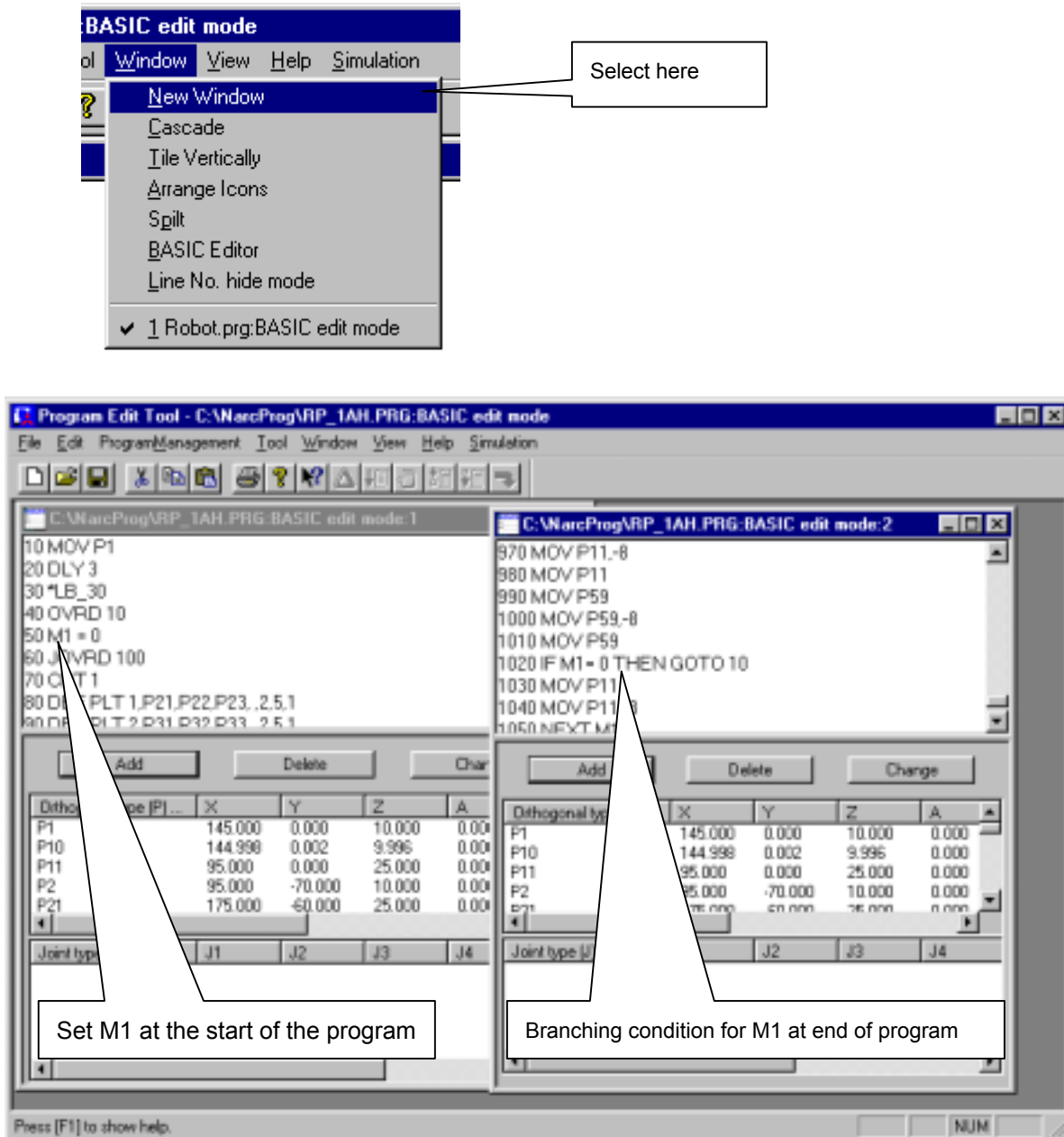
(Displaying the program editing window)

The upper side is the program command statement editing window, and the lower side is the position variable editing window.

To change the position of the upper/lower screen division, drag the boundary line with the mouse. This is handy for enlarging the command editing screen when there are many command statement lines, etc.

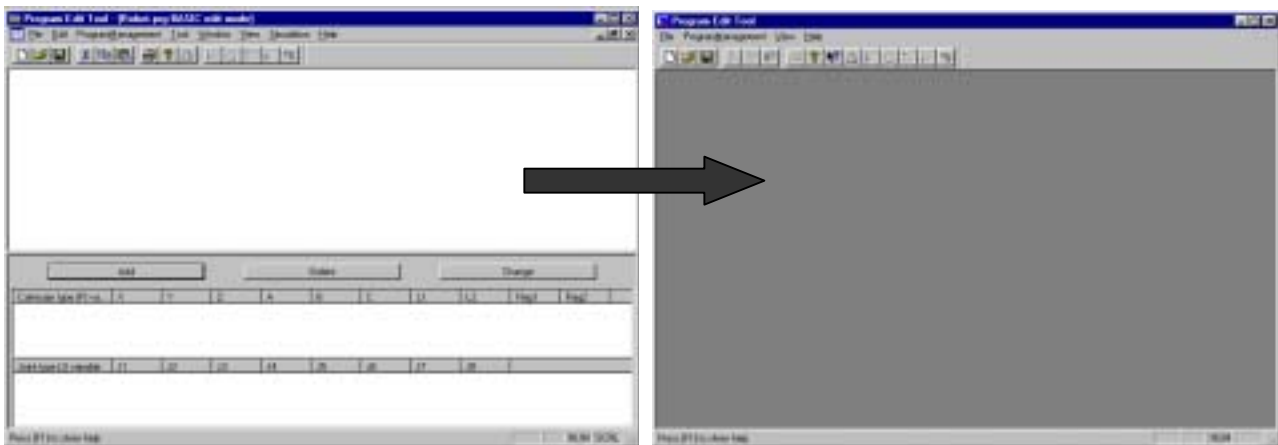
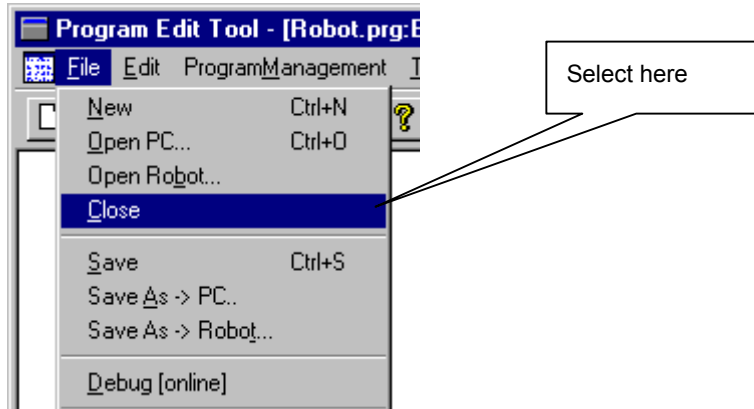
5. Explanation of each tool

To edit the program on multiple editing windows, select [New Window] under the [Window] menu. This function is handy for viewing both the head and end of the program when editing a program having many lines.



(Multiple editing windows)

To close the program editing window, select [Close] under the [File] menu.



5. Explanation of each tool

5.1.2. Editing mode

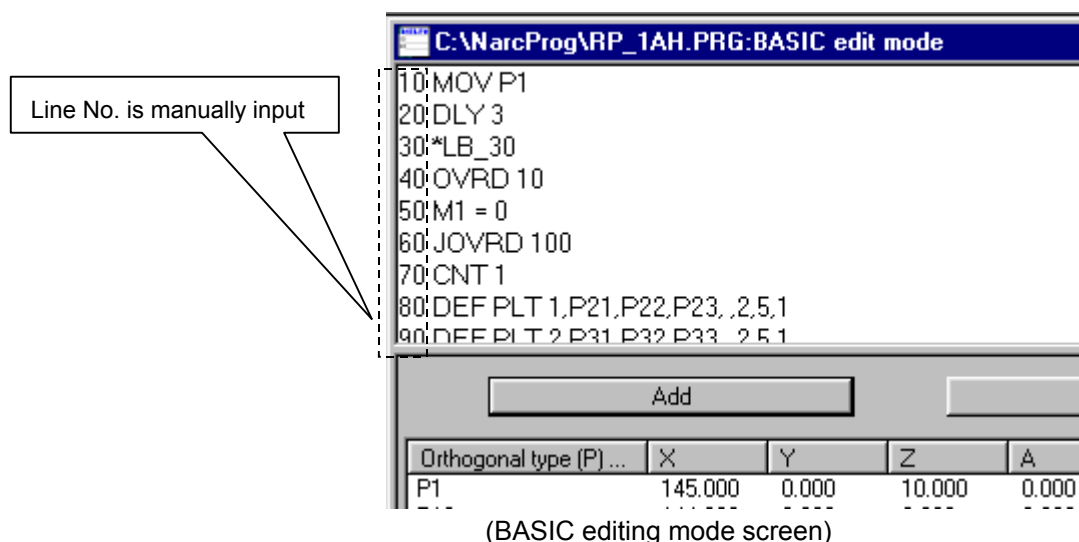
The program editing tool's editing mode is explained in this section.

The program editing window's editing modes include the "BASIC Editor" for persons familiar with the BASIC language editing tool used with the old controller, and the "Line No. hide mode" for persons familiar with Windows wordprocessing operations.

When the normal program editing window is started up, the "BASIC Editor" is entered. These two editing modes can be changed freely at any time.

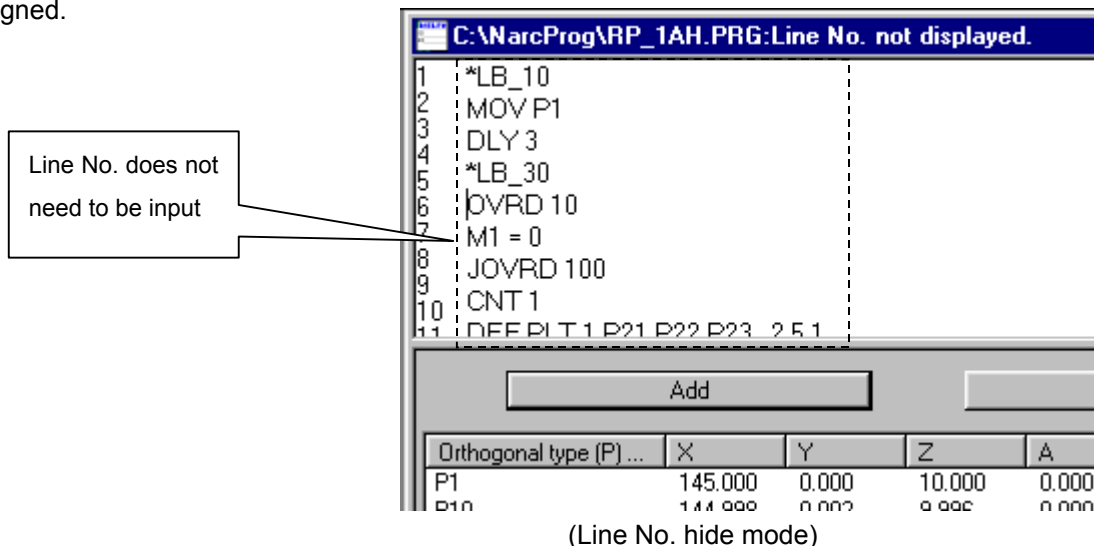
BASIC editing mode

This mode shows the robot program line Nos., and is intended for users familiar with the BASIC language editing tool used with the old controller. The robot program can be edited at the required line No. The AUTO function and renumbering function can be used to edit the line No.



Line No. hide mode

This mode does not show the robot program line Nos., and is intended for users familiar with Windows wordprocessing operations. The program commands can be edited without inputting the line No. The stop position is shown on the left edge. When the edited program is saved, the line Nos. are automatically assigned.



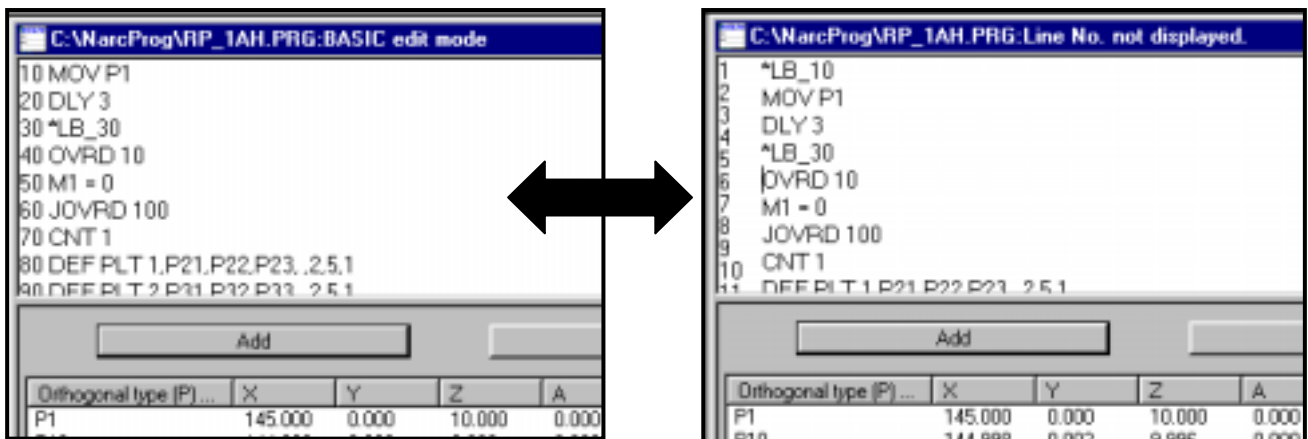
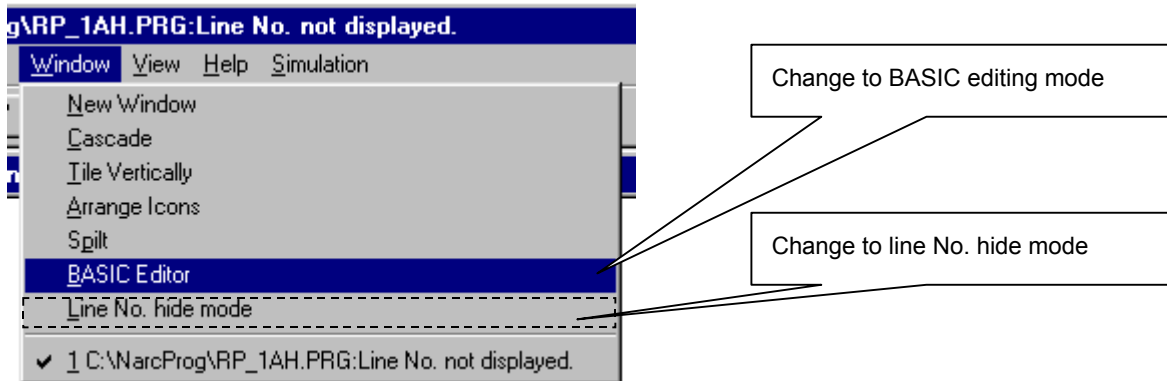
Line No. hide mode cannot be selected Movemaster-Command.

Changing the editing modes

The editing mode can be changed as explained below.

→ When [BASIC Editor] is selected from under the [Window] menu, the currently active editing program window will change to the BASIC editing mode.

→ When [Line No. hide mode] is selected from under the [Window] menu, the currently active editing program window will change to the line No. hide mode. When changing from the BASIC editing mode, the line no. assigned to the program command statement argument will be automatically converted into a label. Note that once the line No. is converted into a label it cannot be returned to a line No.



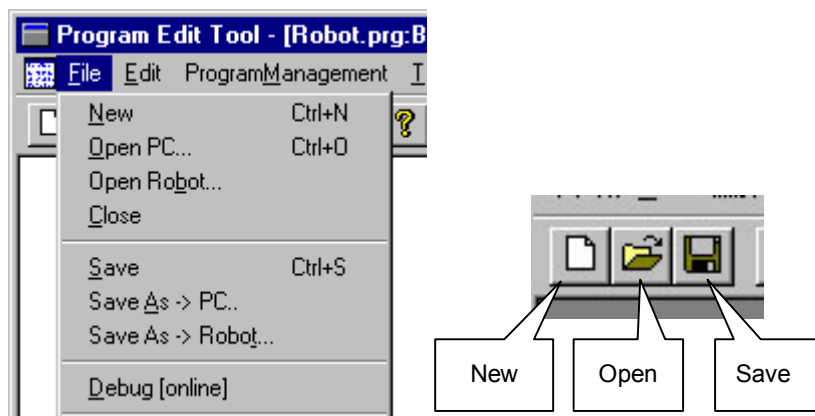
(State of mode conversion)

5. Explanation of each tool

5.1.3. Opening, closing and saving a program

The methods for opening and saving a program are explained in this section.

The operations related to opening, closing and saving an editing program are all carried out from the [File] menu or tool chip (buttons under menu).



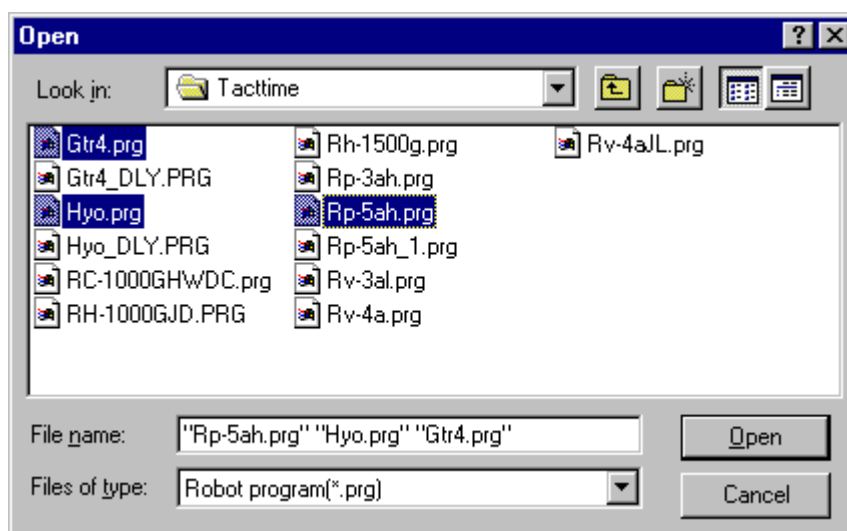
(([File] menu and tool chips)

Opening a program

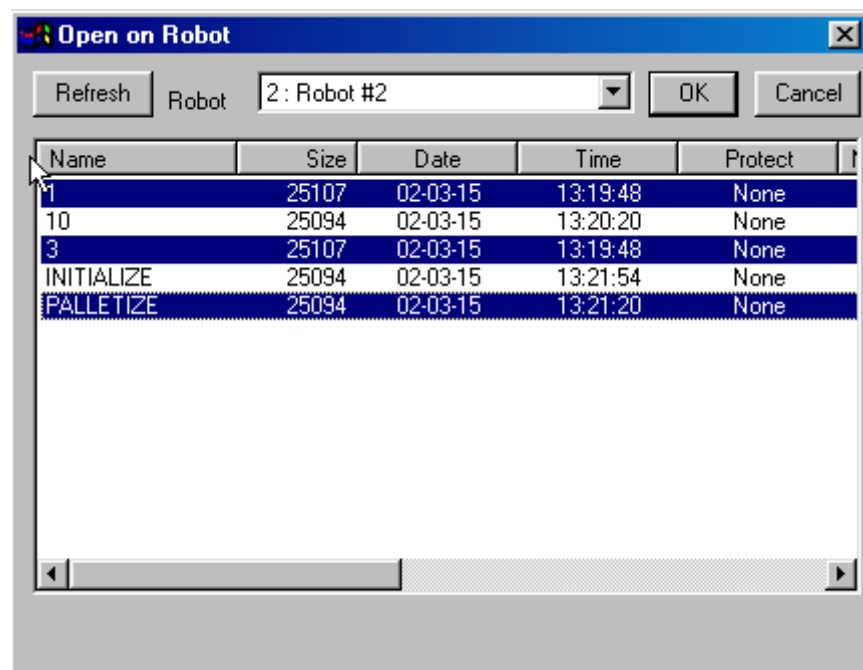
A program in the controller can be opened with the normal open method and with the debug open method.

Normal open : The contents of the program changed with editing are not saved unless "Save" is executed. There is no particular limit to the number of program files that can be opened. To open a program, select [Open PC] or [Open Robot] under the [File] menu.

A window shown below appears when [Open PC] is selected. It is also possible to open several files simultaneously by pressing the [Ctrl] key and making selection using the "mouse."



The following window appears when [Open Robot] is selected. It is also possible to open several files simultaneously by pressing the [Ctrl] key and making selection using the "mouse."

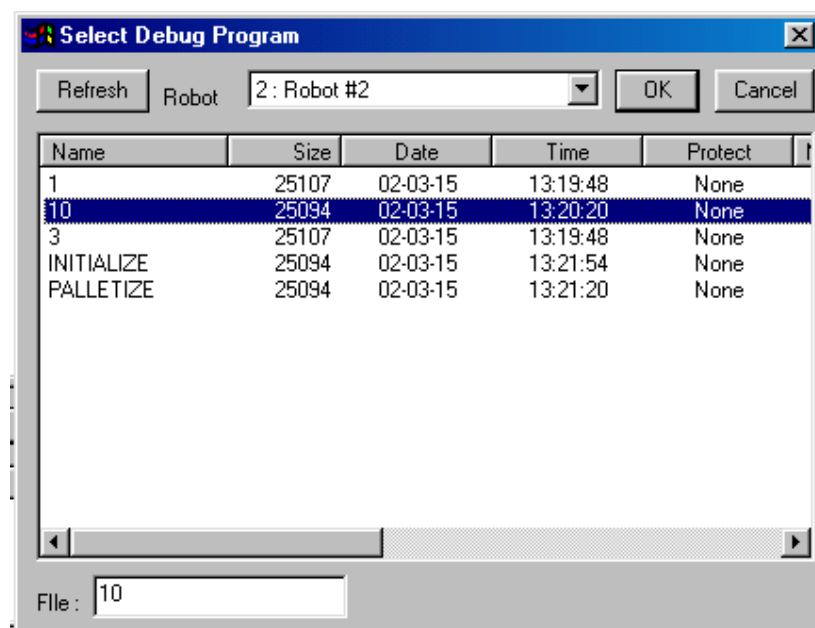


Updates to the latest information when [Refresh] button is clicked.

Refresh the information on this window when changing a file with teaching box, etc. or when connecting to a different robot controller.

If multiple robot controllers have been connected, the robot controllers can be switched by manipulating the combo box labeled "Robot."

Debug open : The contents of the program changed with editing are reflected immediately. The program file contents will be changed, even when "Save" is not executed. Depending on the opened program, the robot can be directly operated (step execution, direct execution). Only one program can be opened, and the editing mode is limited to the BASIC editing mode. To open a program, select [Debug] under the [File] menu.



5. Explanation of each tool

Closing and saving a program

Select the [Save As -> PC] or [Save As -> Robot] of the [File] menu in order to save an edited program.
Then the saving window appears accordingly. Name the file and save it.

Precautions for saving program in controller

When writing (saving) the robot program in the controller, first the program having the same name from the controller is deleted, and then the new program is written in.

With this, if an error occurs or the communication is canceled while transmitting the program from the personal computer to the controller, the original program in the controller will be erased.

In this case, the program can be recovered with the following procedures.

【Countermeasure】

- * If the program to be transmitted is being edited with the personal computer, remove the cause of the error, and then save again.

- * If the program editing has been exited:

A backup file is created in the folder where the personal computer support software is installed. (If the folder was not changed during installation, it is C:\Program Files\MelFa\.)

The name of this backup file has a Temp added before the name that was to be saved. (Ex.: Temp000)

Correct the file name with Explore, etc., and open the file with program editing, and then save the file.

[Correcting the file name]

Temp000

↓

000.prg

Delete Temp from Temp000, and add ".prg" as the extension.

5.1.4. Editing the program

The methods of editing a program are explained in this section.

Program language

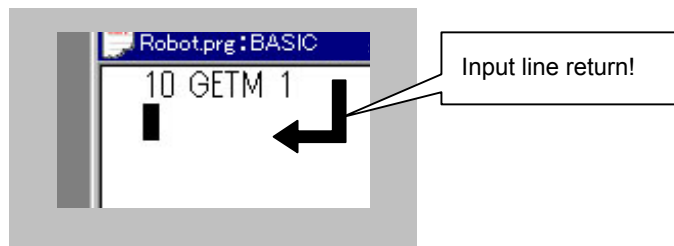
The MELFA-BASIC IV, or MOVEMASTER command language is used. Refer to the controller's language instruction manual for details on the command grammar and format, etc.

The following example shows the case when the MELFA-BASIC IV language has been selected.

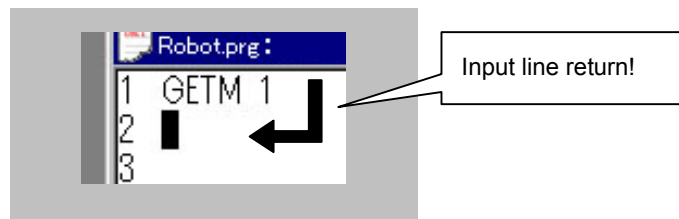
Registering a program command statement

Input the command statement as shown below.

For BASIC editing mode: Input the command statement after the line No., and input by pressing the ENTER key.



For line No. hide mode: Input only the command statement, and input by pressing the ENTER key.



Caution : If the ENTER key is not pressed to input the command statement, it will not be recognized as a robot program.

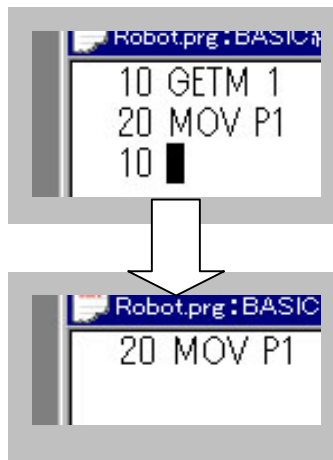
Caution : All characters except comment text and character string data (enclosed by ") are automatically converted into uppercase characters.

5. Explanation of each tool

Deleting a program line

Use the following procedure to delete a designated line No. line from the program.

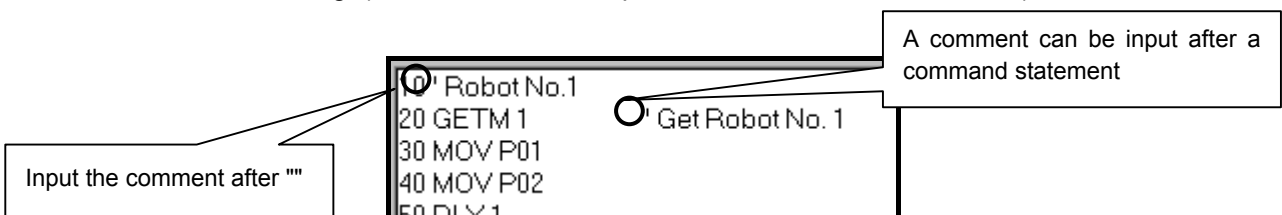
For BASIC editing mode: Describe only the line No., and then press the ENTER key.



For line No. hide mode: Delete the command statement by pressing the Backspace key or Delete key.

Describing a comment statement

To write a comment statement, input " ' " at the head of the command statement, and then input the comment character string. (A comment can be input after the command statement.)

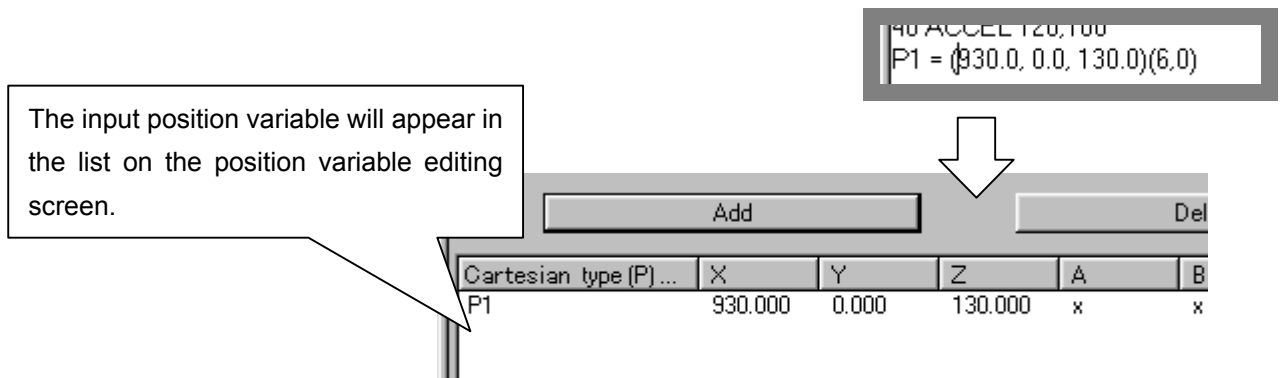


(Example of inputting a comment statement)

Inputting a position variable (Only for BASIC editing mode)

By inputting a position variable definition following the robot program grammar as shown below, the P type and J type position variable can be input.

(When using the line No. hide mode, input from the position variable editing window.)

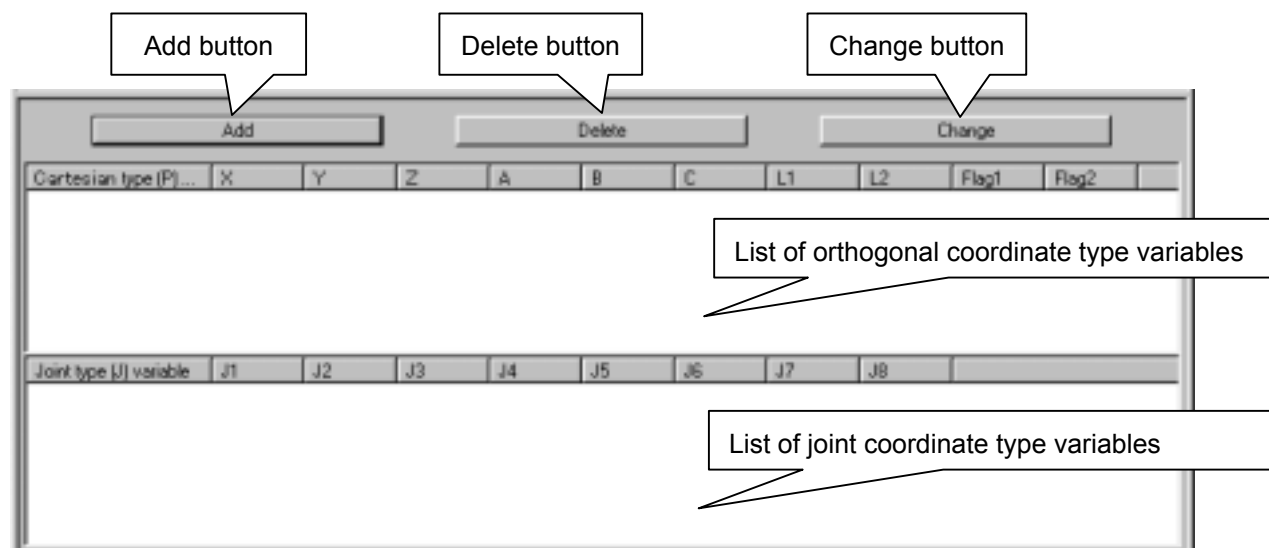


5. Explanation of each tool

5.1.5. Editing the position data

The methods of editing the position variables are explained in this section.

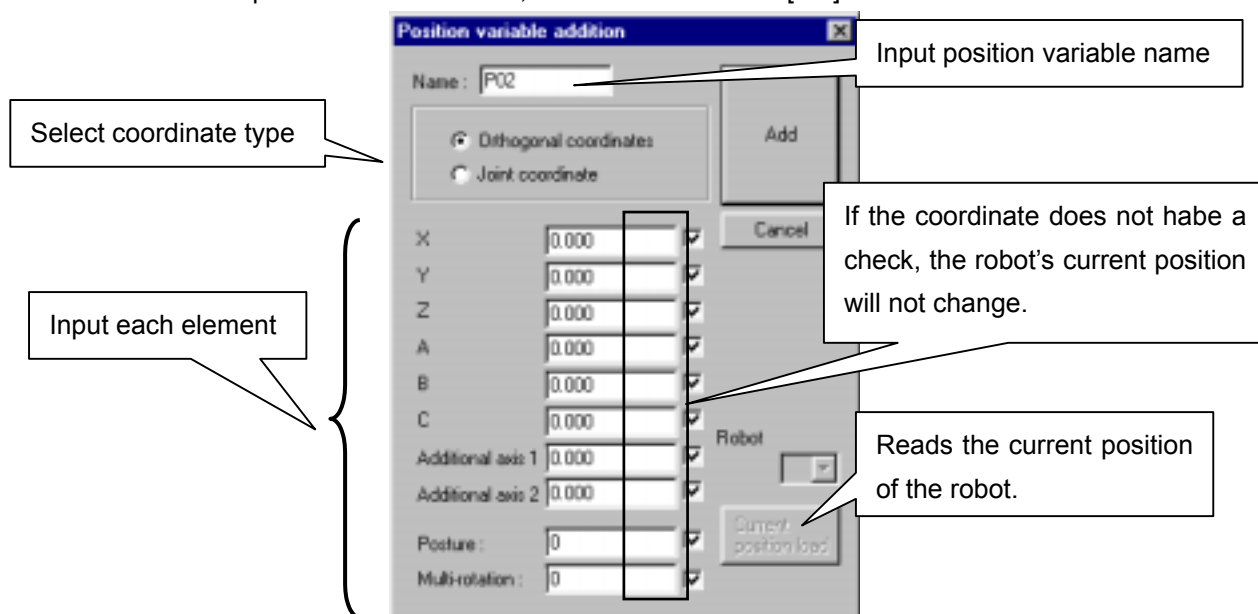
The position data is edited on the position data editing screen. The list on the top is for the orthogonal coordinate type variables, and the list on the bottom is for the joint coordinate type variables. Only the name of the array variable is displayed.



(Position variable editing screen)

Adding a position variable

Click on the "Add" button. The following position variable addition dialog will appear. Select the orthogonal coordinate type or joint coordinate type. The element value with a checked box is defined. Input each element value and position variable name, and then click on the [OK] button.

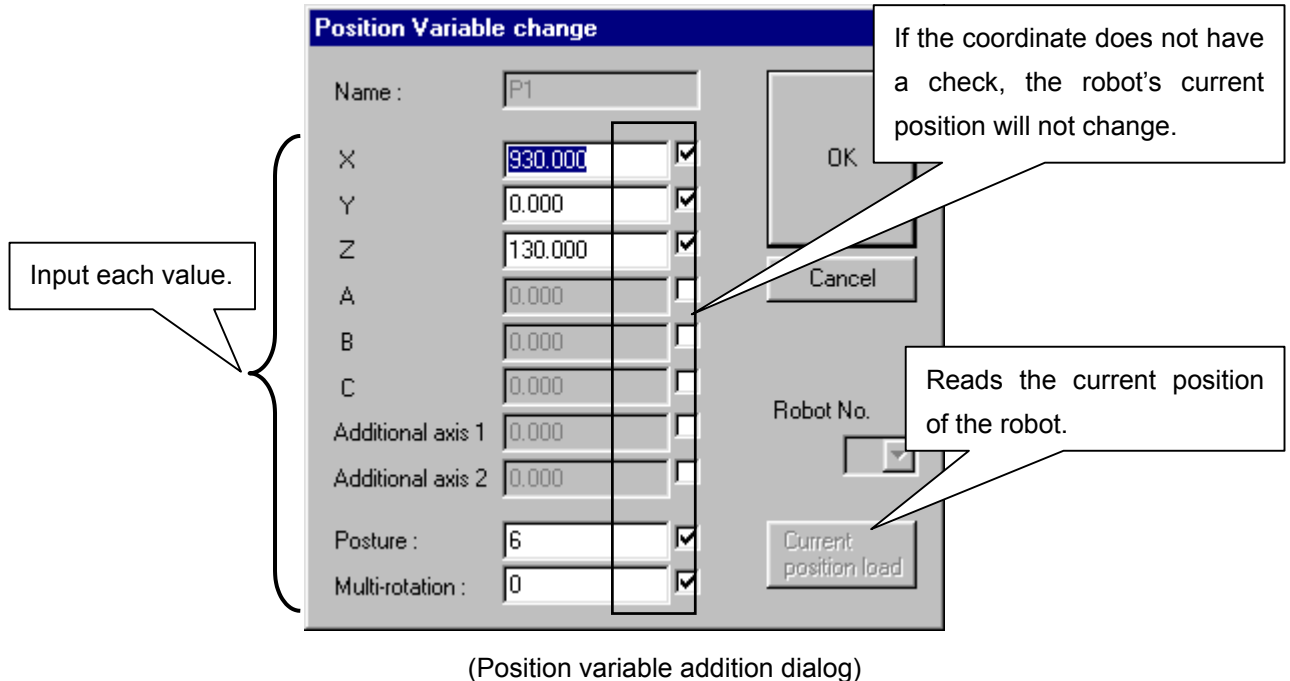


(Position variable addition dialog)

(Caution: The unit of the element being used differs according to the robot being used. Refer to the instruction manual of the respective robot.)

Changing the position variable

Select a variable from the position variable list and click on the "Change" button, or double-click on the position variable to be changed. The following dialog for changing the position variable will appear. The element value with a checked box is defined. Input each value and then press the [Update] button to change the value.



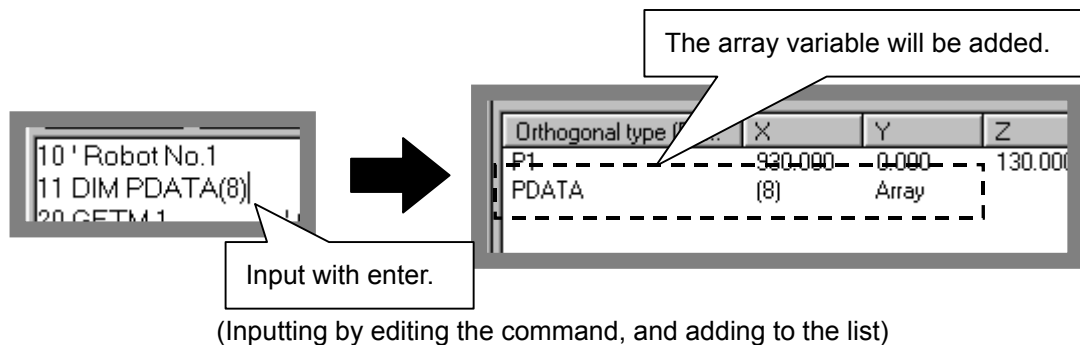
(Caution: The unit of the element being used differs according to the robot being used. Refer to the instruction manual of the respective robot.)

Deleting a position variable

Select a variable from the position variable list, and click on the "Delete" button. The selected position variable will be deleted.

Adding a position variable array

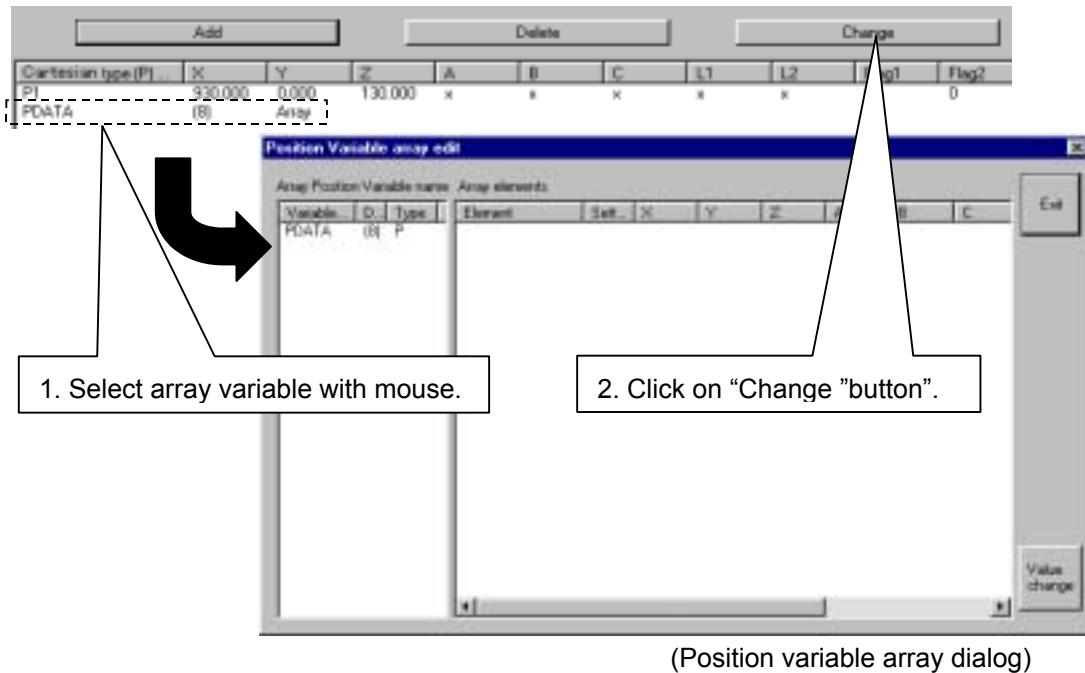
To add a position variable array variable, input a DIM command statement that defines the position variable array variable in the command editing window. The name and No. of array elements of the input position array variable will appear in the position variable list.



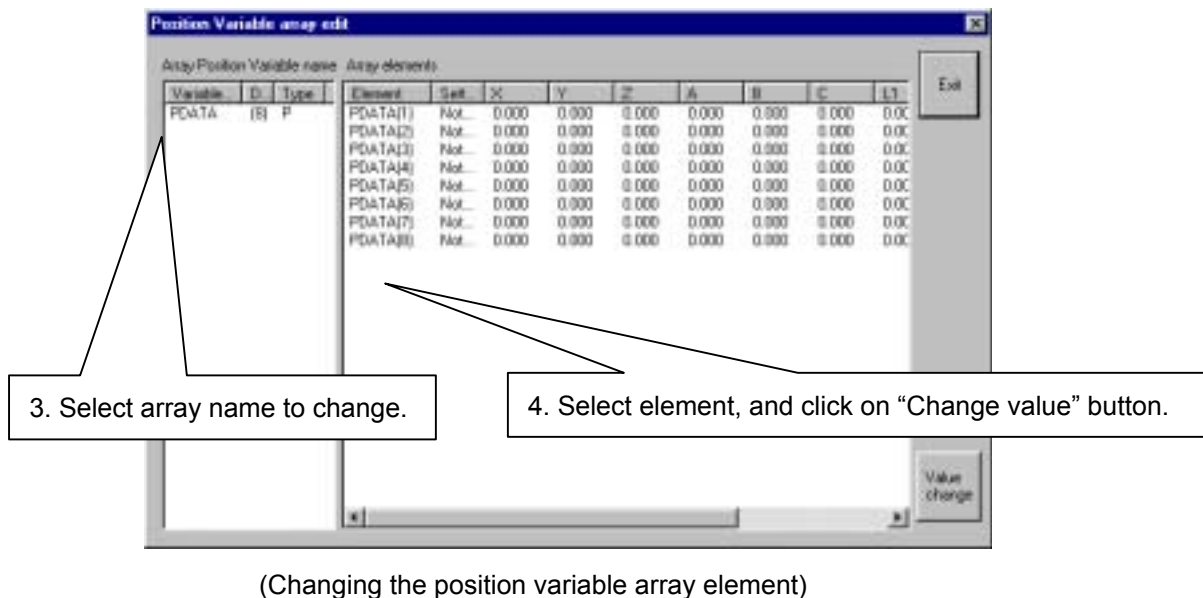
5. Explanation of each tool

Changing the position array variable

Select the array variable from the position variable list and click on the "Change" button, or double-click on the position array variable to be changed. The following dialog for changing the position variable array will appear.

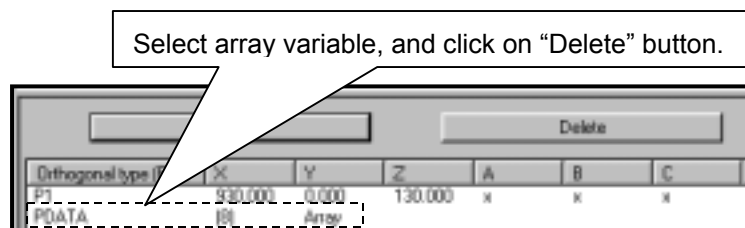


Select the position variable to be edited from the list on the left, and then select the element to be changed from the array element list on the right. Then, click on the "Change value" button, or double-click on the position array variable to be changed. A position variable change dialog will appear in the same manner as for the normal position variable.



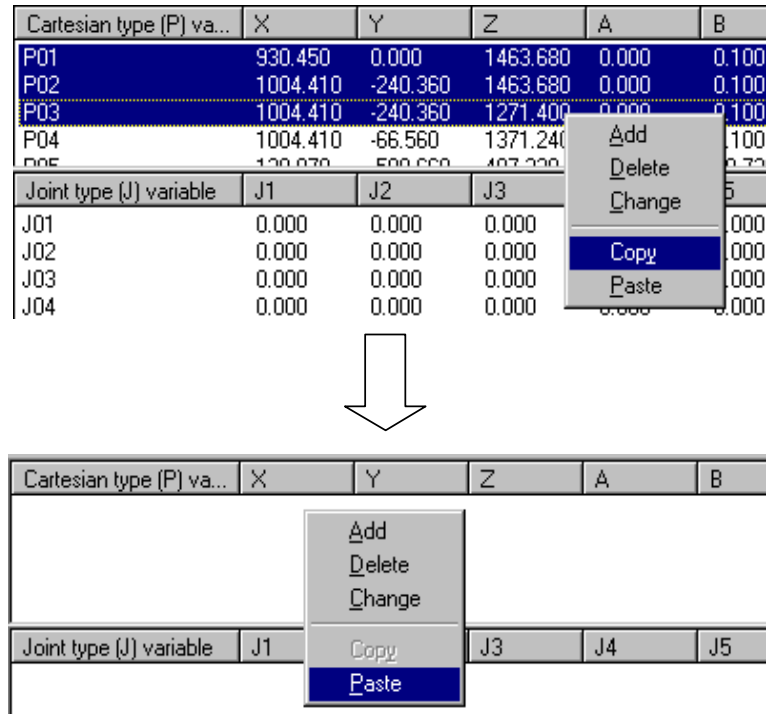
Deleting a position array variable

Select the array variable from the position variable list, and click on the "Delete" button. All elements of the selected array variable will be deleted.



Copying and pasting the position array variables

If a variable is selected from the position variable list and right-clicked with the mouse, the menu is displayed. If "Copy" is selected from the menu, the contents of the variable are copied into the clipboard. If the mouse is right-clicked and "Paste" is selected on the desired position edit screen, the variable can be pasted on another edit screen.



(Copying and pasting the position array variables)

If the same variable name is found, the overwrite confirmation dialog is displayed. Select whether to overwrite or copy after changing the name.

The array variables can be copied, however, it is necessary to define the array variables of the position variables by the DIM command on the command edit window in advance.

Please note that this software does not display the message of the overwrite confirmation when the array position variable is overwritten.

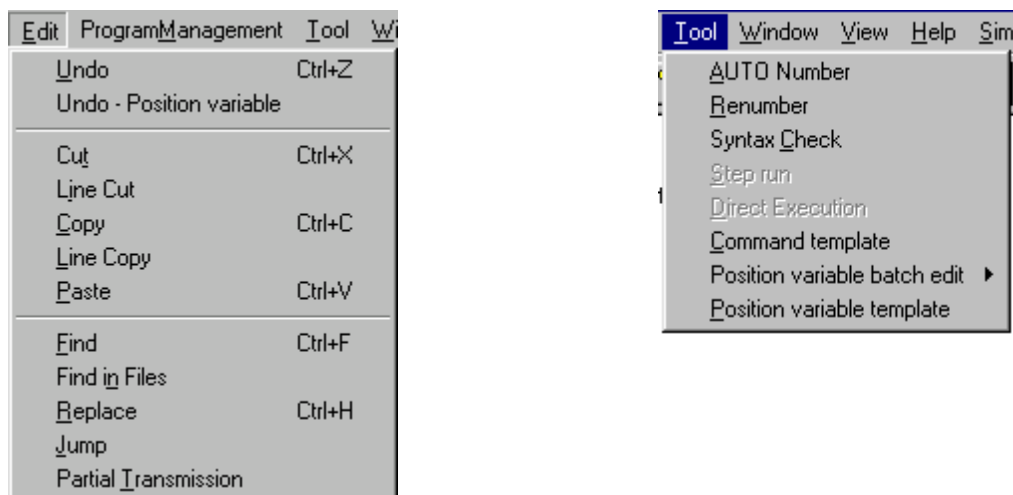


5. Explanation of each tool

5.1.6. Editing auxiliary functions

The editing auxiliary functions, helpful for editing commands, are explained in this section.

The editing auxiliary functions such as copy, cut, insert, search, replace and jump can be used from the [Edit] menu and [Tool] menu.



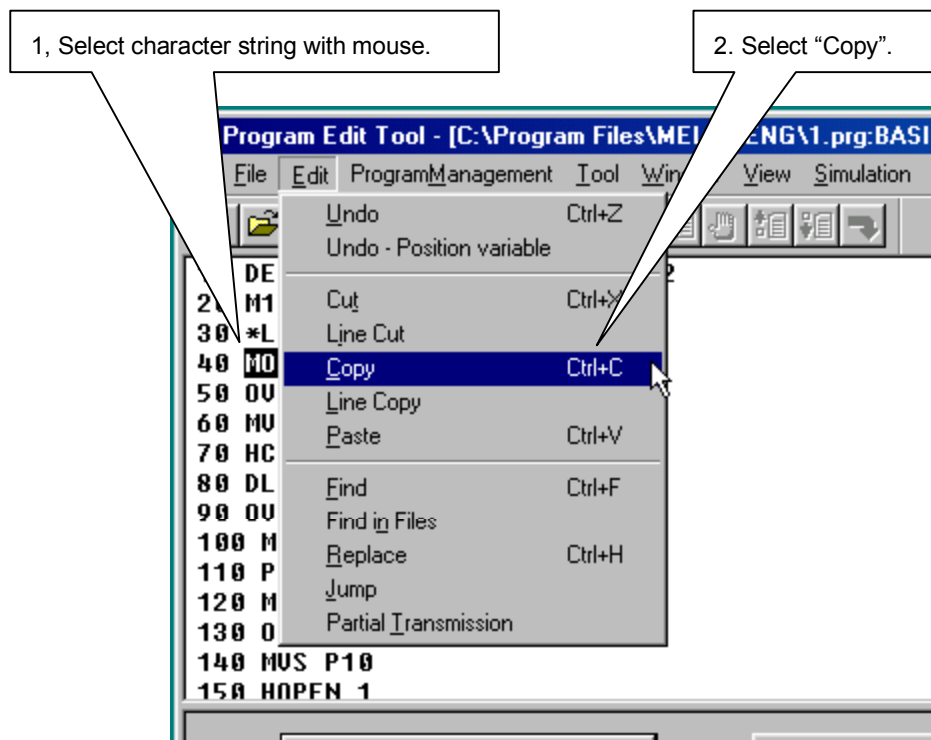
(Edit and Tool menus)

Copy

The character string selected with the mouse is copied. When a range over several lines is selected with the BASIC editing mode, line copy will be executed.

To copy, select [Copy] under the [Edit] menu.

By using the paste function described later, the copied character string can be pasted into another section of the program.

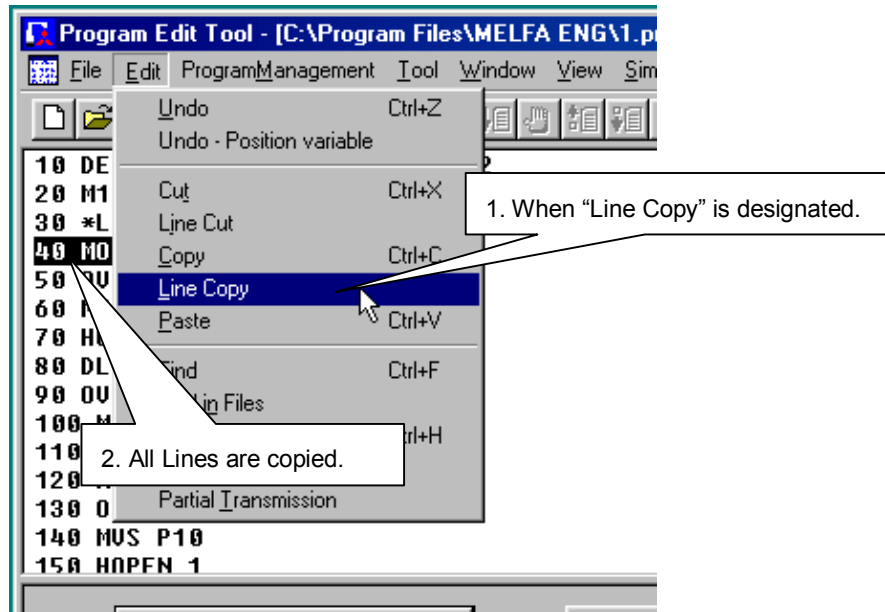


Line copy (Only BASIC editing mode)

The lines containing the selected range are copied.

To carry out line copy, select [Line Copy] under the [Edit] menu.

By using the paste function described later, the copied character string can be pasted into another section of the program.



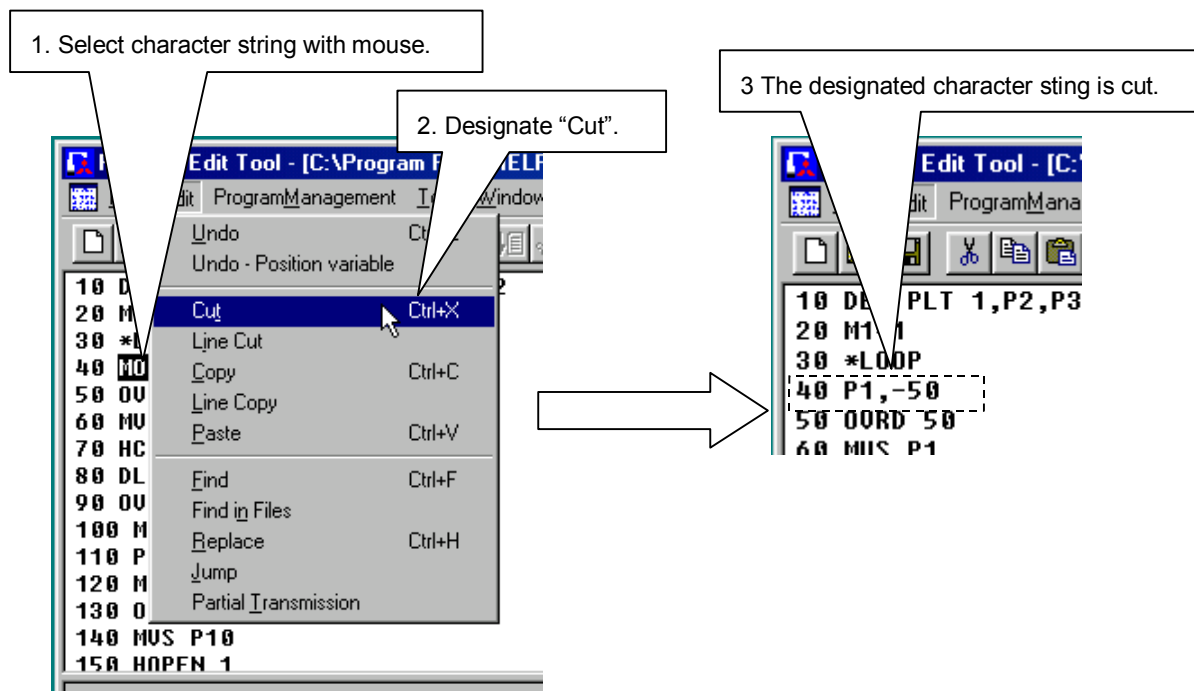
(State of line copy)

Cut

The character string selected with the mouse is cut. When a range over several lines is selected with the BASIC editing mode, line cut will be executed.

To cut, select [Cut] under the [Edit] menu.

By using the paste function described later, the cut character string can be pasted into another section of the program.



(State of cutting)

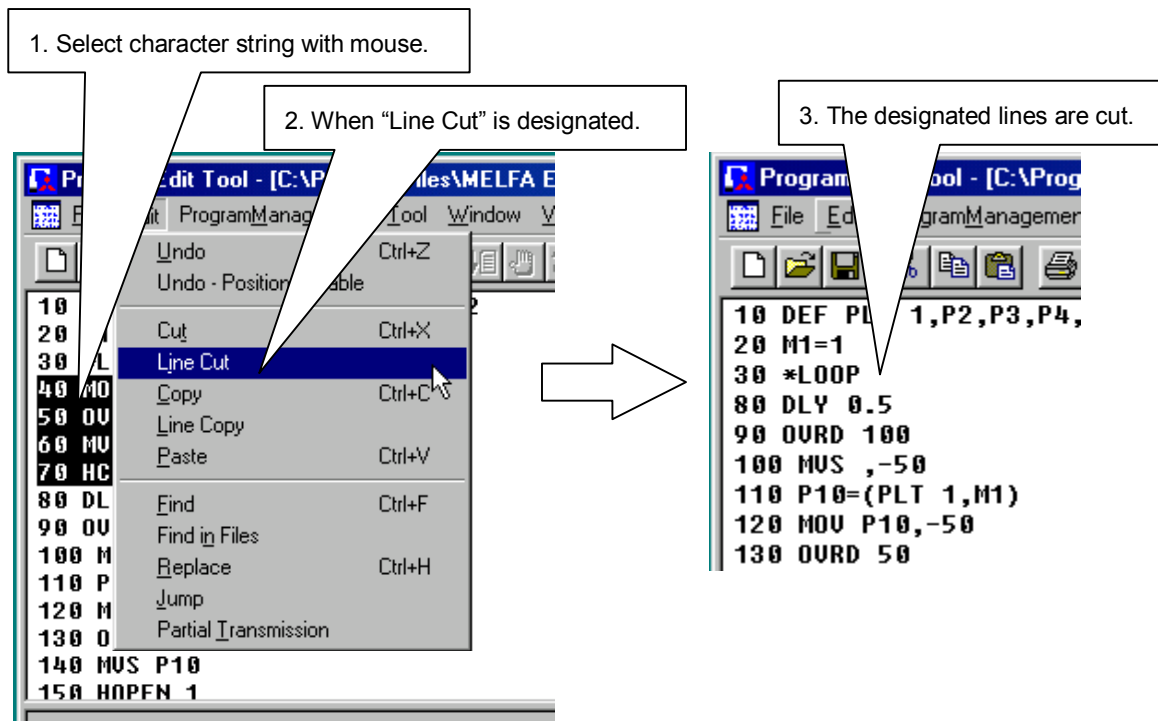
5. Explanation of each tool

Line cut (Only BASIC editing mode)

The lines containing the selected range are cut.

To carry out line cut, select [Line Cut] under the [Edit] menu.

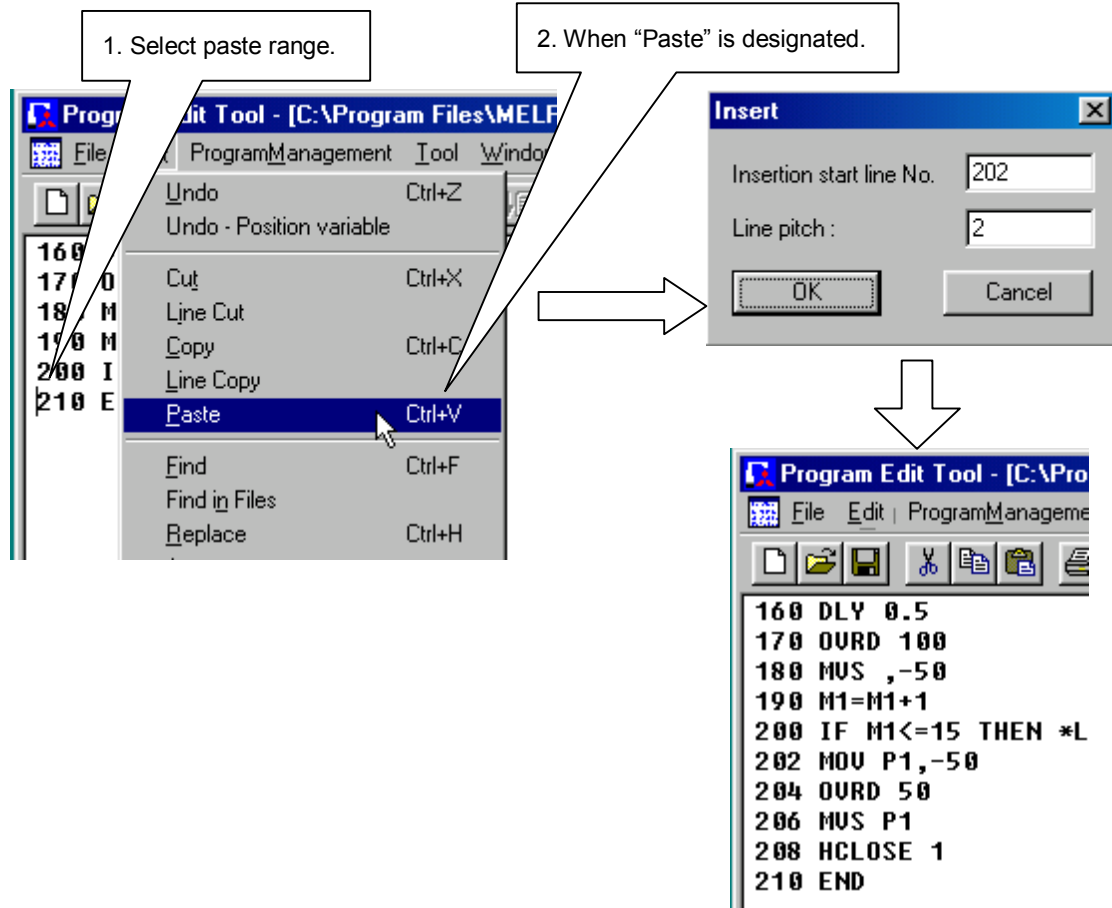
By using the paste function described later, the cut character string can be pasted into another section of the program.



Paste

The character string or line that has been copied or cut is pasted into the selected range. When line paste is carried out, the line Nos. are automatically reassigned so that the lines fit into the paste range. (Only in BASIC editing mode)

To carry out paste, select [Paste] under the [Edit] menu.



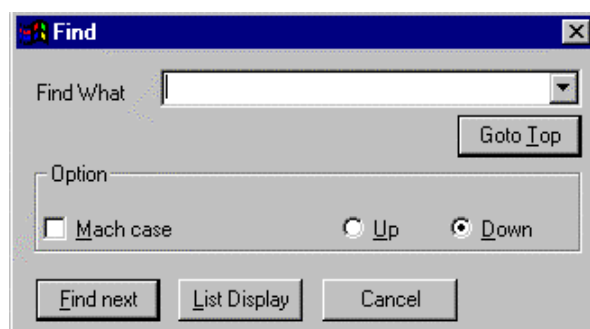
(Paste dialog and state of pasting)

5. Explanation of each tool

Search

The designated character string is searched for. The search range can also be designated.

Select [Search] under the [Edit] menu to display the Search screen.

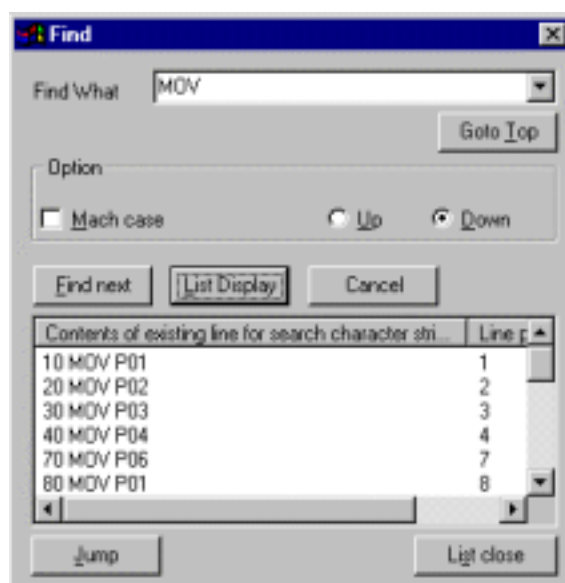


(Search dialog)

[Search Next] : Each time this button is clicked on, the next character string will be searched for.

[List display] : The results of searching the designated range will appear in the list.

If the list display is clicked, the window is enlarged automatically. By double-clicking the search result, or clicking [Jump] after selecting, the list moves to the corresponding command line.



(Search result list display)

Find In Files

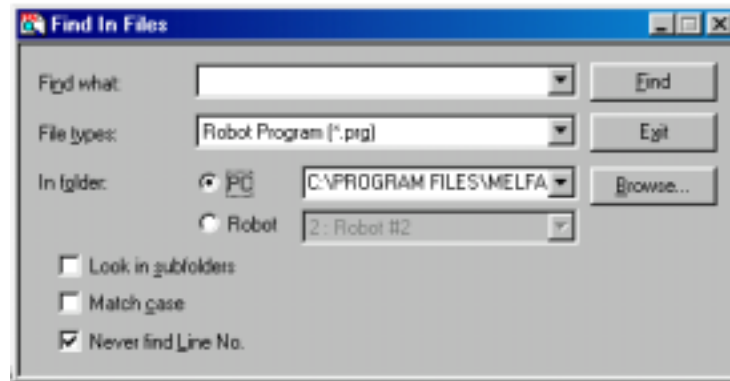
The specified character string is searched from files. (This function corresponds after a version C2.)

To display the [Find In Files] screen, select [Find In Files] from the [Edit] menu.

The same screen can also be displayed by clicking the [Find In Files] button in Program Manager.

The targets of searching from robot controller are only the robot programs.

And the searching from robot controller takes time for a while, because of communicating with it.



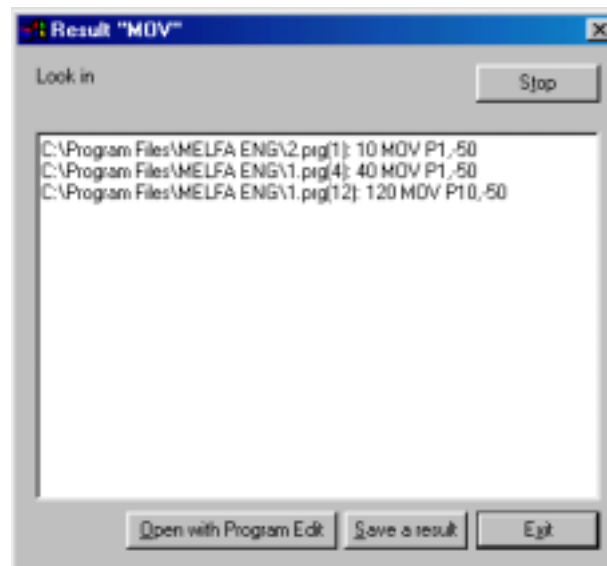
(Find In Files dialog)

[Never find Line No.] : Checking this does not search line numbers from robot programs at all.

When a search is started, a window displaying search results appears. The result is displayed as follows.

C:\Program Files\MELFA\1.prg(4) 40 MOV P1,-50

File name Line number from the top Contents



(Search result list display)

[Stop] : Aborts the search.

[Open with Program Edit] : Opens the program displayed on the line, which has been selected from the search result list, using the Program Edit tool.

Any of the displayed programs can also be opened using the Program Edit tool by double-clicking the desired line.

[Save a result] : Saves the search result list in a file.

Once the search is completed, the [Open with Program Edit] button and the [Save a result] button can be performed.

5. Explanation of each tool

Replace

The designated character string is replaced. The replacement range can also be designated.

Select [Replace] under the [Edit] menu to display the Replace screen.



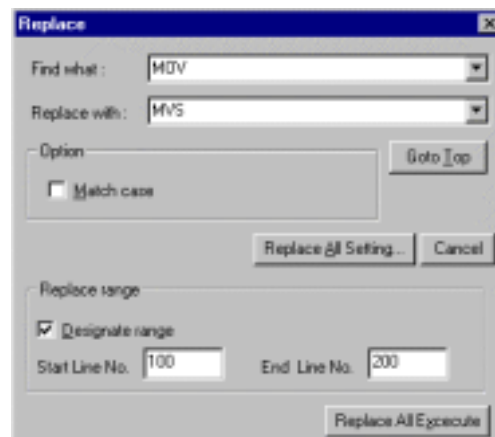
(Replacement dialog)

[Search Next] : Each time this button is clicked on, the next replacement target character string will be searched for.

[Replace Next] : Each time this button is clicked on, the next replacement target character string will be replaced.

[Replace All Setting] : If this button is clicked, the items for designating the range are displayed in order to replace all of the designated character string.

[Replace All Setting] can be used to replace the designated character string by designating the replacement range.



(Replace All)

⚠ CAUTION

****1 Cautions for replacement work***

Do not change numeric values in PC support software prior to version C1.

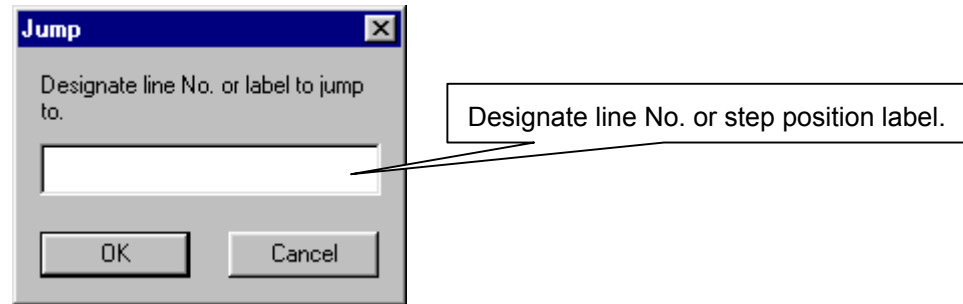
If numbers are replaced, the line Nos. will also be replaced, and the program contents will change.

(For example, if 40 is replaced with 30, the same command as line No. 40 will be overwritten on line No. 30.)

If numbers are inadvertently replaced, carry out [Undo] under the [Edit] menu before carrying out any other operation.

Jump

The program jumps to the designated label, line No. (BASIC editing No.) or step position (line No. hide mode). To carry out jumping, select [Jump] under the [Edit] menu.

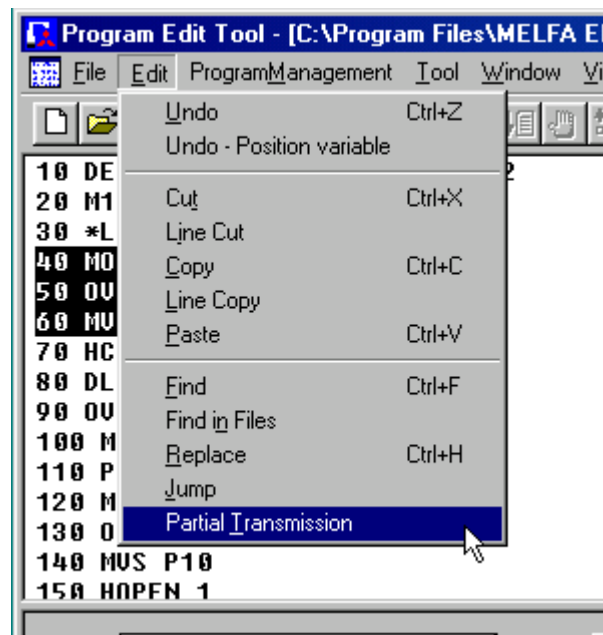


(Jump)

Partial Transmission

The selected program line is written into the robot controller.

This is convenient to reflect the contents of modification in which only a portion of the program was modified in the robot controller. However, exercise caution since only the selected portion will be written. After selecting the line to be written into the robot controller, select [Partial Transmission] in the [Edit] menu.

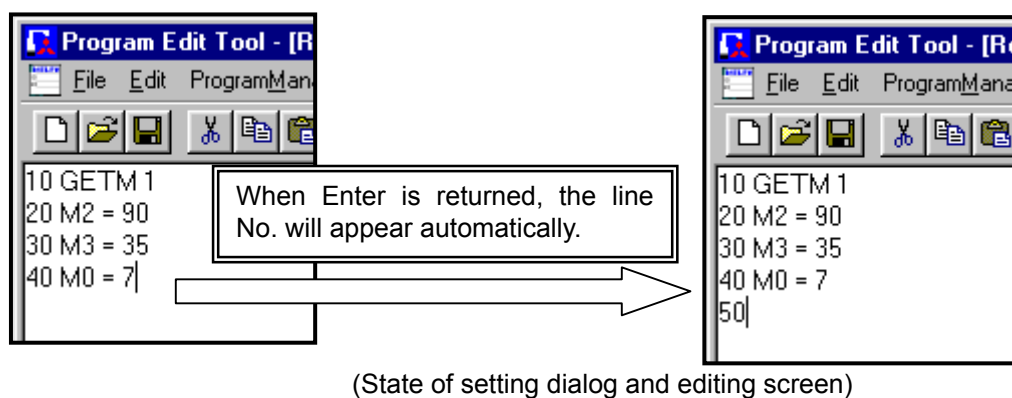
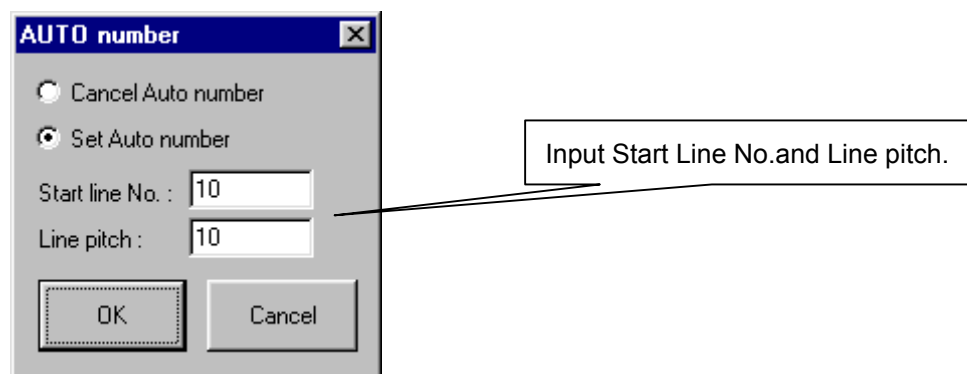


5. Explanation of each tool

Automatic numbering (Only BASIC editing mode)

This function automatically displays the next line No. each time the Enter key is pressed. The start line No. and line pitch can be designated with the setting dialog.

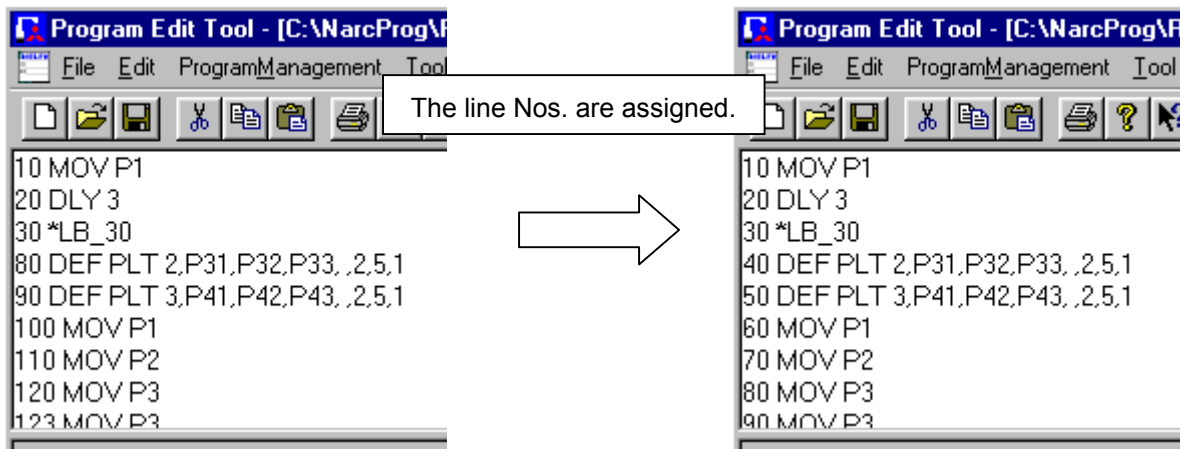
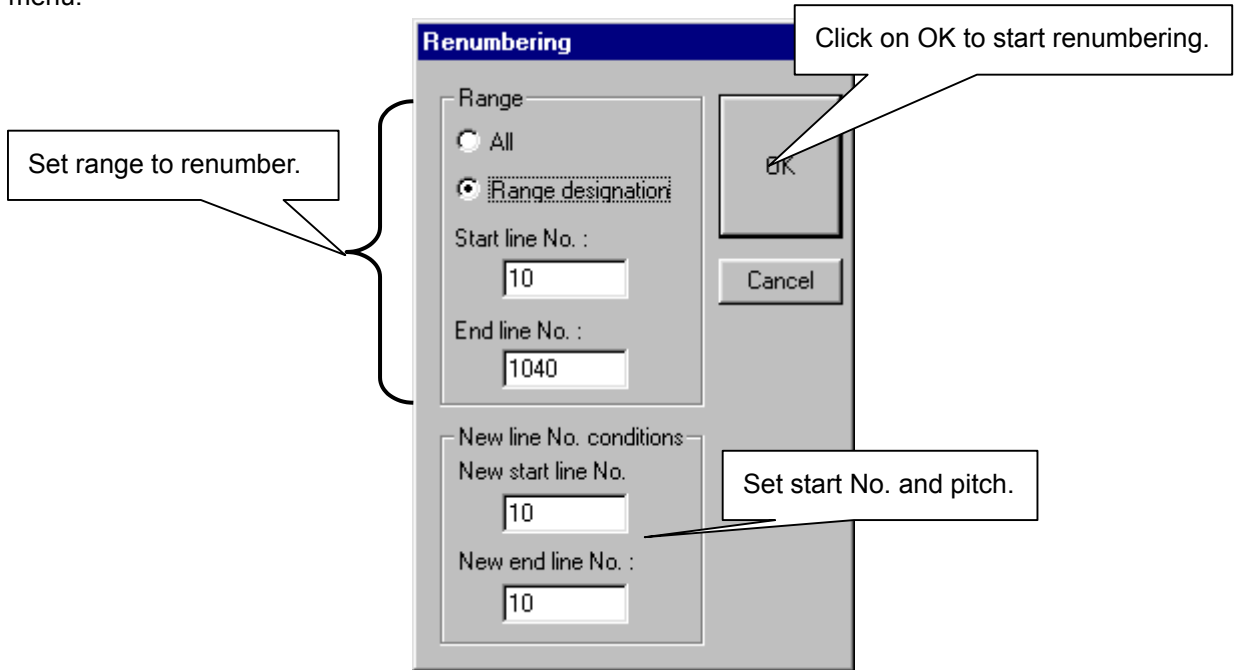
The setting dialog will appear when [AUTO Number] is selected under the [Tool] menu.



Assigning line Nos.

The line Nos. can be assigned in a batch within a designated range. The start/end line No. , new start line No. and line pitch can be designated with the setting dialog.

Renumbering is possible by displaying the renumber setting dialog with [Renumber] under the [Tool] menu.



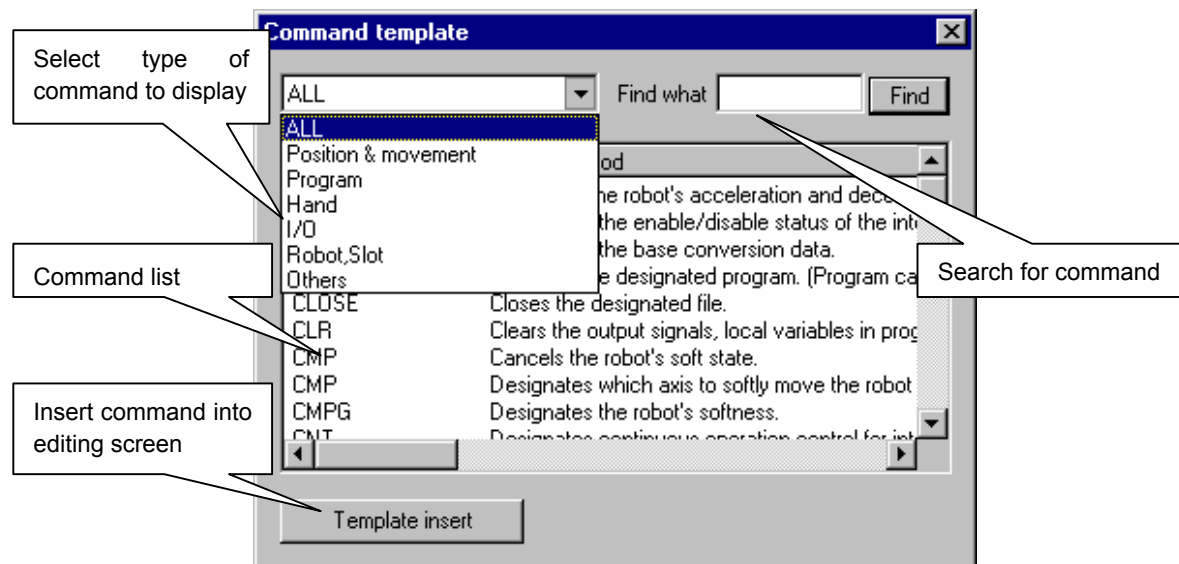
(State of setting dialog and renumbering operation results)

5. Explanation of each tool

Command template

The MELFA BASIC IV language command list is displayed. When a command in the list is selected and the [Insert] button is clicked on, or when the command is double-clicked on, the command can be inserted into the program command editing screen. Displays per command type and searching of commands with character strings are possible.

Display the command template screen by selecting [Command template] under the [Tool] menu.

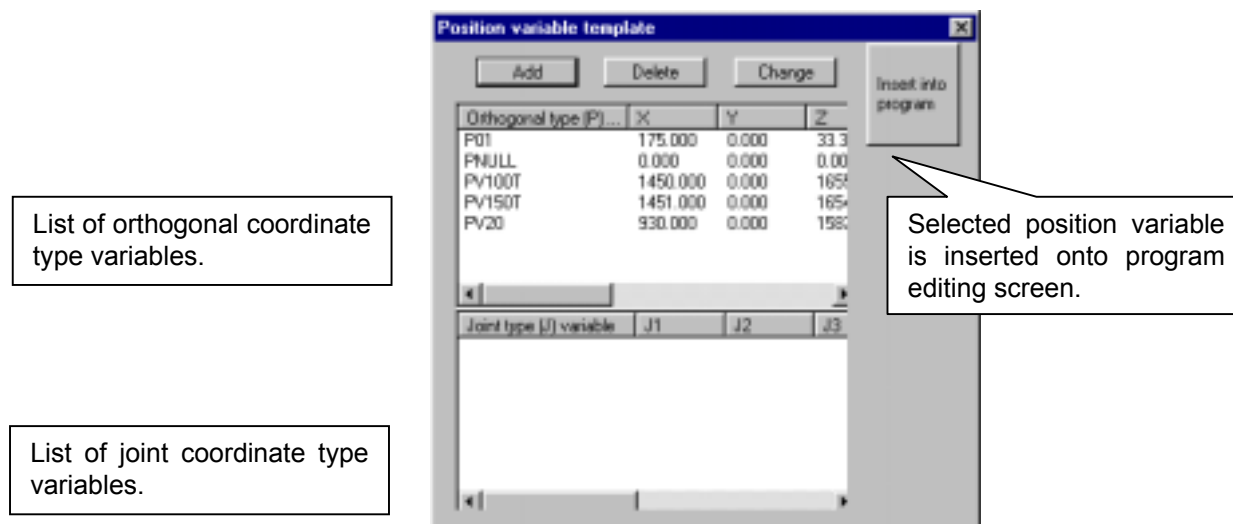


(Image of command template)

Position variable template

Position variables used frequently regardless of the program can be stocked. When a position variable in the list is selected and the [Insert] button is clicked on, or when the variable is double-clicked on, the position variable on the template can be additionally inserted into the program being edited.

Display the position variable template screen by selecting [Position variable template] under the [Tool] menu.



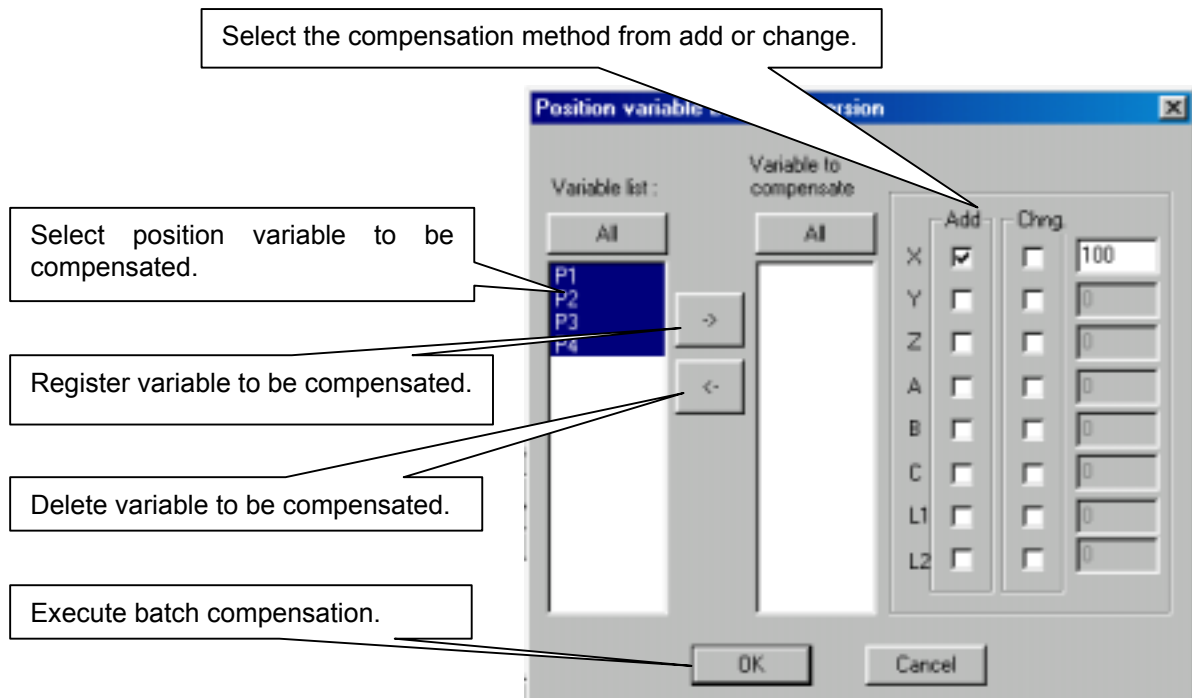
(Position variable template)

The position variables registered on the position variable template are saved even after the program editing tool is exited.

Position variable batch change

The position variables in the program being edited can be changed as a batch, or the values can be added as a batch. For example, 100 can be added to the X elements of P1, P2, P3 and P4 as shown below.

Display the batch change screen by selecting [Position variable batch edit] under the [Tool] menu.

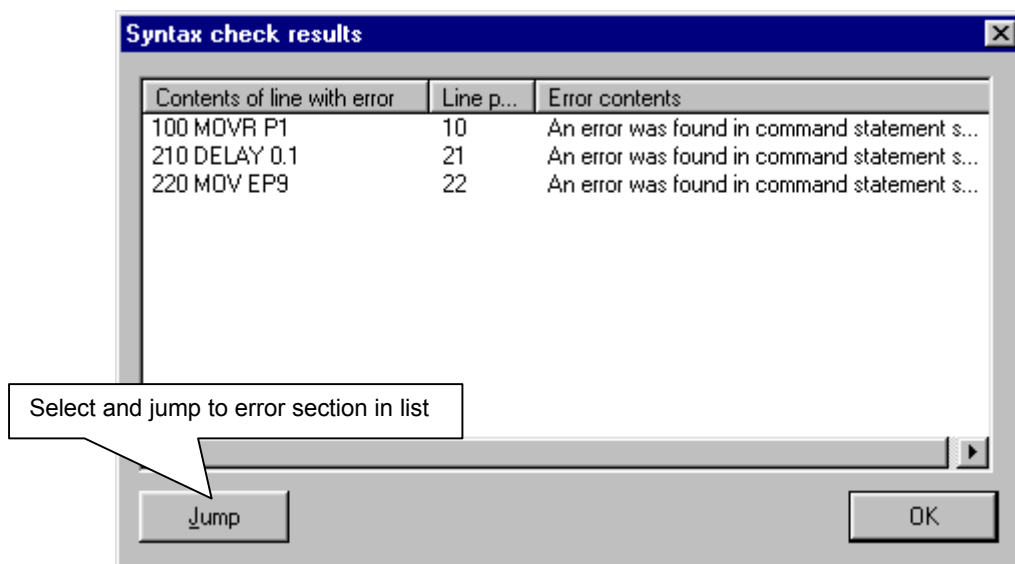


(Image of position variable template)

5.1.7. Syntax check

This function checks whether the edited robot program is grammatically correct. Use this before writing the program into the controller. If there is any error in the grammar, the error section and the details of the error will be displayed.

To use the grammar check, select [Grammar Check] under the [Tool] menu.



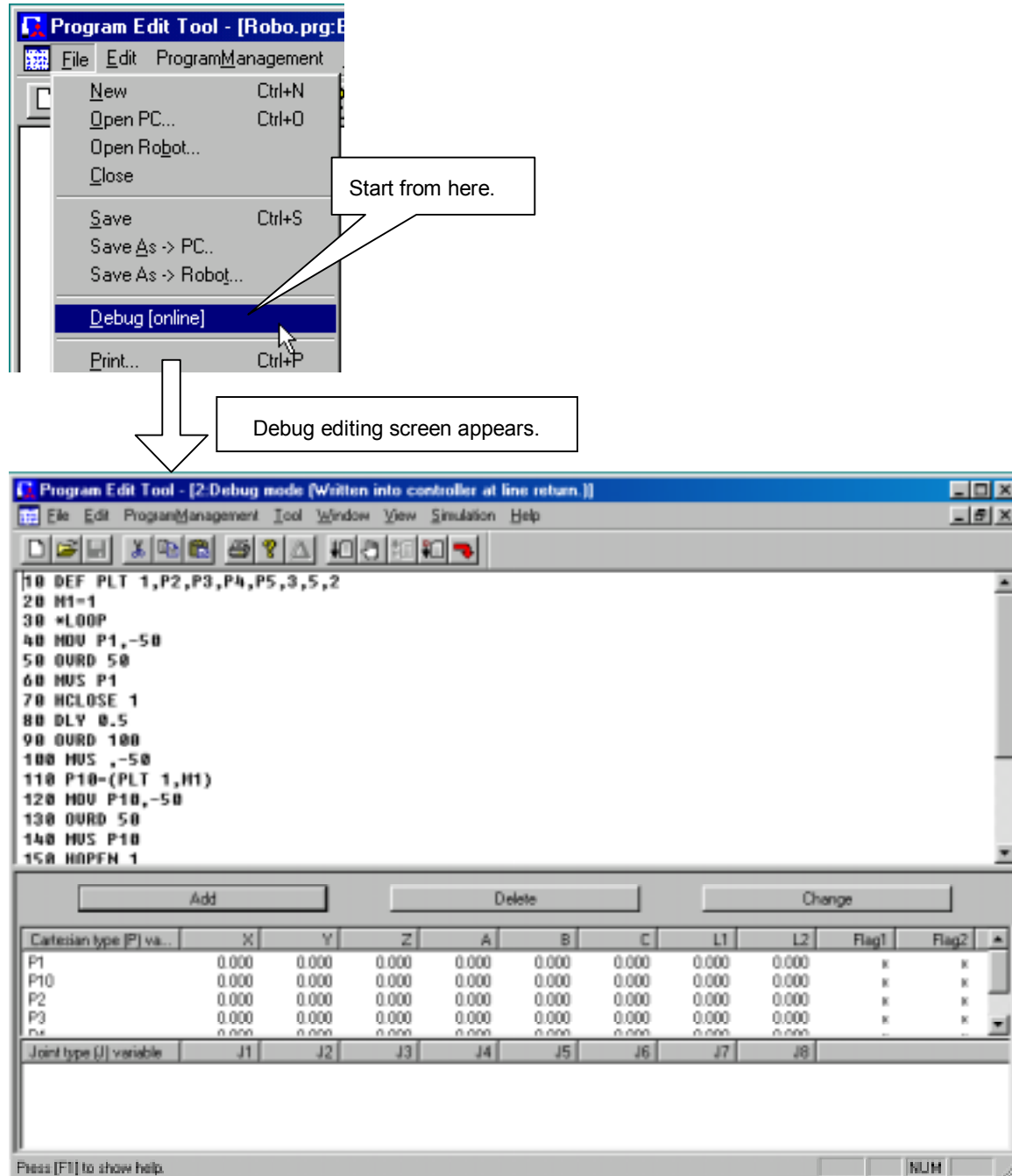
(Display of grammar check results)

5.1.8. Debugging a program

The program can be directly debugged while editing the program and moving the robot. It is also possible to confirm which line of the program is being executed while the robot is moving. Either step execution or direct execution can be used.

To debug the program, select [Debug] under the [File] menu, and open the program.

The editing screen for the program opened with will appear.

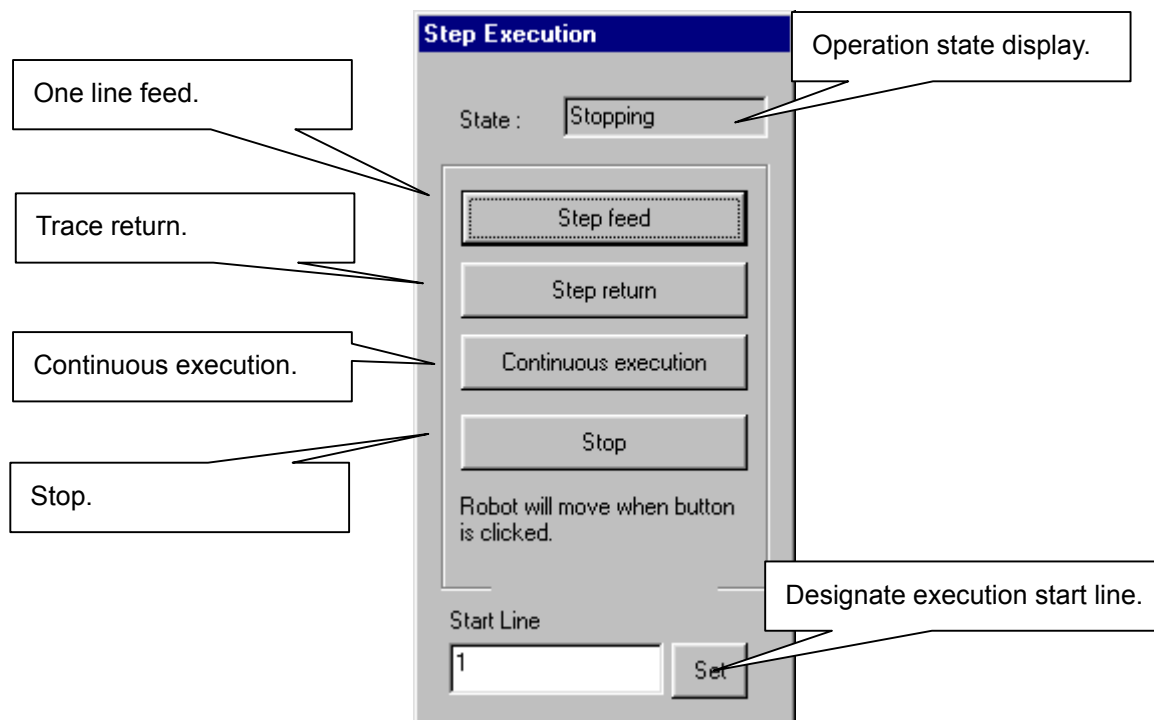


5. Explanation of each tool

Step execution

The program opened with debug can be confirmed by moving one line at a time.

The following dialog will appear when [Step Execution] is selected from the [Tool] menu.



(Step execution)

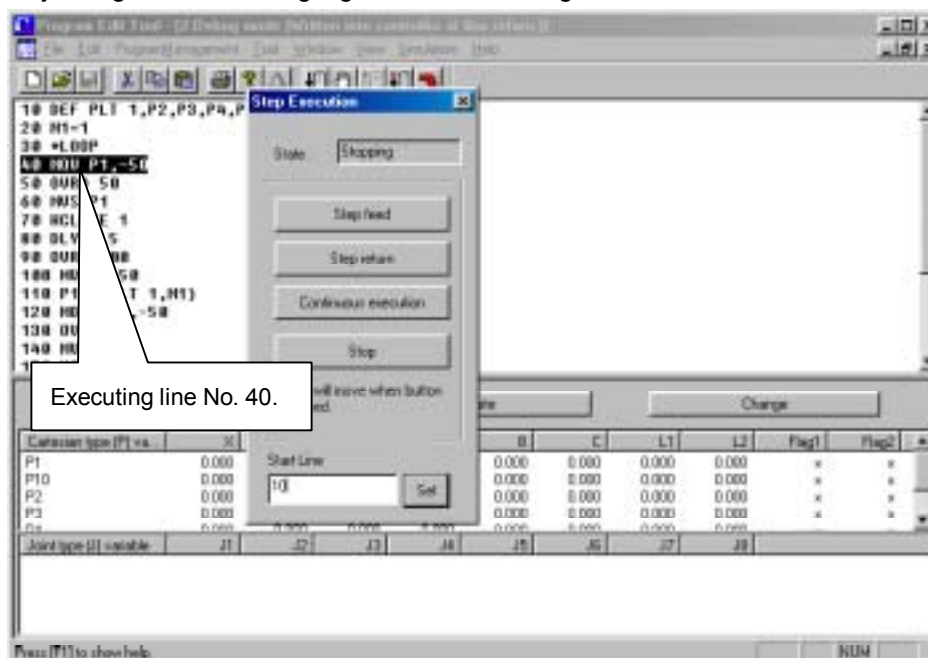
[Step feed] : The program is executed one step at a time.

[Continuous execution] : The robot program is continuously executed.

[Set] : Set the program execution start line No.

Caution) When step execution is carried out, the robot may move at the max. speed, so take special care to safety. Never enter the work area when carrying out step execution.

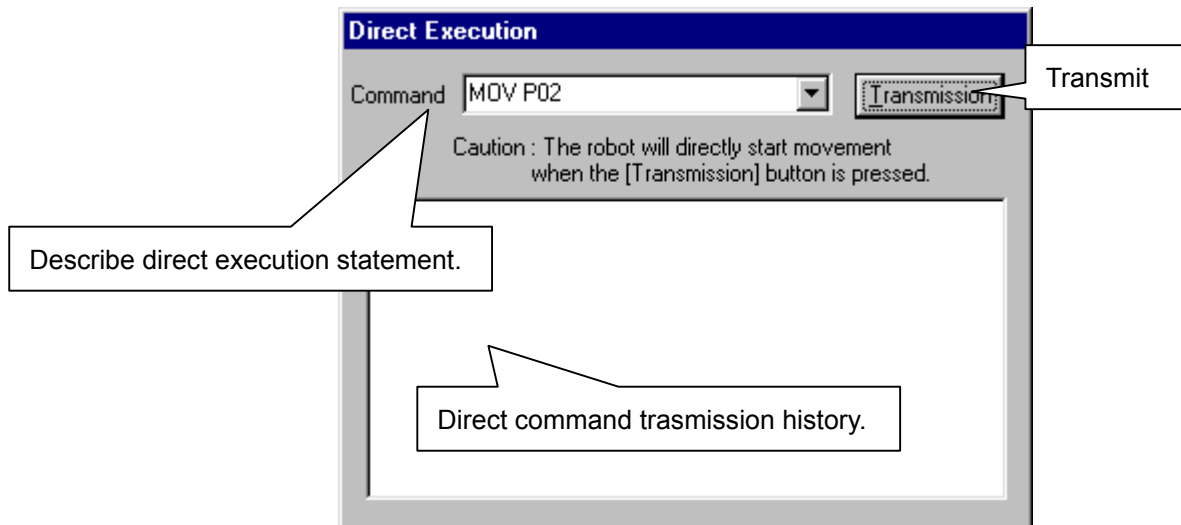
The line currently being executed is highlighted on the editing screen.



Direct execution

The robot can be directly moved by inputting a command statement.

The following dialog will appear when [Direct Execution] is selected from the [Tool] menu.



Caution) When direct execution is carried out, the robot may move at the max. speed, so take special care to safety. Never enter the work area when carrying out direct execution.

5.1.9. Program Management

The program files can be copied, moved, deleted, renamed, protected and the contents compared.

The Program Manager can be used from the [\[Program Management\]](#) menu.

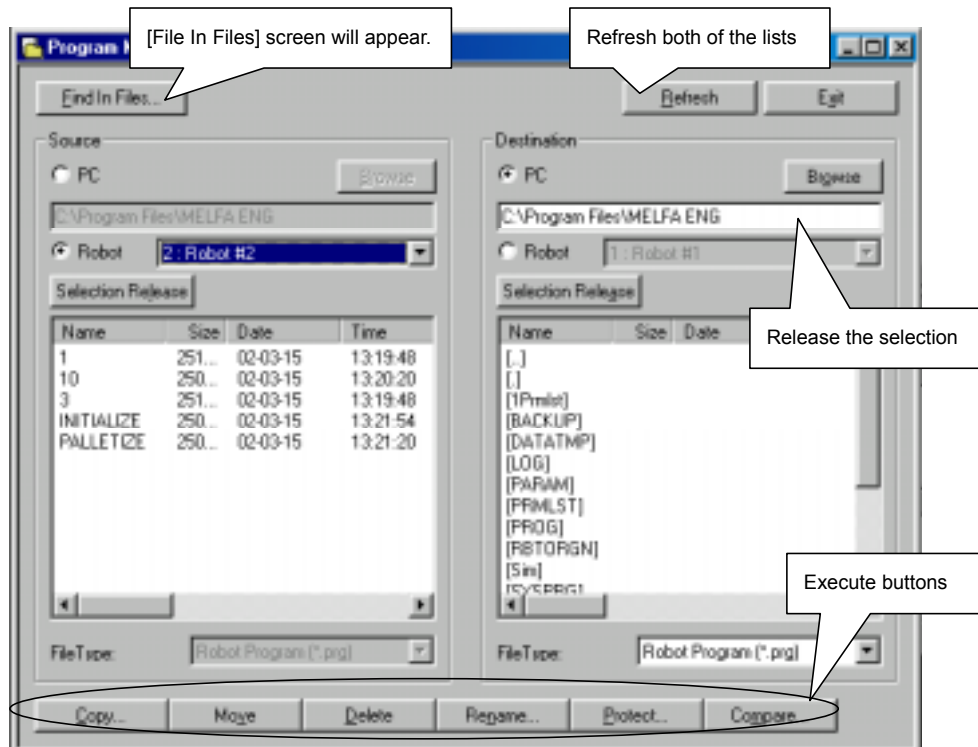


Caution) The targets of any operations in this function are the Robot Program files.

List

The lists of the program files are displayed. For programs in the controller, the "Name", "Size", "Date", "Time", as well as the "Protect information", "No. of lines", "No. of position variables", "Latest cycle time", "Average cycle time", "Operation time", "No. of cycles" and "Comment" are displayed.

- As for the programs in the personal computer and in the controller, when even the contents are the same, the program sizes are different.

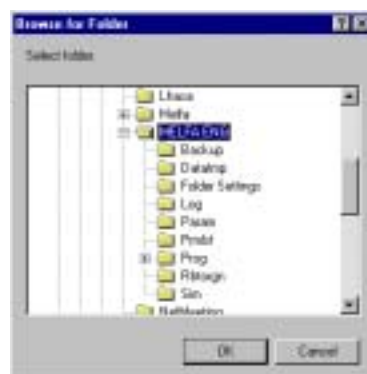


Double-click on a program name in the lists, Program Edit Tool is started up and that program can be edited.

But only **programs in the robot controller** or **"*.prg" files in the personal computer** can be edited. Note that the another files are Opened, then Program Edit Tool cannot behave regularly.

According to the operation, the multiple files can be operated. Selecting the multiple files, click on the files in the lists by the mouse with pressing the [shift] key or the [Ctrl] key. Releasing the selected files in the lists, click on the [Selection Release] button above each lists.

Click on the [Browse] button and select folder at the [Browse for Folder] dialog, and the folder in the personal computer can be selected.



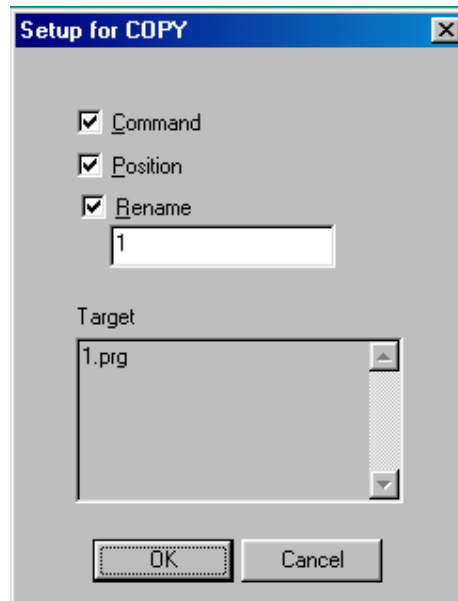
5. Explanation of each tool

(Browse for Folder dialog screen)

Copy

The program files are copied. Copying of the entire program file or only the command statements or only the position variables is possible.

Select the transmission source program names from the list at the left, and designate the transmission destination folder on the right side. The multiple transmission source programs can be selected at the same time, but for copying with changing its name, only one program must be selected. Copying is executed when the [\[Copy\]](#) button is clicked on and [\[Setup for COPY\]](#) dialog is set.



(Setup for COPY dialog screen)

Movement

The program files can be moved.

Select the transmission source program names from the list at the left, and designate the transmission destination folder on the right side. The multiple programs can be selected at the same time. Movement is executed when the [\[Move\]](#) button is clicked on.

Delete

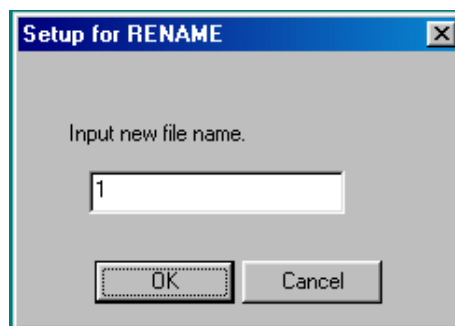
The program files can be deleted. Note that once the program files are deleted, they cannot be recovered.

Select the names of the programs to be deleted from the lists. The multiple programs can be selected at the same time. The programs can be selected at the both lists. Delete is executed when the [\[Delete\]](#) button is clicked on.

Rename

A program file name is renamed.

Select the name of the only one program to be renamed from the lists. The program can be selected at the both lists. Rename is executed when the [\[Rename\]](#) button is clicked on and a new file name is set at the [\[Setup for RENAME\]](#) dialog.



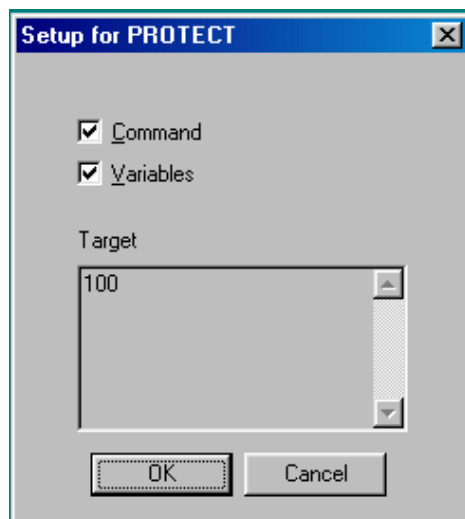
(Setup for RENAME dialog screen)

5. Explanation of each tool

Protect

The program files in the controller can be protected. The entire program file can be protected, or just the command statements or position variables can be protected.

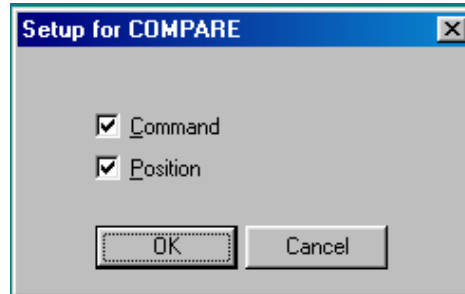
Select the names of the programs to be protected from the lists. The multiple programs can be selected at the same time. The programs can be selected at the both lists. Protect is executed when the [\[Protect\]](#) button is clicked on and [\[Setup for PROTECT\]](#) dialog is set.



(Setup for PROTECT dialog screen)

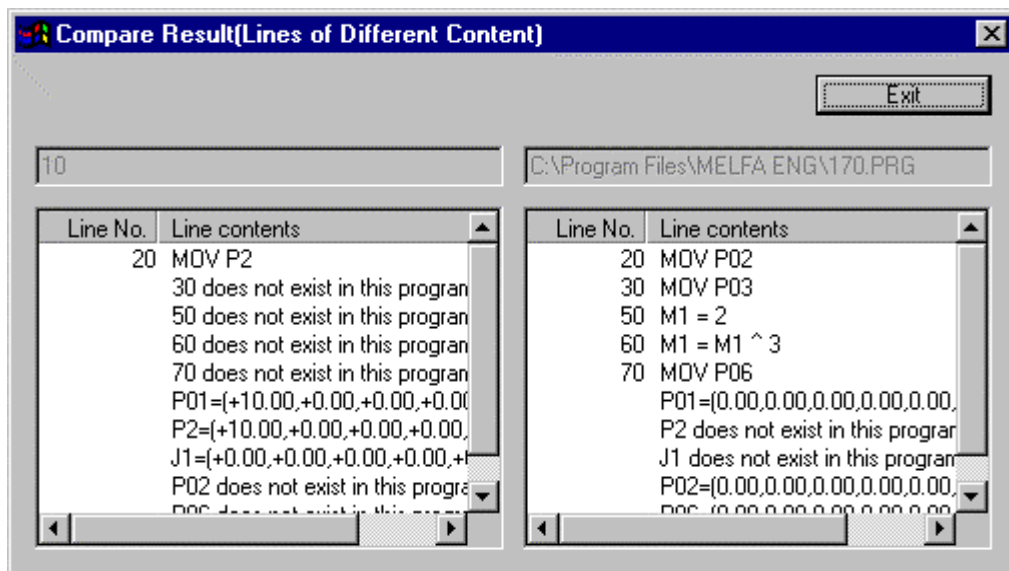
Comparison

The program files can be compared. Comparison of only the command statements or only the position variables is possible. Select the names of the programs to be compared from the left and right lists. A dialog displaying the corresponding comparison results will appear when the [\[Compare\]](#) button is clicked on and [\[Setup for COMPARE\]](#) dialog is set.



(Setup for COMPARE dialog screen)

When both files are the same, the result dialog display nothing.



(Display of comparison results)

5.2. Monitoring tools

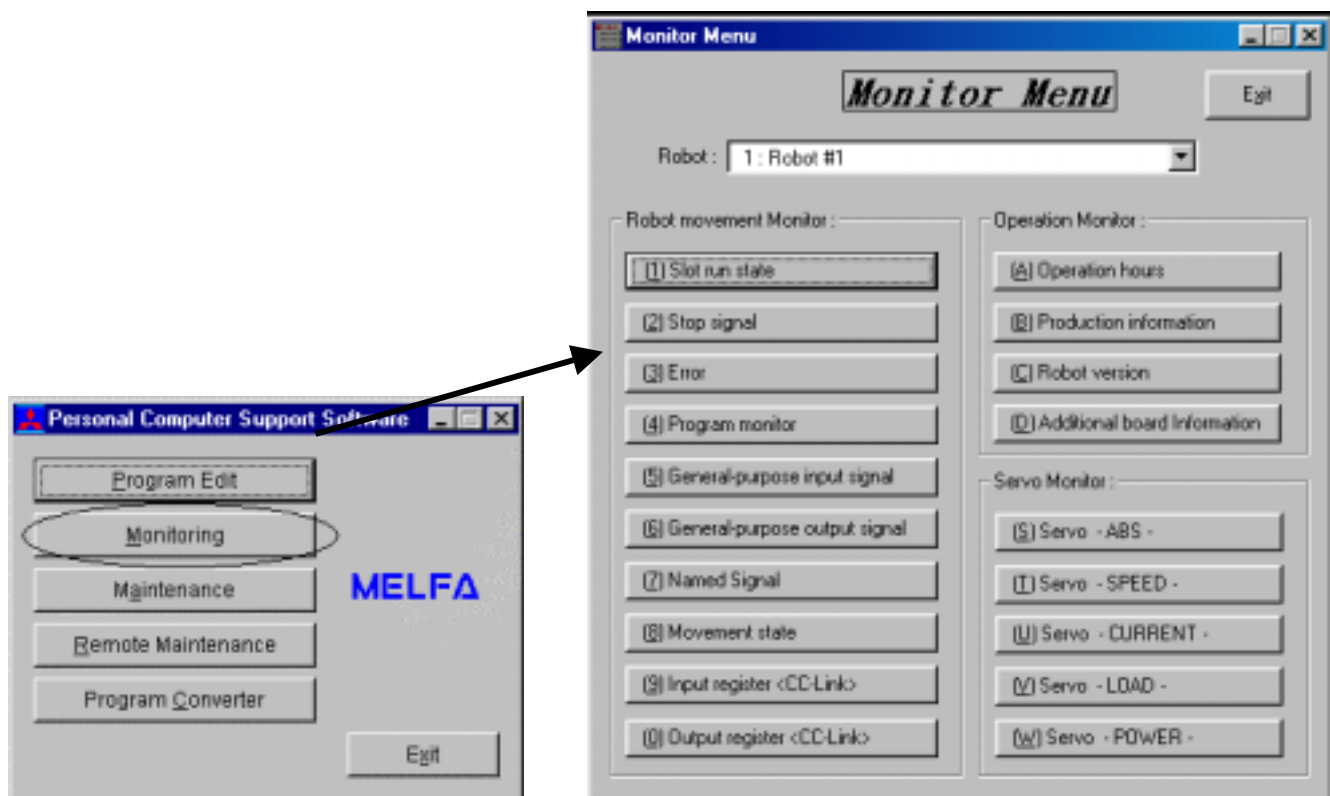
With the monitoring tools, all of the information in the currently connected robot controller can be constantly displayed.

5.2.1. Starting

Select the "Monitoring tool" from the "MELFA Personal Computer Support Software" menu. The "Monitor menu" shown below will appear.

After selecting the robot you want to monitor, click an item with the mouse. The monitor window for each robot appears. (Robot selection is supported in D1 version or later.)

The selected robot controller is displayed in each monitor window title.



Caution) Constant communication is established with the robot controller while each monitoring tool is started. Close any unnecessary windows to reduce the communication load.

5.2.2. Outline of each function and starting methods

(1) Outline of each function

Each monitor function is explained briefly in this section. The monitor functions are largely classified into the following three groups.

1. Robot movement monitor Items related to robot movement are monitored.
2. Operation monitor..... Items related to the robot's operation are monitored.
3. Servo monitor The robot's servo system information is monitored.

	Monitor name	Explanation
Robot movement monitor	Slot run state	The operation state of each slot can be confirmed.
	Stop signal	The stop signal input into the robot controller can be confirmed.
	Error	The currently occurring error can be confirmed. The history of the errors that have occurred can be confirmed.
	Program monitor	The program execution line set for each slot, the contents of the variable used in the program, and the robot current position, etc., can be confirmed.
	General-purpose input signal	The state of the signal input from an external device to the robot controller can be confirmed.
	General-purpose output signal	The state of a signal output from the robot controller to an external device can be confirmed. The signal can also be output forcibly.
	Named signal	The status can be checked by naming the status of the dedicated I/O signal that has been set in the robot controller, as well as each bit or within the range of 32 bits of the general-purpose signal. The signals are set via parameter setting (maintenance tool).
	Movement State	The current position information and hand open/close state of each connected mechanism can be confirmed.
	Input register	The input registers in the CC-Link function can be monitored and pseudo-input.
	Output register	The output registers in the CC-Link function can be monitored and forcibly output.
Operation monitor	Operating time	The robot operation time (power ON, etc.) can be confirmed.
	Production information	The operating time of the program in the robot controller and the No. of program cycles can be confirmed.
	Robot version	The robot controller version can be confirmed.
	Option card information	Information on the option card mounted on the robot controller can be referred to.
Servo monitor	Servo monitor position, speed, current, load, power	The servo system information can be monitored.

(2) Operation method

Mouse operation : Click on the button for the item to be displayed from the Monitor menu.

Keyboard operation : While holding down the [Alt] key, press the alphanumeric character displayed at the left of the button.

(Example: Press the [4] key while holding down the [Alt] key to display the program monitor. Hereinafter, expressed as [Alt] + [4].)

5.2.3. Exiting the monitor menu

Click on the "Exit (X)" button on the screen to quit the monitor menu.

Press [Alt] + [X] to exit using the keyboard.

5.2.4. Each monitor function

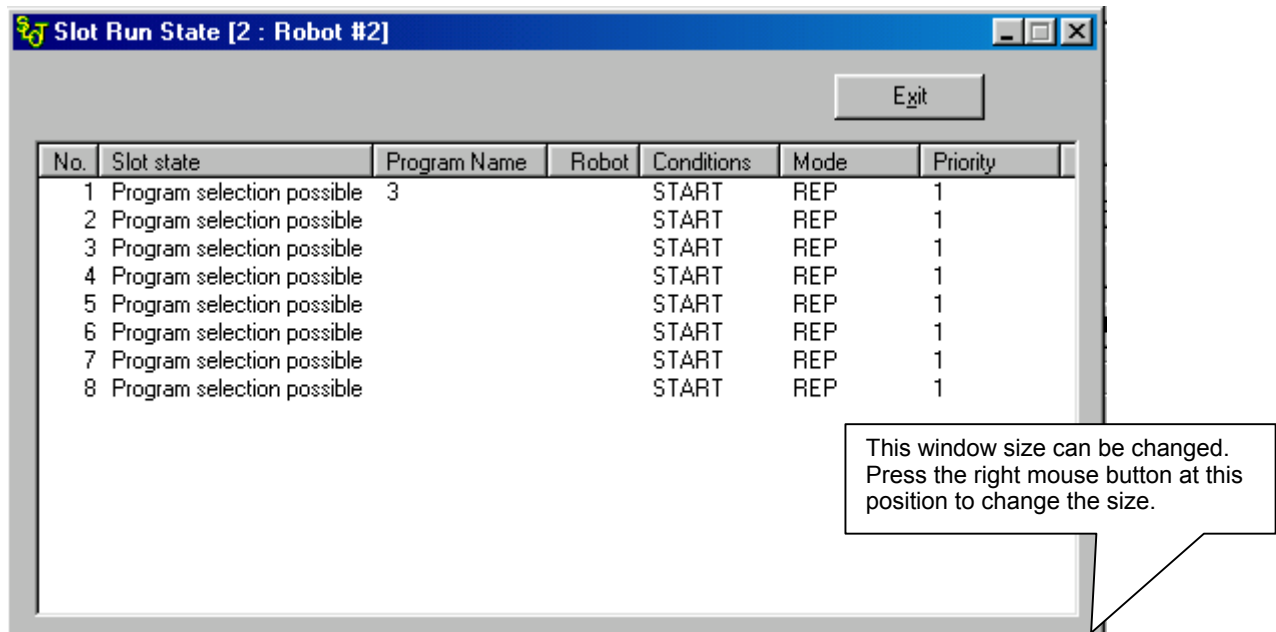
Each monitor function is explained in this section.

5.2.4.1. Robot movement monitor

(1) Slot Run State

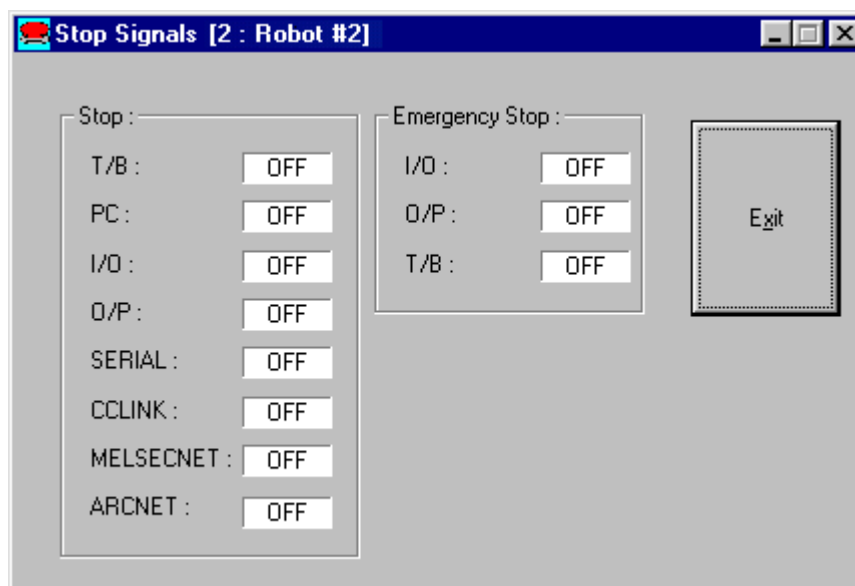
The state of the slots in the robot controller can be monitored.

The No. of displayed slots is determined with the parameters.



(2) Stop Signal

The state of the stop signal (stop, emergency stop) input into the robot controller can be referred to.

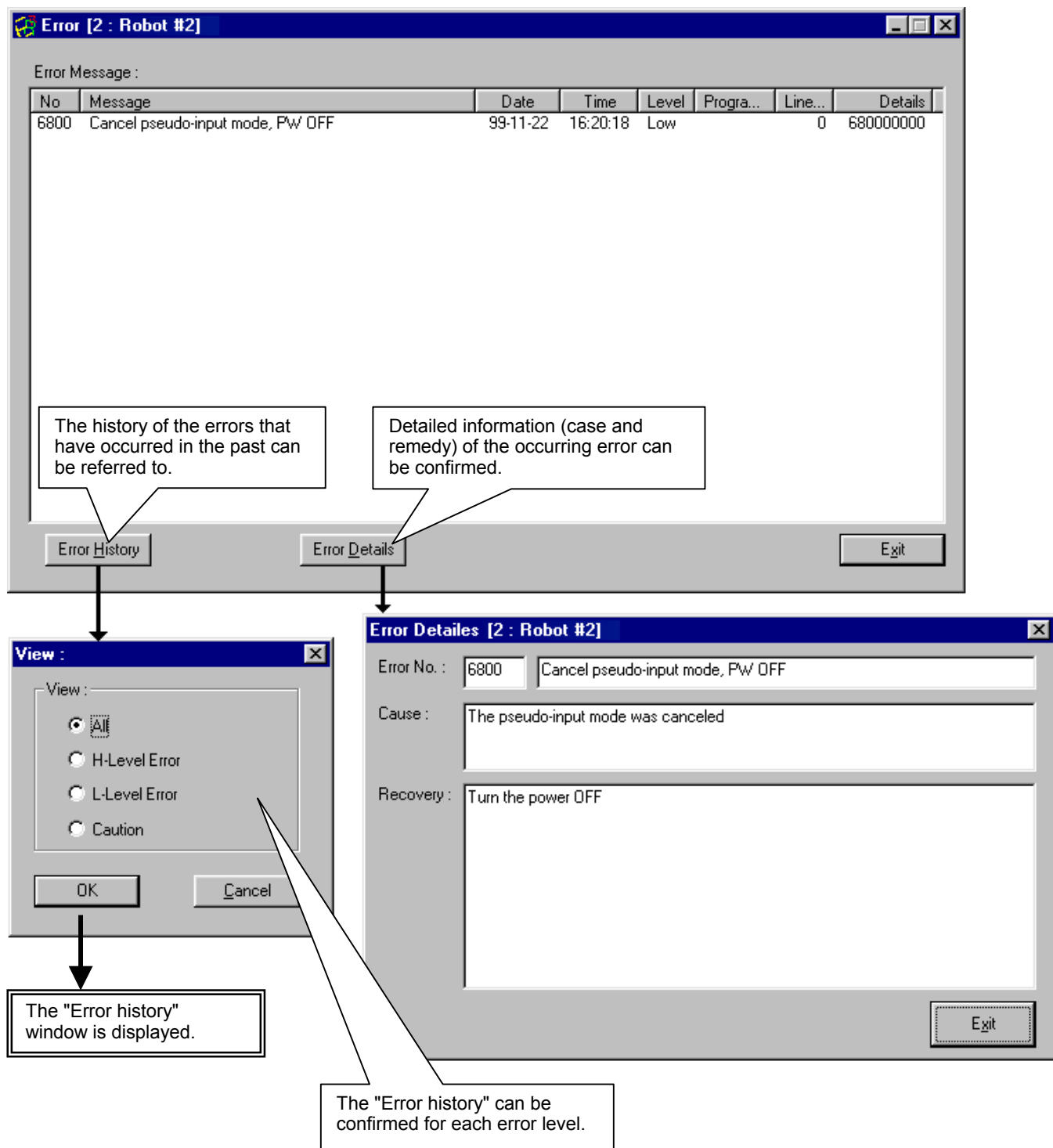


5. Explanation of each tool

(3) Error

The errors currently occurring in the robot controller are displayed.

1. Currently occurring errors



2. Error history

The history of errors that have occurred in the past can be referred to.

This display is not shown at all times. To update the information, press the "Update" button.

The screenshot shows a window titled "Error Histroy [2 : Robot #2]". It contains a "View" section with radio buttons for "All" (selected), "H-Level Error", "L-Level Error", and "Caution". A "Tabulation information" section shows "Start Date : 99-11-16" and "Start Time : 18:42:10". A status bar indicates "No. of registered items : 3". Below this is a table of error history.

No	Message	Date	Time	Le...	Program ...	Line ...	Details
6800	Cancel pseudo-input mode, Pw OFF	99-11-22	16:20:18	Low		0	680000000
3361	Can not load the program(SLT*)	99-11-22	15:37:46	Low	ACCEL...	0	336100000
4800	System error (System base prog)	99-11-22	15:37:46	Low		0	480000000

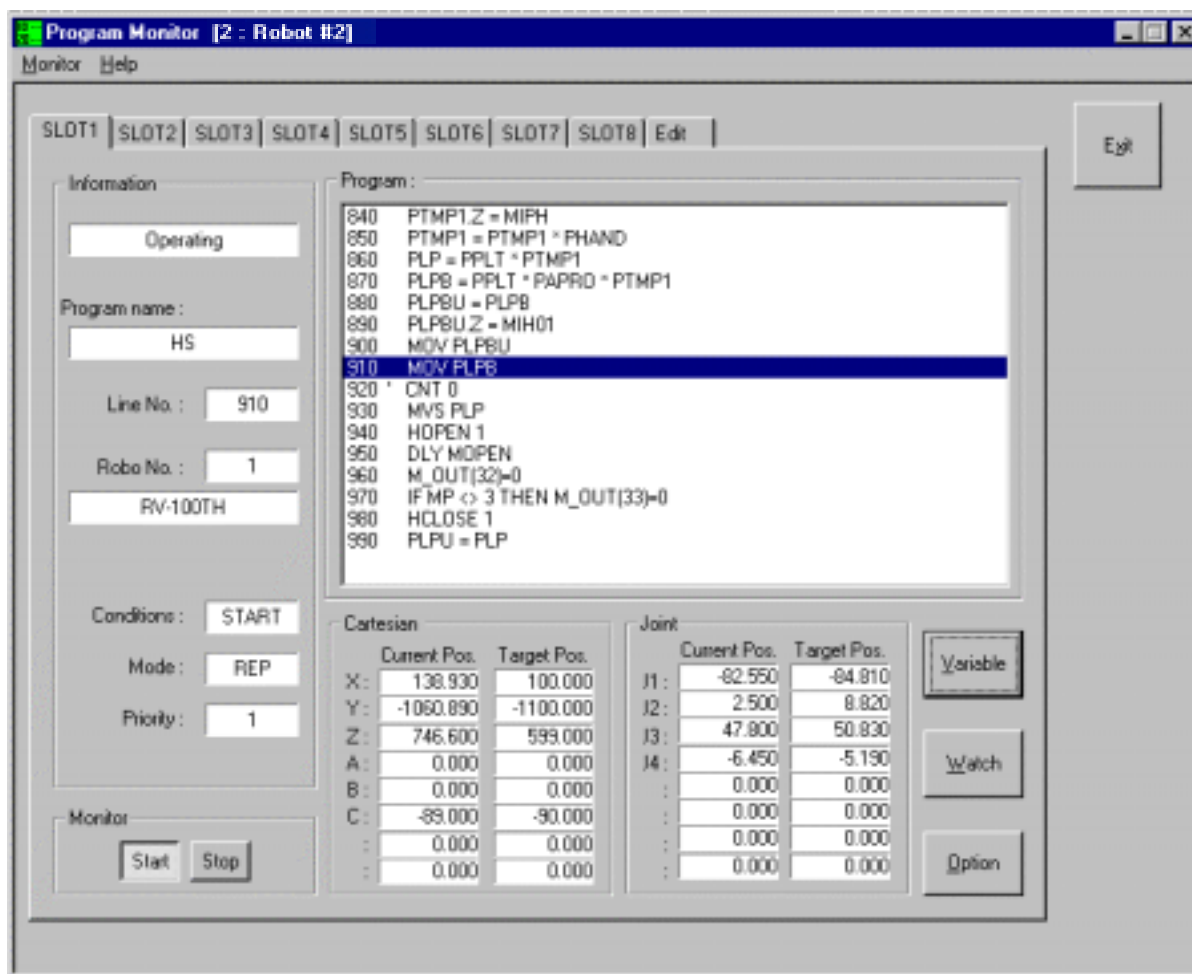
At the bottom of the window are five buttons: "Refresh", "Details", "Save", "Clear", and "Exit". Callout boxes point to these buttons with the following descriptions:

- Refresh:** The displayed information is updated.
- Details:** The cause and remedy of each error can be confirmed.
- Save:** The displayed error history information is saved in a file.
- Clear:** The error history information is cleared.
- Exit:** (No callout box is present for this button).

5. Explanation of each tool

(4) Program monitor

Information of the running program can be monitored.



[Variable] : The value of the variables in the running program can be referred to.
The variable to be displayed can be selected.

[Watch] : The constant display window for the variables used in the running program is displayed.

[Option] : The No. of execution lines of the program being executed can be changed.

As the program monitor uses a large amount of information, when monitoring the variables with "Watch", the update of the display data may be delayed from the actual information.

When only the variables are to be monitored, press the monitor [Start] and [Stop] buttons on this screen, stop the display program monitor, and then watch.

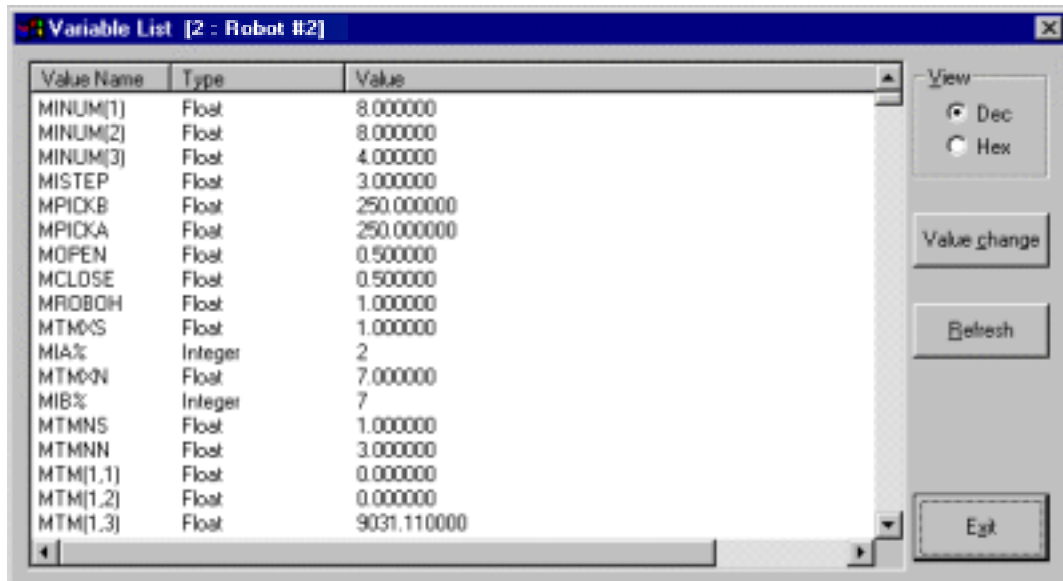


Variable List

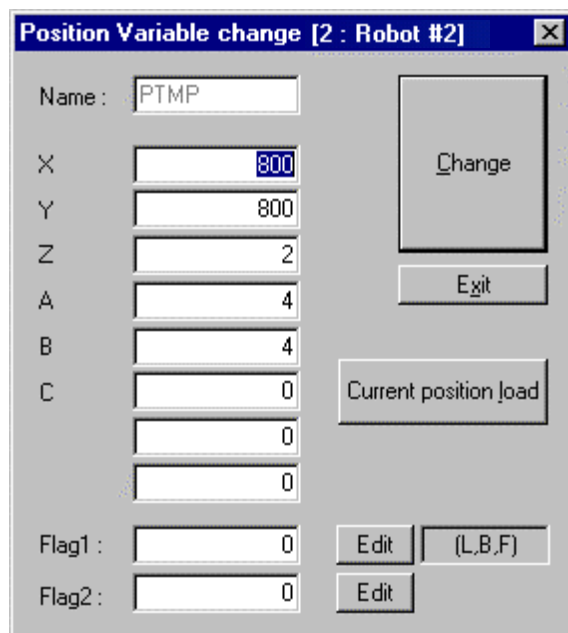
When the [Variable] button is clicked on, the following type of window will appear.

In this variable list, all variables used in the program allocated in the selected slot are displayed.

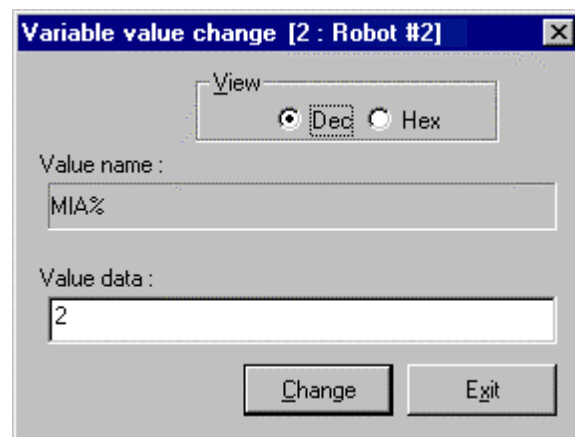
The value of the variable displayed here is not shown at all times. To update the information, press the [Refresh] button.



When the variable name is selected and the [Value Change] button is clicked on, the window for making changes will appear.

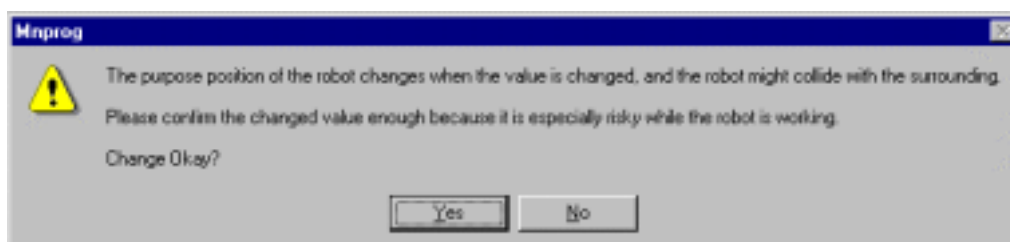


(Location)



(Integer, Float, String)

Please click [Change] button after changing the value. However, Please note that the movement of the robot might change by the change enough when you change the value.

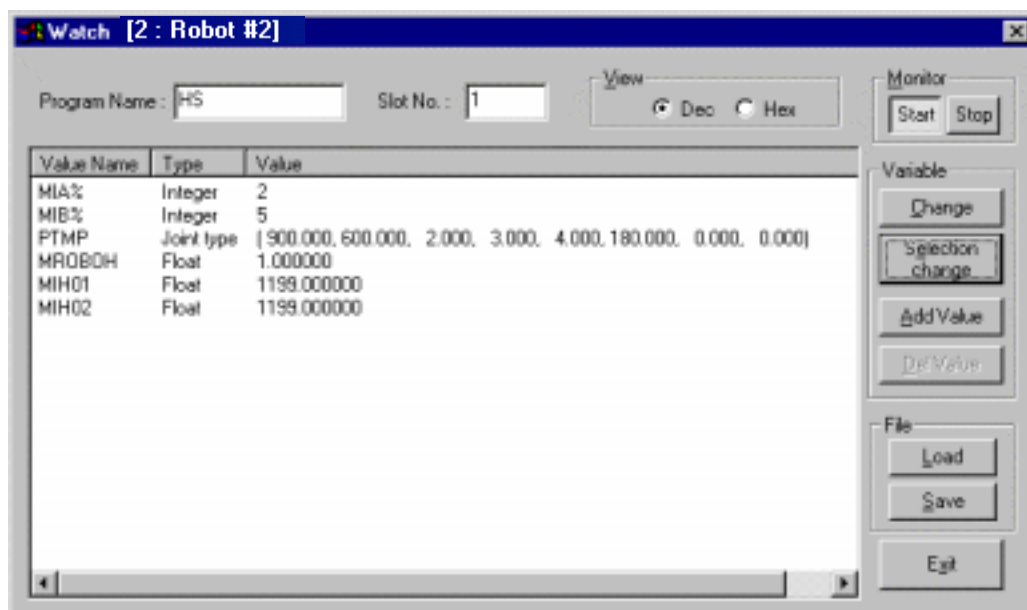


5. Explanation of each tool

Watch monitor

With the watch monitor, the variables to be constantly displayed can be selected.

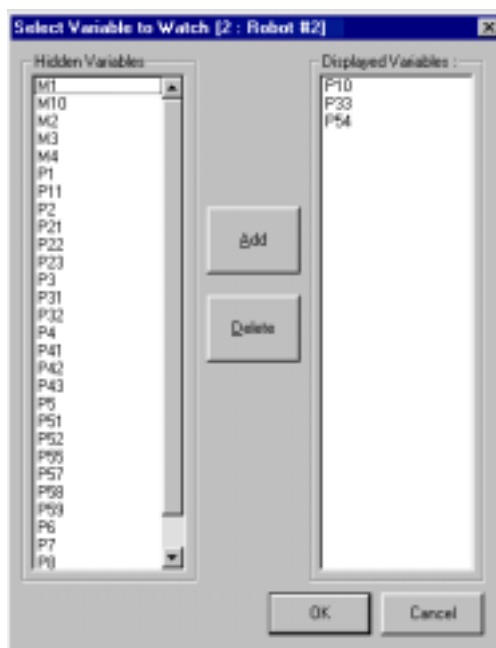
When the [Watch] button is clicked on, the watch monitor window and a window for selecting the variables to be displayed will appear.



[Change] The value of the variable being watched now is changed. Variable identifier to which the value is changed is clicked double or [Change] button is clicked while selected. Variable value change window is displayed.

notes) Please note that the movement of the robot might change by the change enough when you change the value.

[Selection change] The watched variable can be selected from the variable list used by the program. When [Selection change] button is clicked, the screen like the right is displayed. [Add] button is clicked after the variable watched on this screen is selected, and the variable monitored to [Displayed Variables] is selected. If the watched variable is selected, [OK] button is clicked. The variable which is not displayed here can be specified by the [Add Value].



- [Add Value]** The variables to be monitored can be designated.
Input the variable in Variable Name, select the variable type, and then click on the [OK] button.



- [Load]/[Save]** The list of the variable monitored by Watch is preserved in the file or the monitored variable can be read from the file. (This function corresponds from Ver.C1.)

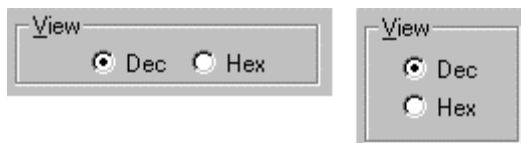
Variable name, the type, and the value being watched now can be saved in the file by clicking [Save] button.

Variable name and the variable type are read from the saved file when [Load] button is clicked, and it is possible to add [Watch] as a monitored variable.

About the hexadecimal number display

The variable displayed with "Variable List" and "Watch" can be switched to the hexadecimal number/the decimal number.

Please select the type which wants to be displayed with "View" on each screen.



Please refer to the following for the variable which can be displayed by the hexadecimal number.

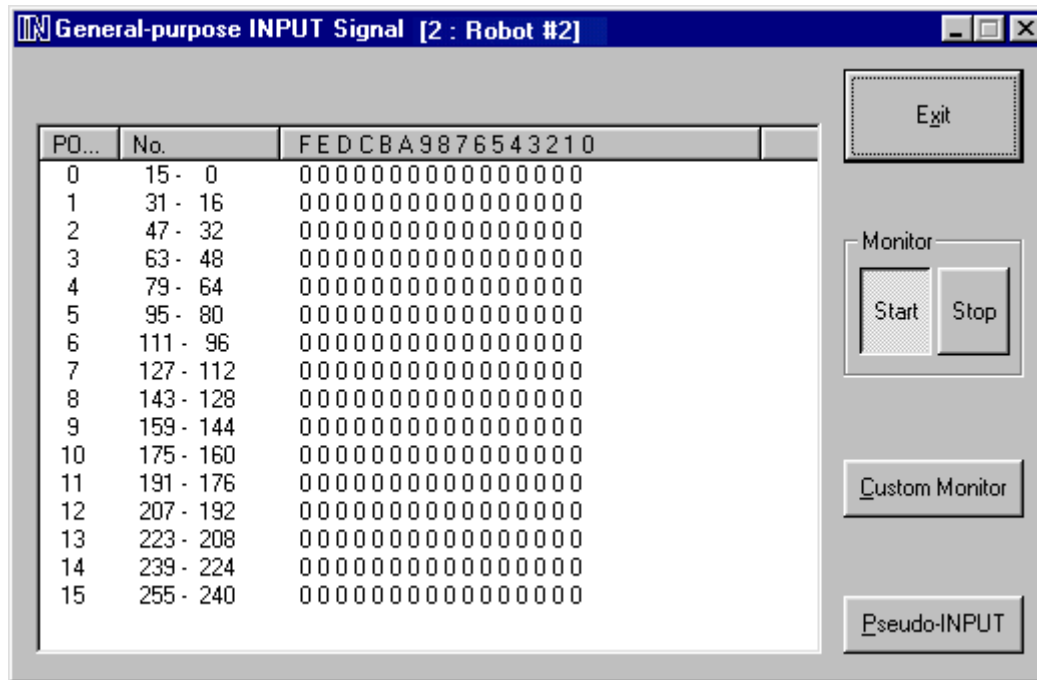
Integer	The displayed variable can be switched to the hexadecimal number/the decimal number.
Float	When it is 0 below the decimal point, it is possible to switch to the hexadecimal number/the decimal number. However, the value is the one within the range of -9999999-9999999.
String	The hexadecimal number is not displayed.
Location	The hexadecimal number is not displayed.

As for the value displayed by the hexadecimal number, &H is added to the head of the value.
(This function corresponds from Ver.C1.)

5. Explanation of each tool

(5) General-purpose input signals

The status of the signals input into the robot controller can be monitored.

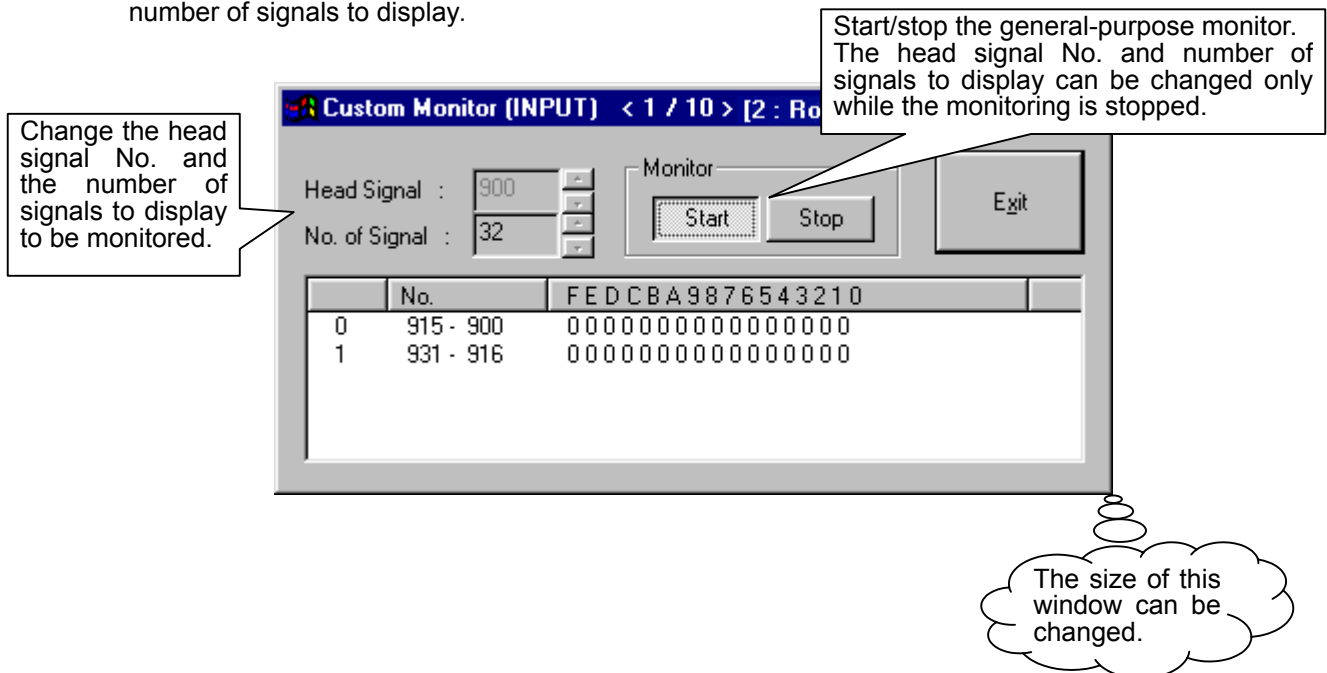


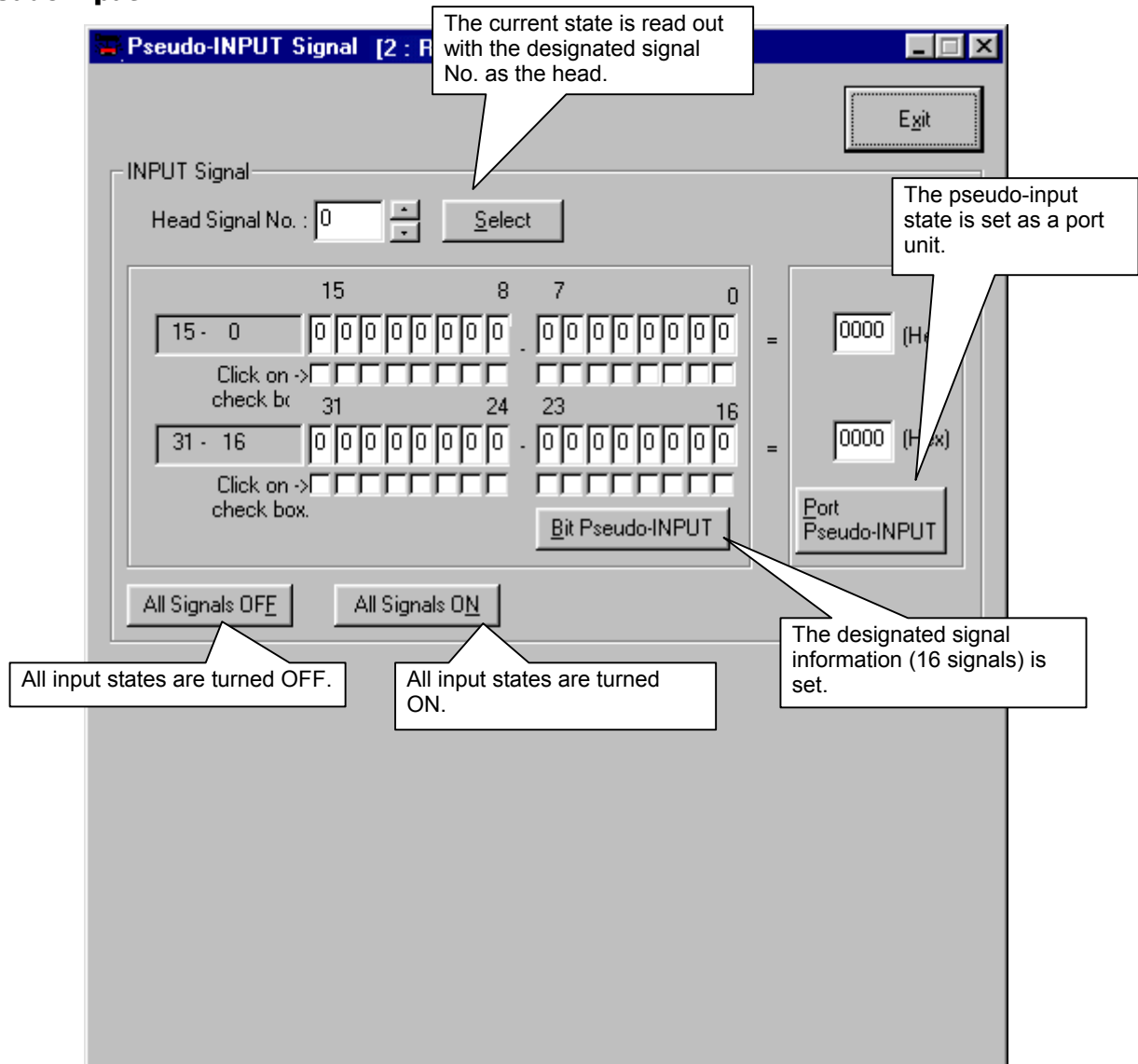
When the [Pseudo-input] button is clicked on, the robot controller will enter the "pseudo-input mode", and the following window will appear.

Caution: To cancel the pseudo-input mode, the robot controller power must be turned ON again.

General-purpose monitor

With the general-purpose monitor, monitoring can be carried out by designating the head signal No. and number of signals to display.



pseudo-input

In the pseudo-input mode, the state input from the following window is interpreted as the input signal instead of the robot controller general-purpose input signals.

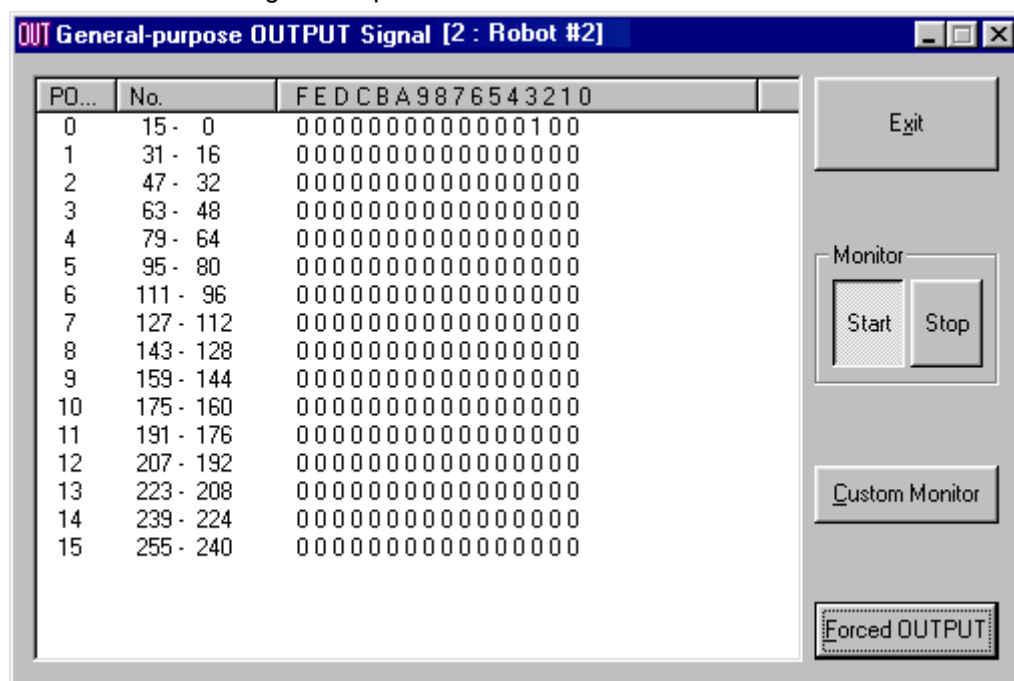
1. First the signal to be pseudo-input signal is read.
16 signals can be set simultaneously. Input the head No. of the signal to be read, and then click on the "Select" button.
2. The input state of the 16 signals is displayed with the designated signal at the head. Set the pseudo-input state, and then click on the "Pseudo-input setting" button.
3. 16 signals from the No. designated as the head can be forcibly output as a hexadecimal.
Input the hexadecimal value, and then click on the "Port pseudo-input" button.

The pseudo-input mode is canceled when this button is closed.

5. Explanation of each tool

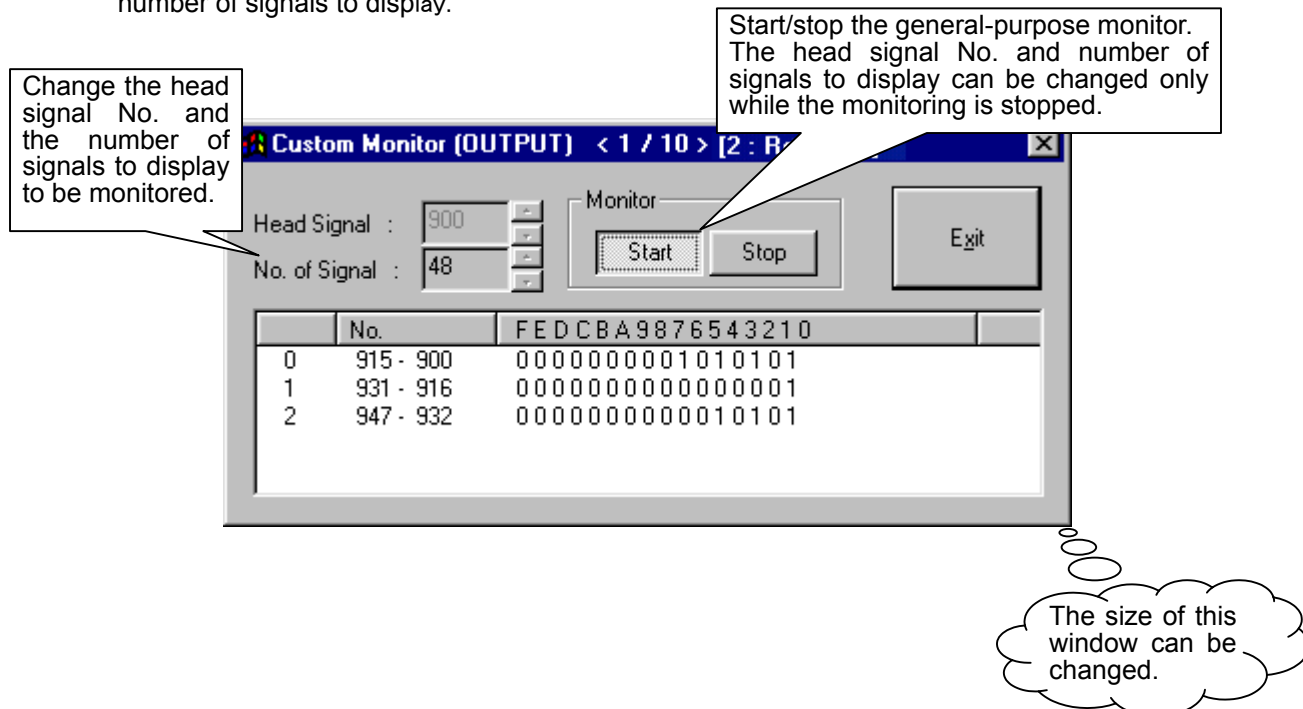
(6) General-purpose output signal

The state of the signals output from the robot controller can be confirmed.



General-purpose monitor

With the general-purpose monitor, monitoring can be carried out by designating the head signal No. and number of signals to display.



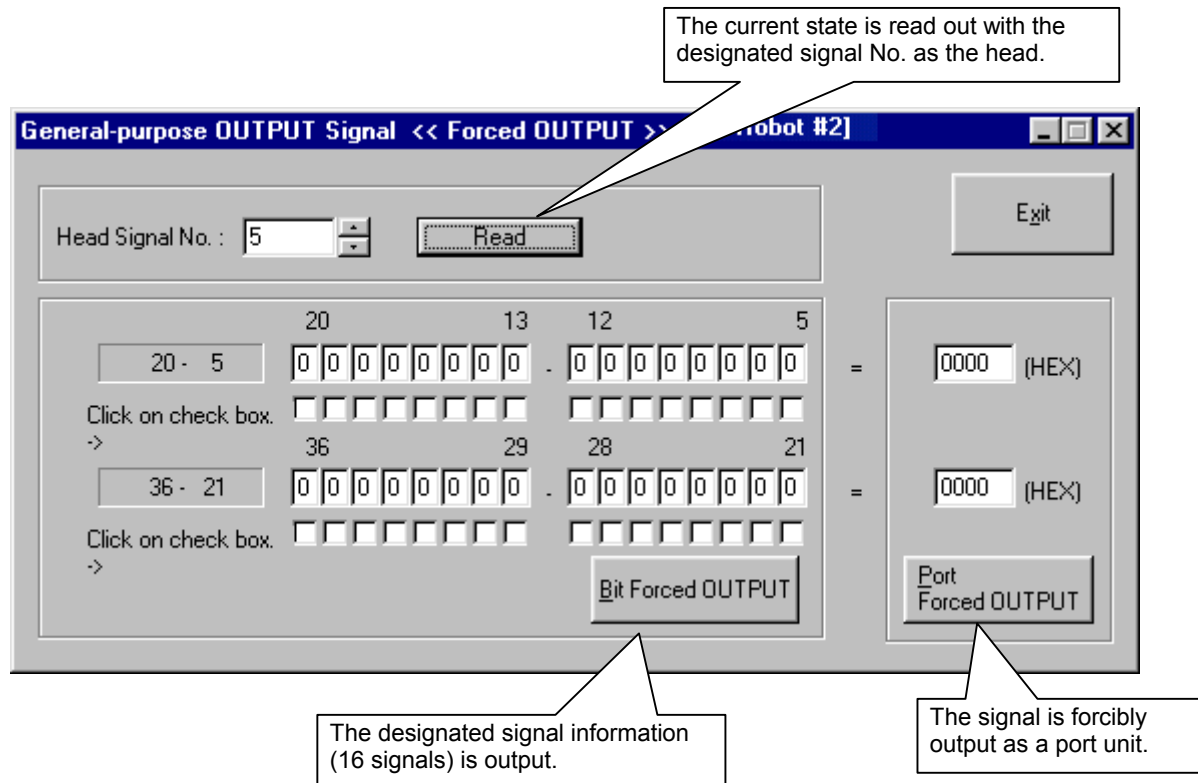
Forced output

The robot controller's general-purpose signals can be forcibly output.

Click on the "Forced output" button. A window for forcibly outputting the signal will appear.

1. First, read out the signal to be forcibly output.

16 signals can be output simultaneously. Input the head No. of the signal to be read, and then click on the "Read" button.



2. The output state of the 16 signals is displayed with the designated signal at the head. Set the output state, and then click on the "Forced output" button.

3. 16 signals from the No. designated as the head can be forcibly output as a hexadecimal.

Input the hexadecimal value, and then click on the "Port output" button.

Caution

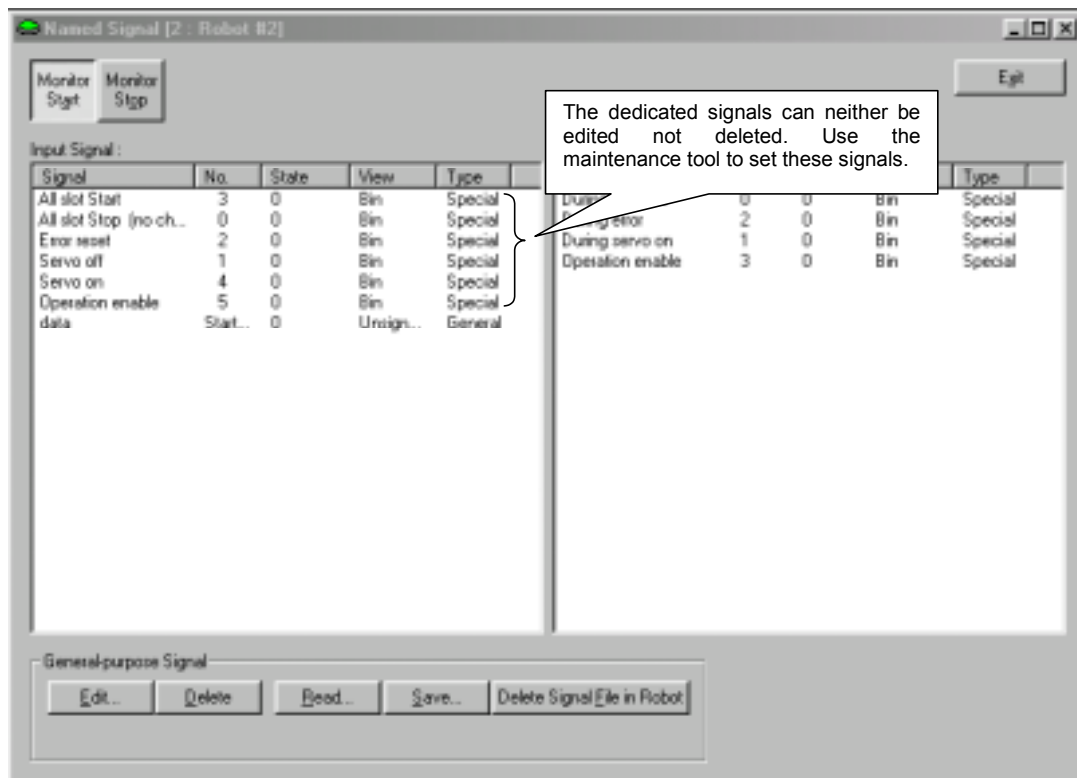
- The signal Nos. assigned (used) with the dedicated output signal cannot be forcibly output.
- Forced output is possible in the [TEACH], [AUTO (OP)] and [AUTO (EXT.)] states, but if even one program is running, output is not possible. (Excluding the ALWAYS program.)

5. Explanation of each tool

(7) Named signal

The status can be checked by naming the status of the dedicated I/O signal that has been set in the robot controller, as well as each bit or within the range of 32 bits of the general-purpose signal. The signal file in the robot controller is loaded at startup. If, however, it is not found, the previously used file is loaded.

The signals are set via parameter setting (maintenance tool).



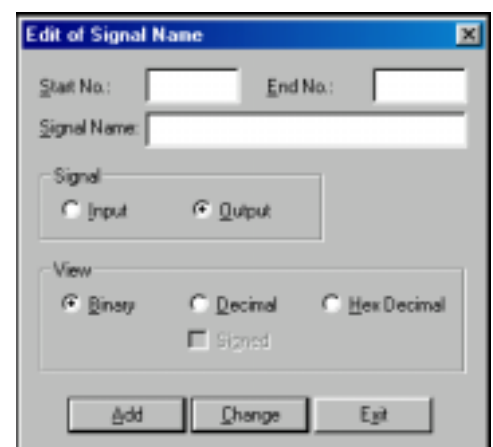
[Edit]/[Delete]

The I/O signals you want to monitor can be added or edited. Enter the range of the signals you want to monitor in the Start No. and End No. boxes, and name it. If the signal you want to monitor is one bit, enter only the start No. For multiple bits, enter the numbers so that the start No. is smaller than the end No. (If reversed, an error will occur.)

Binary, decimal or hexadecimal notation can be selected as the display method. For decimal notation, signed display using the most significant bit as a signed bit can also be performed.

Once the entry is finished, click the [Add]/[Change] button. If the [Add] button is clicked when a signal name is being selected, it is inserted to the selected line. If a signal name is not being selected, it is added at the end of the list.

You can delete unwanted signal names from the list by selecting them and clicking the [Delete] button.

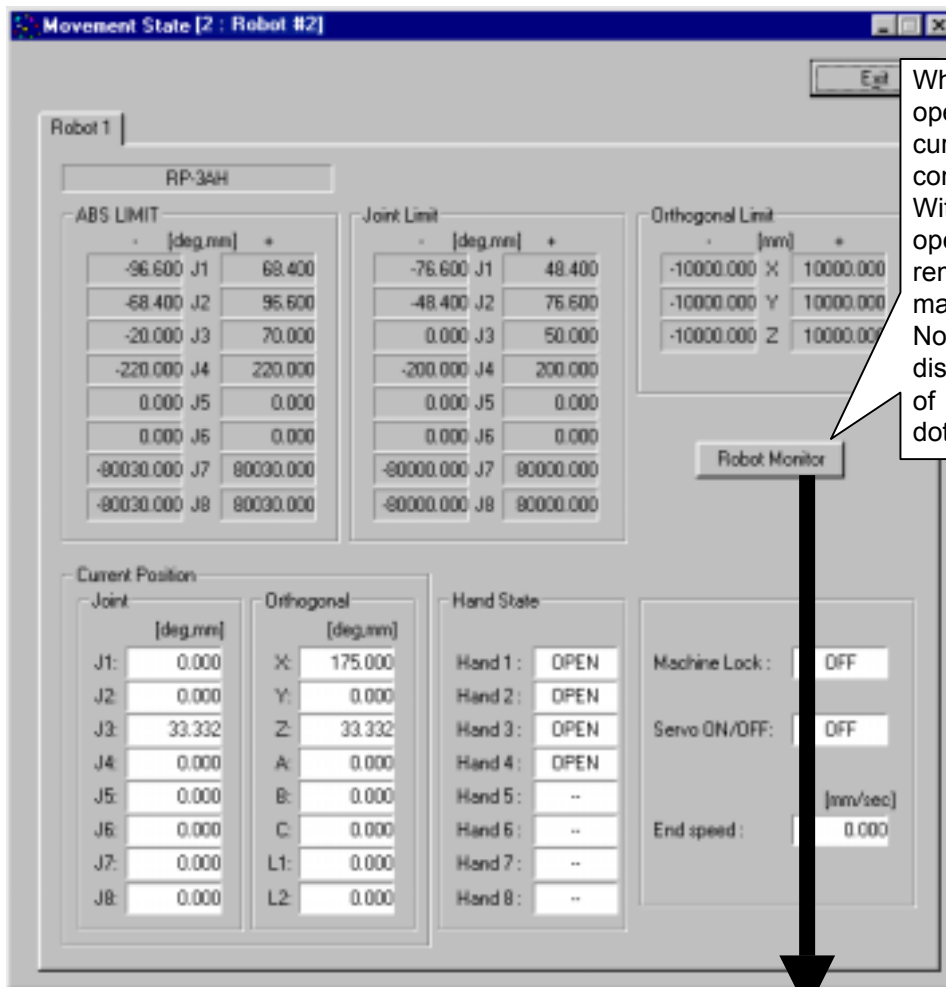


[Load]/[Save]

The edited result can be saved on or loaded to a personal computer or robot controller. Specify the save destination/load destination and click the [OK] button. If the save destination is a personal computer, a file name can be specified. However, if the save destination is a robot controller, the result will be overwritten on the previous information.

(8) Movement State

With the operation confirmation, the robot's movement range, current position and hand open/close state, etc., can be confirmed.



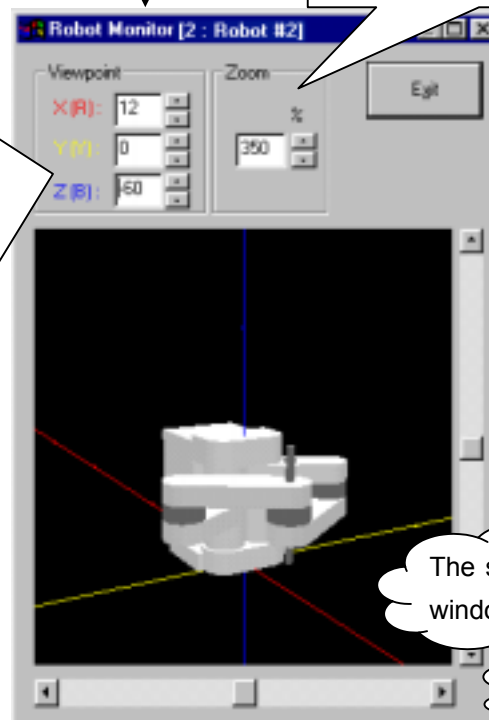
When this button is clicked on, the operation of the currently connected robot can be confirmed with graphics. With this function, the robot operation can be confirmed from a remote place using remote maintenance. Note that when this robot is being displayed with graphics, monitoring of the section enclosed in the dotted line will stop.

The size of the robot display can be changed. It can also be changed with the [Shift] key + left mouse button (up and down).

The view point can be changed by rotating the designated coordinate axis.

The viewpoint can also be changed by the following mouse operations:

- Left button (left and right) → Rotation of Z-axis center
- Left button (up and down) → Rotation of X-axis center
- Left + right buttons (left and right) → Rotation of Y-axis center
- Right button (up, down, left and right) → Move



5. Explanation of each tool

(9) Input register

This screen cannot be referred to if the CC-Link option card is not mounted on the robot controller.

The values of the CC-Link function input registers can be monitored.

No.	DEC	HEX
6000	0	0x0000
6001	0	0x0000
6002	0	0x0000
6003	0	0x0000
6004	0	0x0000
6005	0	0x0000
6006	0	0x0000
6007	0	0x0000
6008	0	0x0000
6009	0	0x0000
6010	0	0x0000
6011	0	0x0000
6012	0	0x0000
6013	0	0x0000
6014	0	0x0000
6015	0	0x0000

Monitoring of the registers shown on the left can be started and stopped.

The register to be monitored can be changed. (For example, monitoring of only No. 6000 is possible.) When monitoring is stopped, this button will become valid.

The input registers are pseudo-input.

The size of this window can be changed.

Monitor setting

When "Stop" is selected for the monitor state, the "Monitor Setting" button will be a validated.

The Number selected here will be the register to monitor.

The No. selected in the "information List" will move to the "Display List".

All Numbers in the Information List" will move to the "Display List".

The Number selected in the "Display List" will move to the "Information List".

All Numbers selected in the "Display List" will move to the Information List".

Caution : When a large amount of information is monitored, the communication size with the robot controller will increase, and it may take time to update the information. It is recommended to monitor only the required registers with the monitor setting.

Pseudo-input

With the pseudo-input mode, the values input from the following window will be interpreted as the input register values instead of the registers input from an external source.

INPUT Signal

Head Signal No. : 0 Select

Click on -> check box. 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 = (Hex)

Click on -> check box. 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 = (Hex)

Bit Pseudo-INPUT

All Signals OFF All Signals ON

INPUT Register

Head Register No. : 6000 Select

No. of Register : 10

☒ Decimal ☐ Hexadecimal

6000	0	6004	0	6008	0
6001	0	6005	0	6009	0
6002	0	6006	0		
6003	0	6007	0		

Register Pseudo-INPUT

Refer to the pseudo-input of the general-purpose input signals.

The input registers are pseudo-input.

【Operation methods】

- 1) First, read out the registers to be pseudo-input. Up to 16 sequential registers can be set simultaneously. Input the head No. of the register to be read and the number of registers to read, and then click on the "Select" button.
- 2) The values of the designated number of registers will appear with the designated register at the head.
- 3) Set the register value, and click on the "Register Pseudo-Input" button.

The pseudo-input mode is canceled when this window is closed.

5. Explanation of each tool

(10) Output register

This screen cannot be referred to if the CC-Link option card is not mounted on the robot controller.

The values of the CC-Link function input registers can be monitored.

No.	DEC	HEX
6000	88	0x0058
6001	0	0x0000
6002	0	0x0000
6003	0	0x0000
6004	74	0x004A
6005	0	0x0000
6006	0	0x0000
6007	0	0x0000
6008	0	0x0000
6009	0	0x0000
6010	0	0x0000
6011	0	0x0000
6012	0	0x0000
6013	0	0x0000
6014	0	0x0000
6015	0	0x0000

Monitoring of the registers shown on the left can be started and stopped.

The register to be monitored can be changed. (For example, monitoring of only No. 6000 is possible.) When monitoring is stopped, this button will become valid.

The output registers are forcibly output.

The size of this window can be changed.

Monitor setting

When "Stop" is selected for the monitor state, the "Monitor Setting" button will be a validated.

The Number selected here will be the register to monitor.

The No. selected in the "information List" will move to the "Display List".

All Numbers in the Information List" will move to the "Display List".

The Number selected in the "Display List" will move to the "Information List".

All Numbers selected in the "Display List" will move to the Information List".

Caution : When a large amount of information is monitored, the communication size with the robot controller will increase, and it may take time to update the information. It is recommended to monitor only the required registers with the monitor setting.

Forced output

The registers can be forcibly output.

OUTPUT register Forced output [2 : R...]

Exit

Head Register No. : 6000

Number of Register (<=16) : 8

Read

☐ Hexadecimal ☒ Decimal

6000	0
6001	0
6002	0
6003	0
6004	0
6005	0
6006	0
6007	0

Write

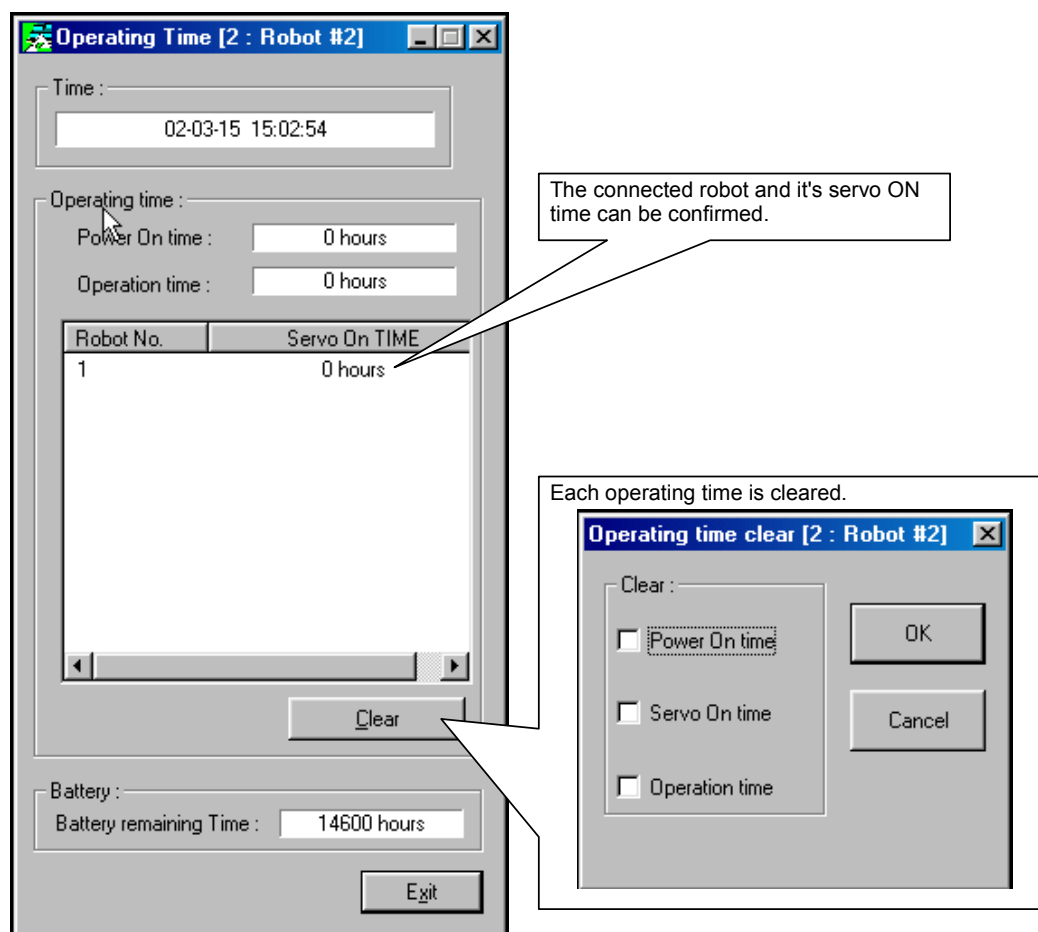
【Operation methods】

- 1) First, read out the registers to be forcibly output. Up to 16 sequential register can be output simultaneously.
Input the head No. of the register to be read and the number of registers to read, and then click on the "Read" button.
- 2) The designated number of registers will appear with the designated register at the head. Set the value, and click on the "Forced Output" button.

5.2.4.2. Operation monitor

(1) Operating time accumulation

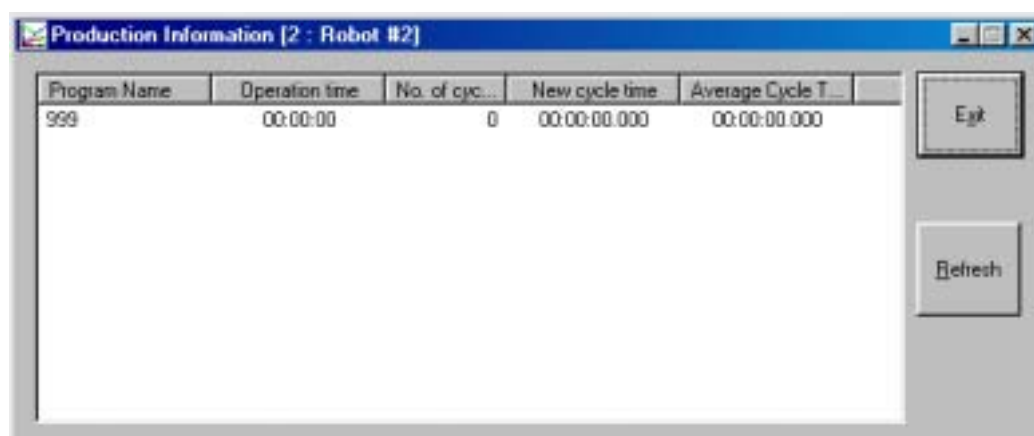
The robot operating time, and battery usage time, etc., can be confirmed.



(2) Production information

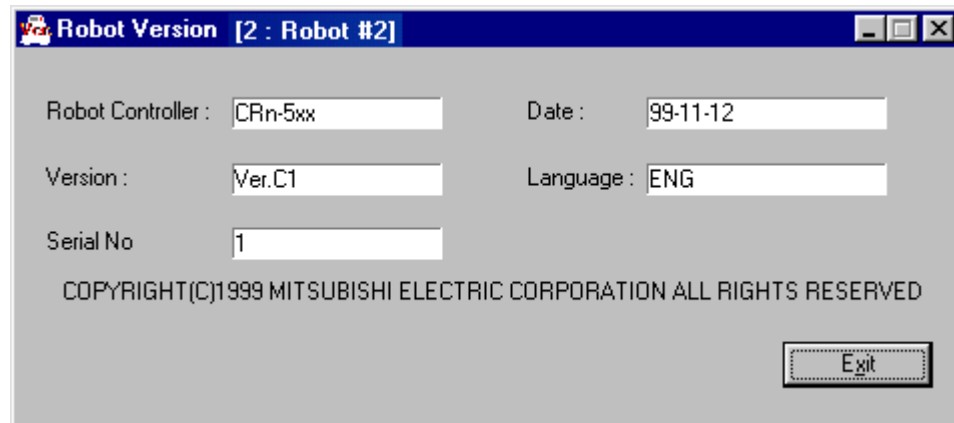
The latest cycle, operation time, No. of cycles and average cycle time for each program in the robot controller can be confirmed.

The production information is not constantly displayed. Click on the "Update" time as necessary.



(3) Robot version

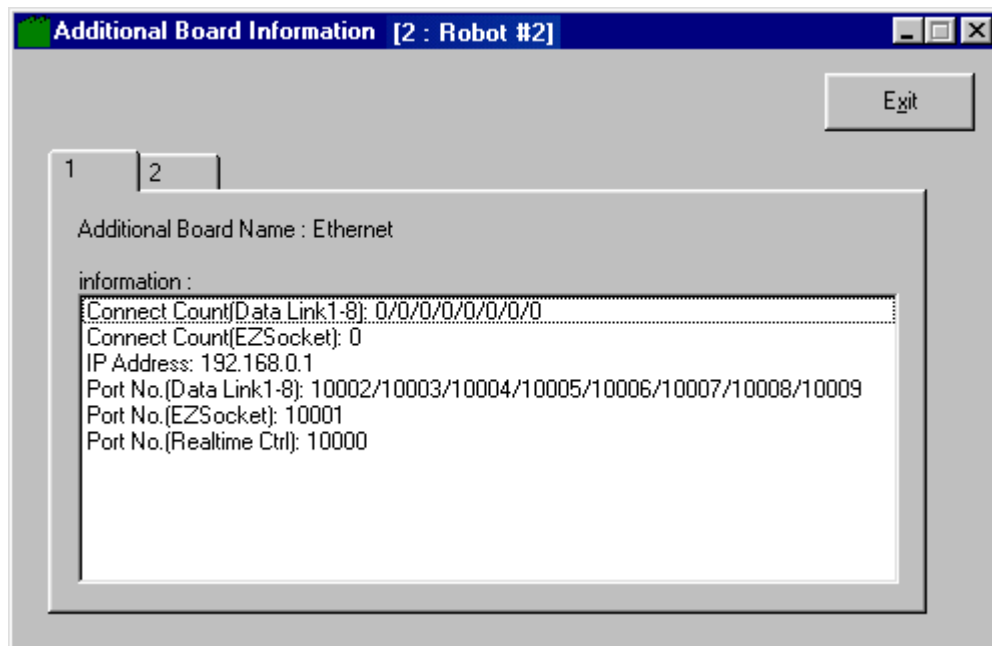
The robot controller version can be confirmed.



(4) Additional Board Information

Information on the option card mounted on the robot controller can be confirmed.

Note that this screen cannot be referred to if robot controller is not provided with a slot for mounting the option card.



5. Explanation of each tool

5.2.4.3. Servo monitor

The servo system is monitored.

(1) Position (ABS)

The state of the currently connected robot encoder can be monitored.

The screenshot shows the 'Servo Monitor << ABS >> [2 : Robot #2]' window. It features a tab labeled 'Robot 1' and a sub-tab 'RP-3AH'. The window is divided into six sections, each with a table of data for joints J1 through J8:

- Position Feedback [pulse]**: All values are 0.
- Position in one rotation [pulse]**: All values are 0.
- Full Command [pulse/IT]**: All values are 0.
- Position Droop [pulse]**: All values are 0.
- Max. Position Droop [pulse]**: All values are 0.
- Position Command [pulse]**: J3 is 17066, all other joints are 0.

(2) Speed

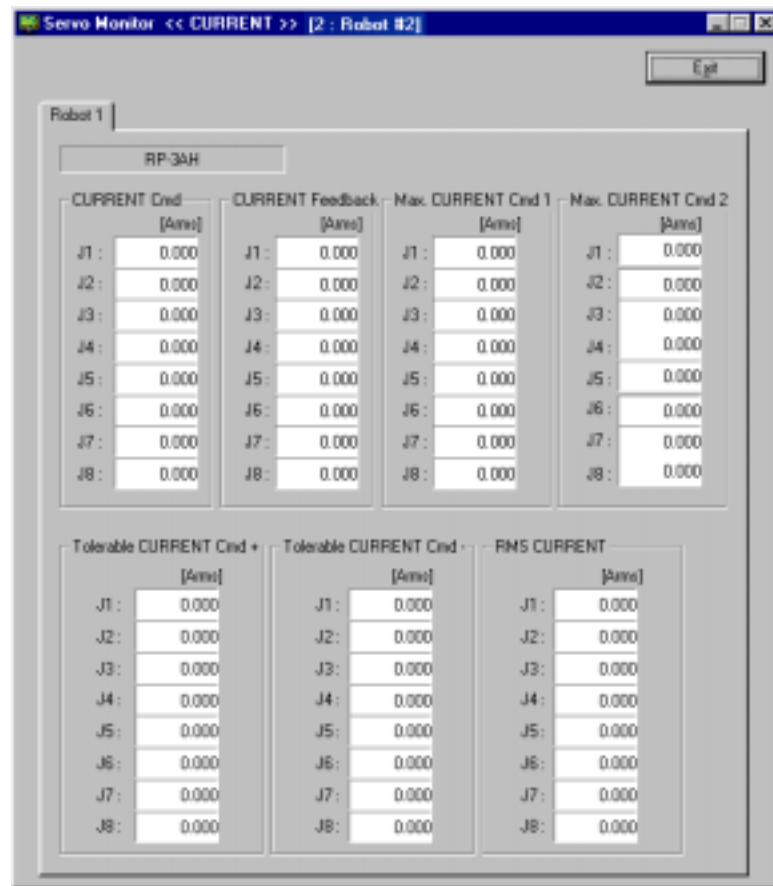
The motor speed, etc., of each robot axis can be monitored.

The screenshot shows the 'Servo Monitor << SPEED >> [2 : Robot #2]' window. It features a tab labeled 'Robot 1' and a sub-tab 'RP-3AH'. The window is divided into three sections, each with a table of data for joints J1 through J8:

- Speed Feedback [rpm]**: All values are 0.000.
- Speed MAX [rpm]**: All values are 0.000.
- Speed Command [rpm]**: All values are 0.000.

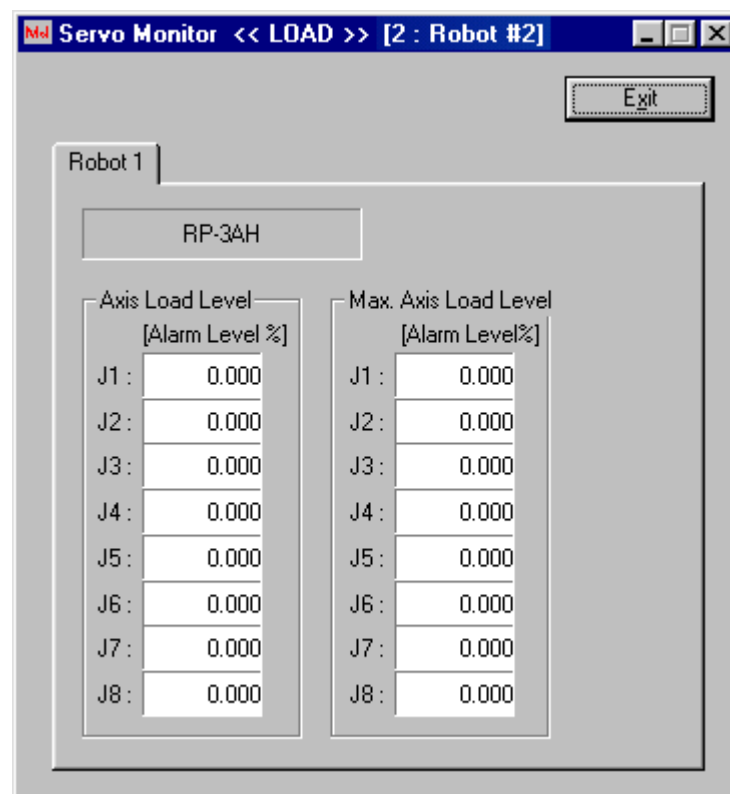
(3) Current

The current state of each robot axis can be monitored.



(4) Load

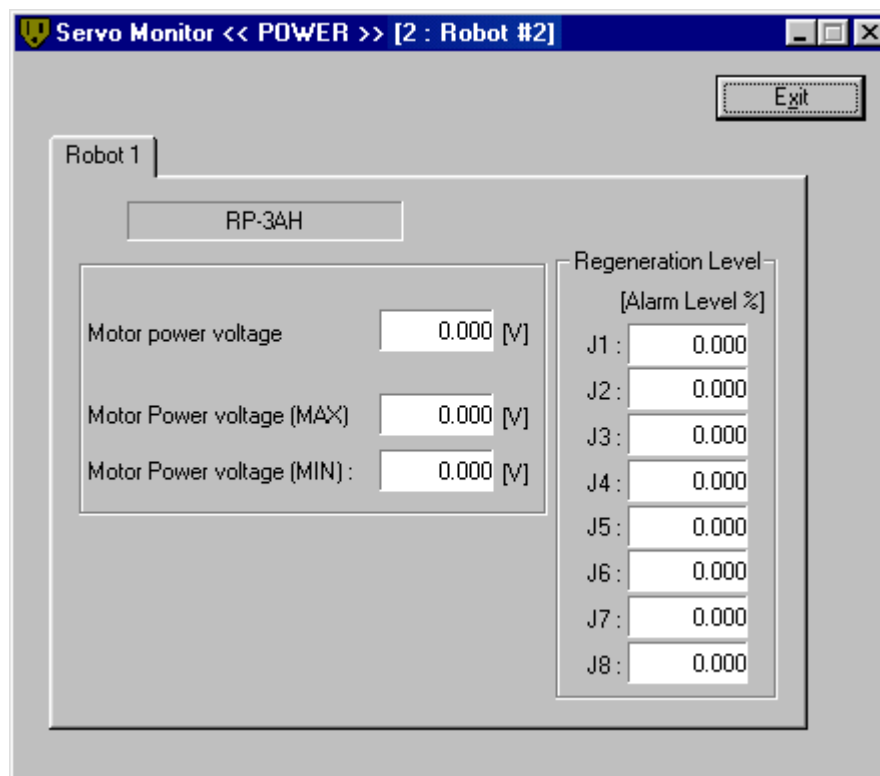
The load state of each robot axis can be monitored.



(5) Power

5. Explanation of each tool

The power state of the robot's main circuit can be monitored.



5.3. Maintenance tool

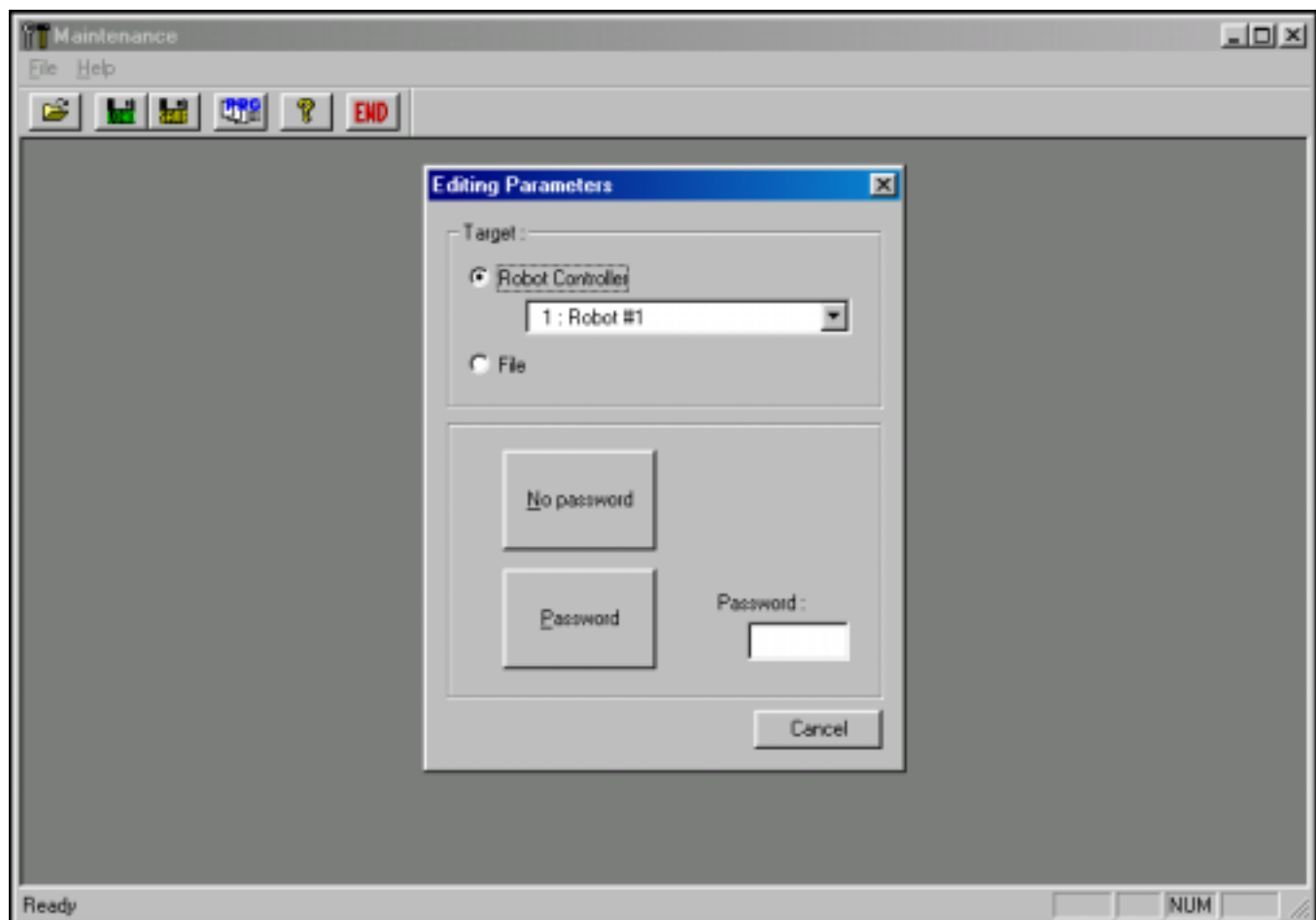
With the maintenance tools, the parameter information set in the robot controller can be referred to and rewritten.

The following operations can be carried out with the maintenance tools.

1. Connect with the robot controller and refer or change the parameter information in the robot controller.
2. Download parameter information in robot controller to personal computer.
3. Off-line edit the parameter information downloaded from the robot controller.
4. Upload off-line edited parameter information to robot controller.
5. Backup information in robot controller.
6. Print of parameter in robot controller.

5.3.1 Starting

Select "Maintenance tool" from the menu. The following "Maintenance tool" window will appear.



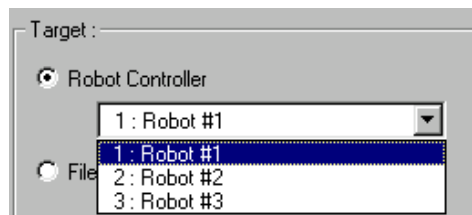
5.3.2. Editing Parameters

When Maintenance is started, or when [File] – [Open] are selected from the menu, a password entry window appears.



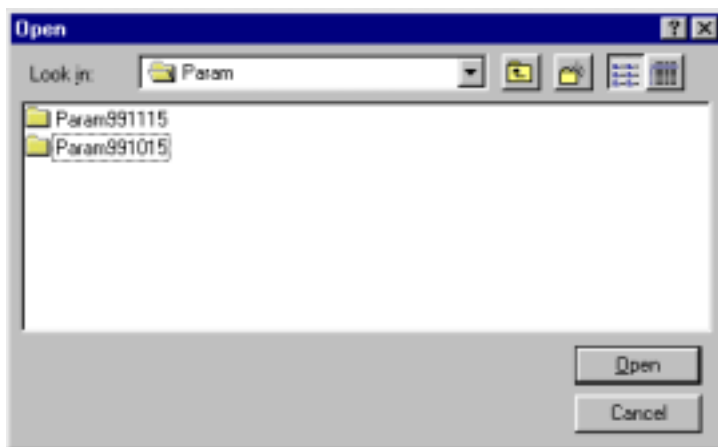
1) Load destination

To connect with a robot controller and perform the maintenance of the parameters in the robot controller, select [Robot Controller]. To edit the parameter information that has been downloaded to a personal computer, select [File]. If a robot controller has been selected as the load destination, select the robot controller you want to connect.



This window will appear when "File" - "Open" are selected from under the menu.

When "File" is selected, a window for selecting the parameter information to edit will appear.

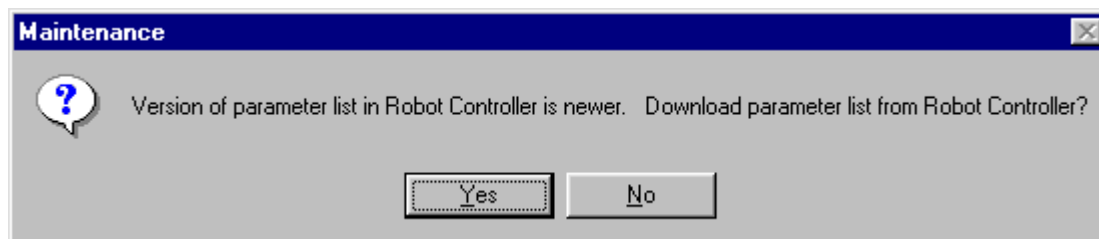


Double-click on the folder containing the parameter information.

2) Password

"Password input" is used by the Mitsubishi service personnel to change the robot's system information. Unconditionally select "No password" here.

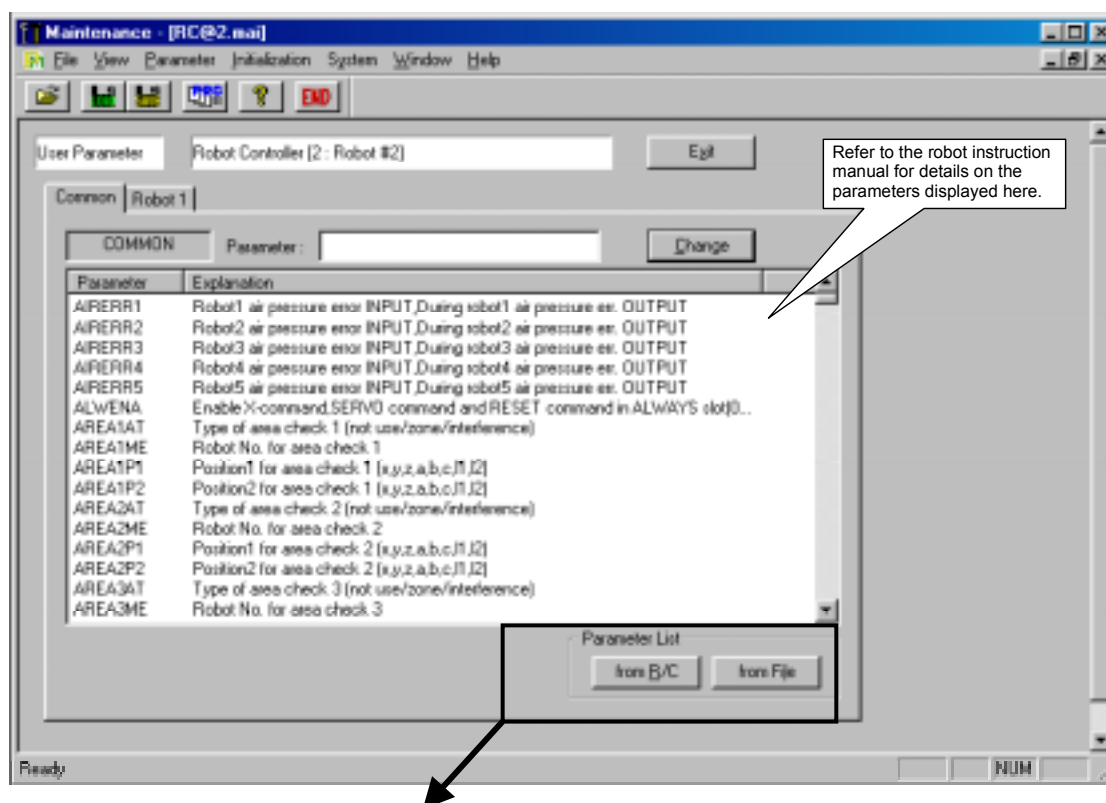
5.3.3. Downloading the parameter list (Controller → Personal computer)



This window will appear in the following cases.

1. When there is no parameter list information in the personal computer.
 2. When the parameter list used in the robot controller is newer than the parameter list already stored in the personal computer.
- * Although it will take approx. five minutes to download the parameter list information, using the latest parameter information is recommended.

If the parameter list information is not read out correctly, the following type of window will appear.



The parameter list can be read from the robot controller or from the personal computer using these buttons.

Reading the parameter list

[From R/C] : The parameter list is read from the robot controller.

[From File] : The parameter list written in the personal computer is read.

5.3.4. Changing the parameters

Click on a parameter displayed in the list, or input the parameter name, and then click on the "Change" button. The designated parameter information in the robot controller or personal computer will appear. After confirming the parameter, the parameter information in the robot controller can be rewritten by clicking on the "Write" button. (When editing the parameter information saved in the personal computer, the "Write" button will change to an "OK" button.)

The screenshot shows a window titled "Parameter" with a close button (X) in the top right corner. Inside the window, there is a text input field containing "AXMREV" and a "Robot:" label followed by a numeric input field containing "0". Below these is a label "Rated speed of motor for additional axis [rpm]". A list of 16 parameters is displayed, each with a number (1 to 16) and a value of 2000. At the bottom right of the list area is a "Write" button, which is disabled (grayed out). Below the "Write" button is an "Exit" button.

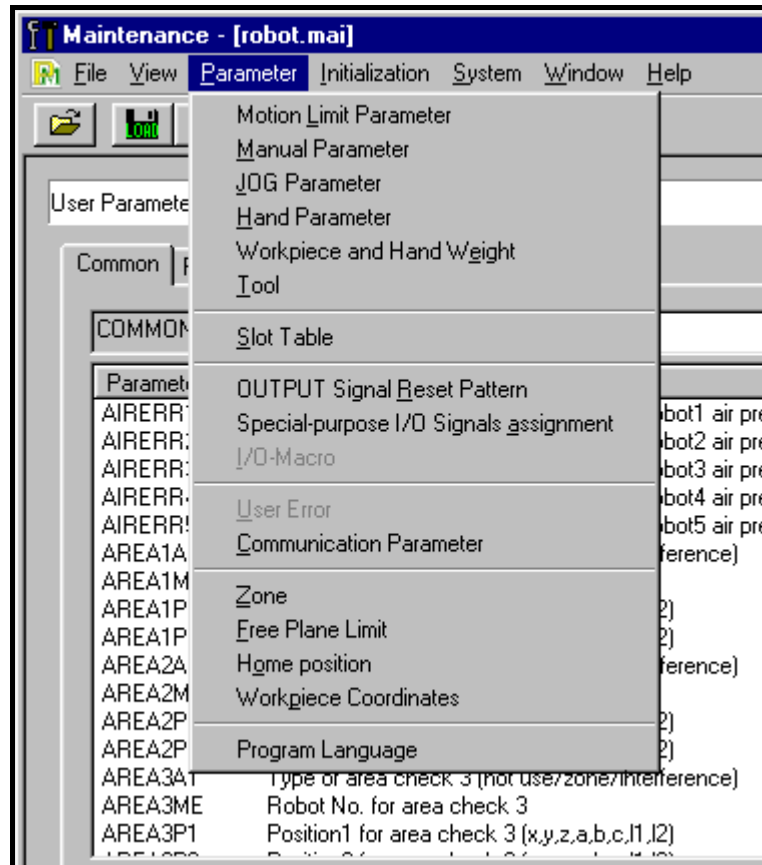
Axis	Rated speed of motor for additional axis [rpm]
1	2000
2	2000
3	2000
4	2000
5	2000
6	2000
7	2000
8	2000
9	2000
10	2000
11	2000
12	2000
13	2000
14	2000
15	2000
16	2000

There are some parameters that cannot be edited. In this case, the "Write" button operation will be disabled.

To validate the rewritten parameter information in the robot controller, the robot controller power must be turned ON again.

5.3.5. Parameter panels

With the maintenance tool, parameter panels grouped as windows for each function are prepared. Select the name of the parameter to be referred to with "Parameter setting" on the menu.



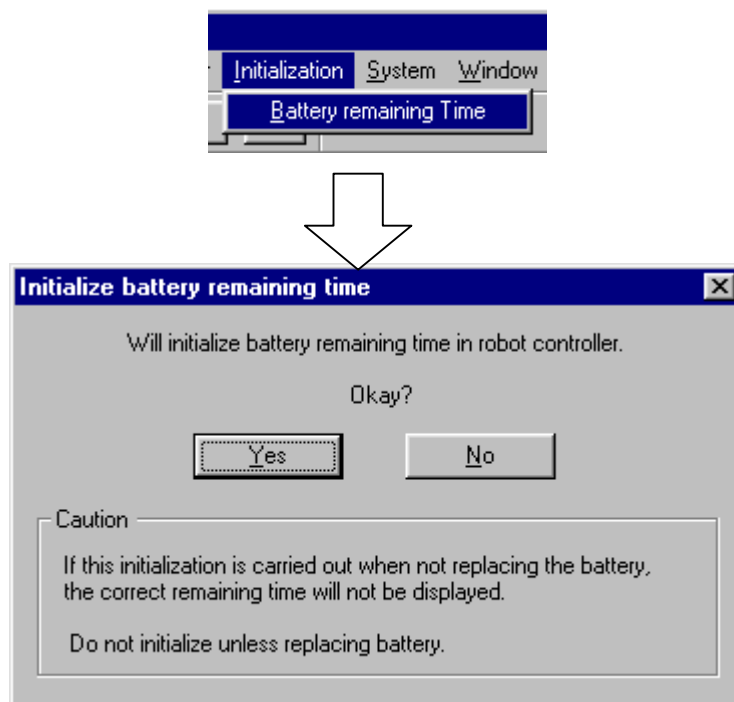
Caution : The language (MELFA-BASIC IV/MOVEMASTER command) to be used by the controller can be changed with the "Robot Language" parameter. Note that the robot models that can use the MOVEMASTER commands are limited. Refer to the Standard Specifications for the model in use to confirm whether the commands can be used.

5.3.6. Initialization

(1) Initializing the battery remaining time

The robot controller battery's remaining time can be initialized. This function is used when replacing the battery.

Select "Initialize" - "Battery remaining time" from the menu.



Caution: If the battery remaining time is initialized when not replacing the battery, the correct remaining time will not be displayed.

5.3.7. System information

The system information can be used to save the robot origin data into a file and to transfer the information in the file to the robot controller.

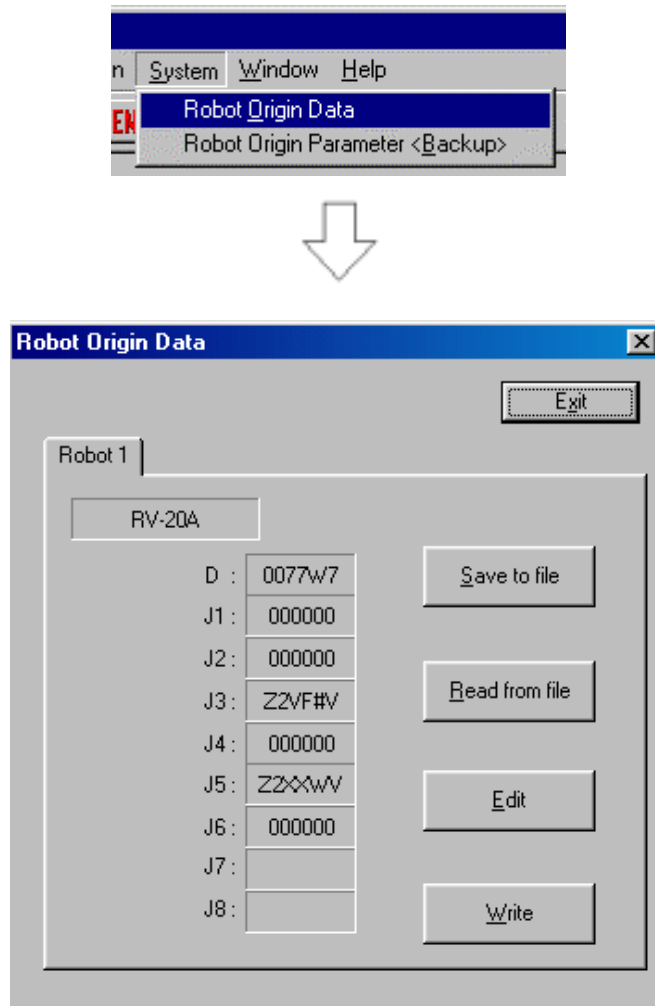
Two types of origin data handling are described in this section, however, the file save format is different in each type. To write the saved information to the robot controller, use the same file save format.

(1) Robot origin data

[Robot Origin Data] can be used to save the robot origin data without additional axes in a file, edit it, or transfer it to the robot controller.

Select "System" - "Robot origin data" from the menu.

Set the R/C mode to AUTO (Ext.).



[Save to file] : Can save the displayed origin data in a file.

[Read from file] : Reads the origin data saved in a file, and displays it on the screen.

[Edit] : Can edit the origin data that is being displayed on the screen.

[Write] : Writes the origin data that is being displayed on the screen into the robot controller.

5. Explanation of each tool

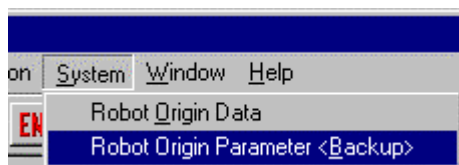
* Robot Controller Operating Mode When Writing Robot Origin Data

When writing robot origin data by using this Software, some restrictions apply to the controller's operating mode depending on the versions of both the Software and the controller. See the table below for a list of restrictions.

	Controller Version G9 or later.	Controller Version G8 or earlier.
Personal Computer Support Software Version C2 or later.	AUTO(Ext.) or Teach mode	AUTO(Ext.) mode
Personal Computer Support Software Version C1 or earlier.	Origin data cannot be rewritten from the Personal Computer Support Software.	

(2) Robot Origin Parameter <Backup>

[Robot Origin Parameter <Backup>] can be used to back up the parameters that comprise the robot origin data. If the robot origin data has additional axes, use this function to back up the origin data.

A screenshot of the 'Robot Origin Parameter <Backup>' dialog box. The dialog box has a title bar with the text 'Robot Origin Parameter <Backup>' and a close button. Inside the dialog box, there is an 'Exit' button in the top right corner. Below the 'Exit' button, there are two input fields: 'Robot No.' with a dropdown menu showing '1' and 'Robot type' with a text box containing 'RV-1A'. Below these fields, there is a section titled 'Robot Origin Data (Parameter):' which contains four input fields: 'MEINSD' with the value '-241210', 'MEINST' with the value '3F', 'MEINSZ' with the value '0, -182044, 218453, 0, 204800, 0,', and 'MEOFFZ' with the value '0, 1820, 2731, 0, 0, 0, 0'. At the bottom of the dialog box, there are two buttons: 'Save to file' and 'Load to R/C'.

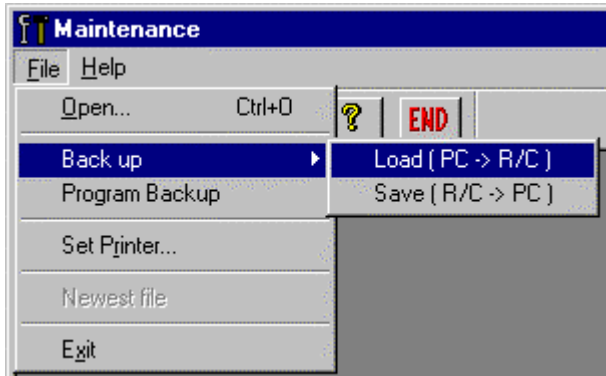
[Save to file] : Can save the contents of the parameters for the origin (four displayed parameters) loaded from the robot controller in a file.

[Load to R/C] : Can transfer the parameters for the origin saved in a file to the robot controller. To transfer the parameters for the origin to the robot controller, set the mode to the [TEACH] mode.

5.3.8. Backup

A backup of the data in the robot controller can be downloaded to the personal computer, and the backup data saved in the personal computer can be uploaded to the robot controller.

(1) Backup



From the menu "File" - "Backup", select "Load (Personal computer → Robot)" or "Save (Robot → Personal computer)".

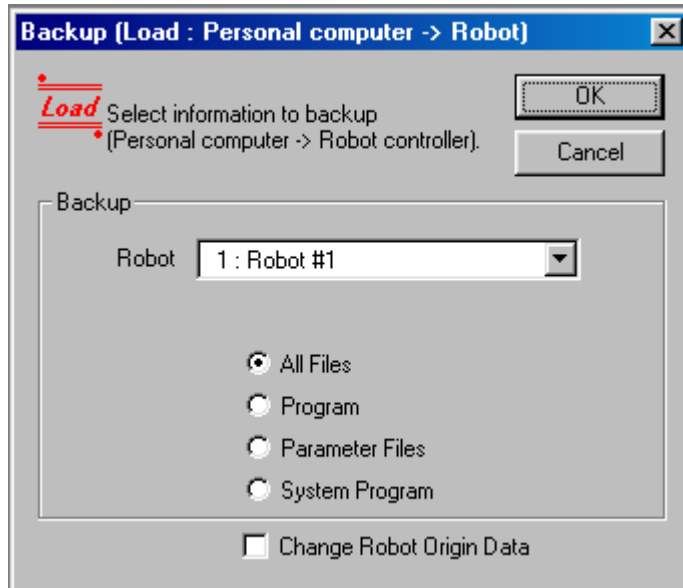
- * Load (Personal computer → Robot)The backup data saved in the personal computer is transmitted to the robot controller.
- * Save (Robot → Personal computer)A backup of the data in the robot controller is saved in the personal computer.

5. Explanation of each tool

(1-1) Load (Personal computer → Robot)

The backup data saved in the personal computer is transmitted to the robot controller.

Select the robot controller at the transfer destination.



<< Backup >>

All Files : Transfers all files (except BKUP.SYS and MECHA.SYS) in the designated folder to the robot controller after all information in the robot controller is cleared (initialized).

Program : Transfers the robot program file in the designated folder to the robot controller.

Parameter Files : Transfers the parameter file in the designated folder to the robot controller.

System Program : Transfers the system base program file in the designated folder to the robot controller.

[Change Robot Origin Data] Valid only when [All Files] or [Parameter Files] is selected under [Backup].

If checked : Replaces the origin information in the robot controller with the contents of the mechanical parameter file to be transferred.

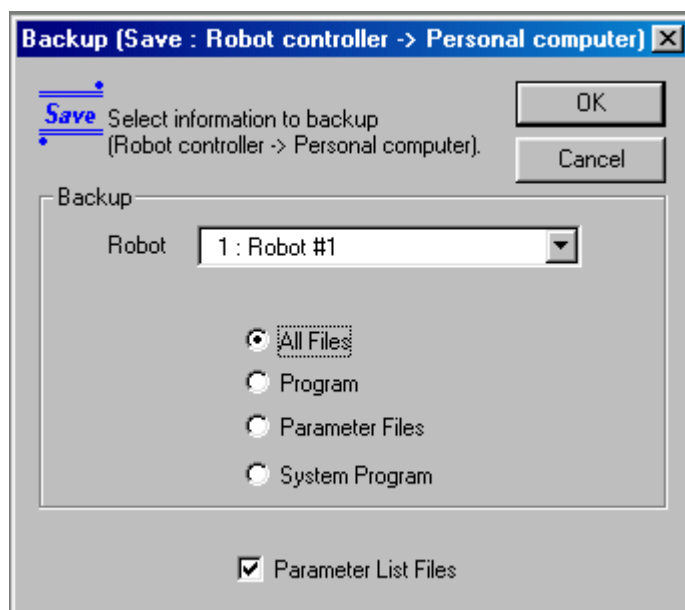
If not checked : Loads the origin information from the robot controller, transfers the information in the designated folder, and then returns the origin information that has been loaded to the robot controller.

If communication is cancelled during a series of load processing, please note that the origin data may have been changed.

(1-2) Save (Robot → Personal computer)

Saves the information in the robot controller in a PC file.

Select the robot controller you want to back up.



<< Backup >>

All Files : Saves all files (robot program, parameter files, etc.) in the robot controller into the designated folder.

Program : Saves the robot program file into the designated folder.

Parameter Files : Saves the parameter files into the designated folder.

System Program : Saves the system base program file into the designated folder.

Note that this Save (Robot → Personal computer) is intended to back up the robot controller, so the program cannot be edited using the program editing tool.

[Parameter List Files] This is used to edit the parameter information saved by backup in offline mode, and is not required for backup. If this is not checked, the time required to save all files will be shortened.

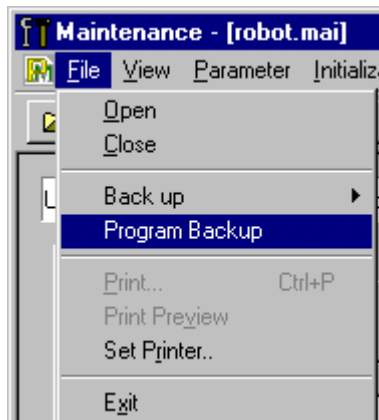
BKUP.SYS and MECHA.SYS files are automatically created in the designated folder. These files contain the saved mechanical information of the robot controller and describe the save format. If these files are deleted or overwritten, please note that offline data editing and data transfer to the robot controller cannot be performed.

5. Explanation of each tool

(2) Program Backup

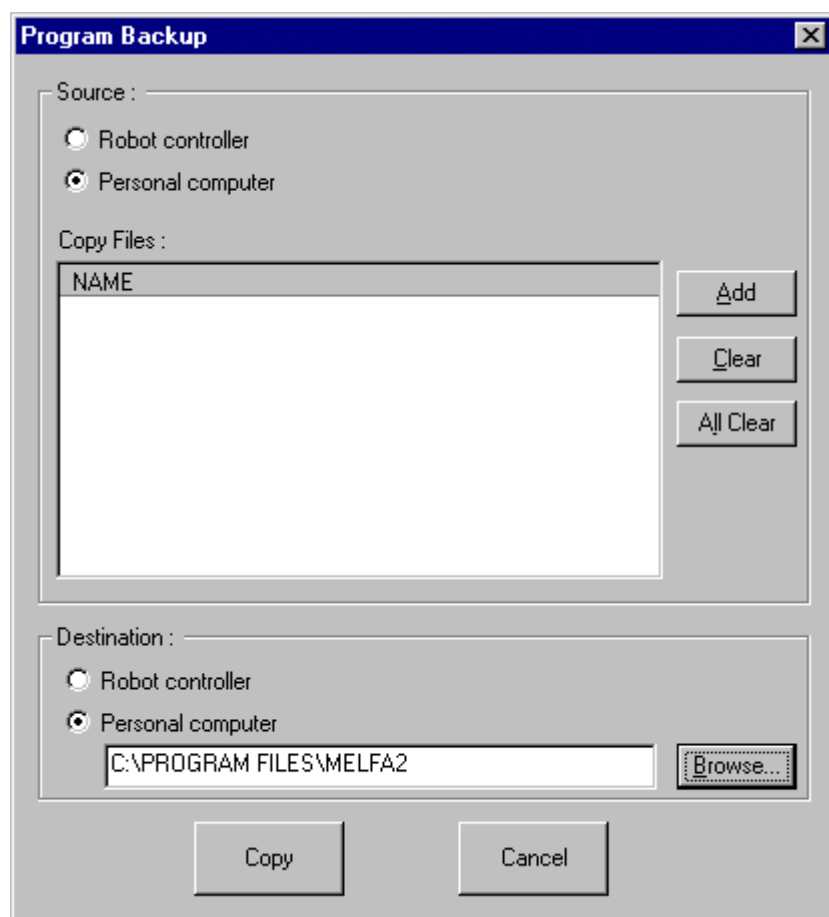
Several robot programs are copied between the R/C and personal computer.

The robot programs copied to the personal computer can be edited with this software.



Select [File] - [Program Backup].

The Program Backup window will open.



Using the [Add] button, set the file to be copied into the copy source area.

The file selected as the copy source can be canceled by selecting with the [Clear] button.

After selecting the copy destination, click on the [Copy] button.

5.3.9. Print of Parameter

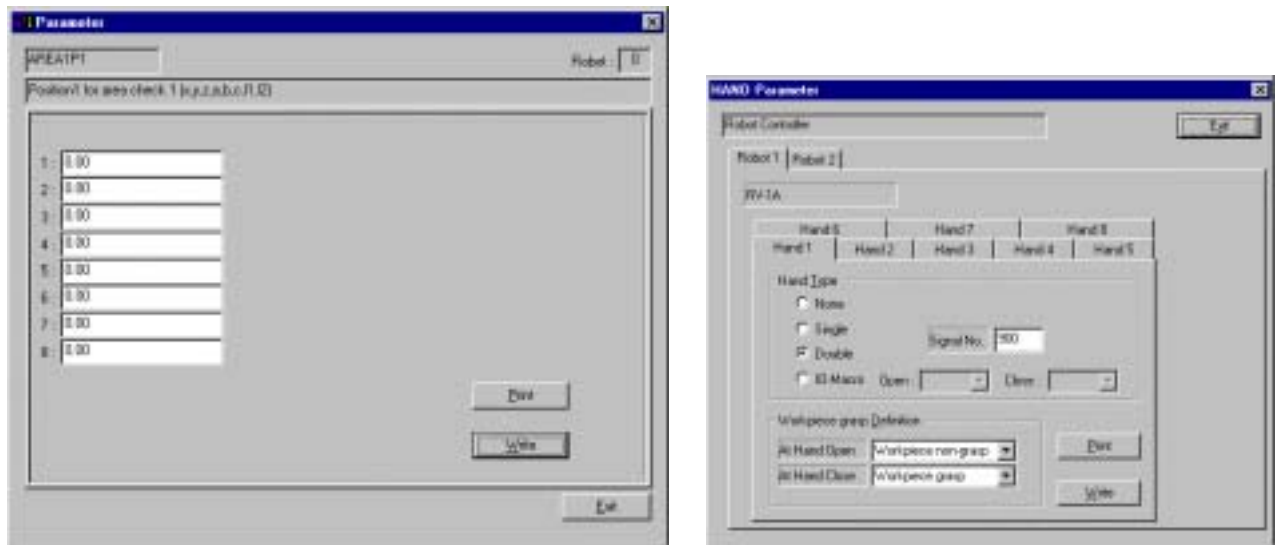
The parameter values in the robot controller can be printed.

With the print function, the parameter values can be output to the printer or saved in a file.

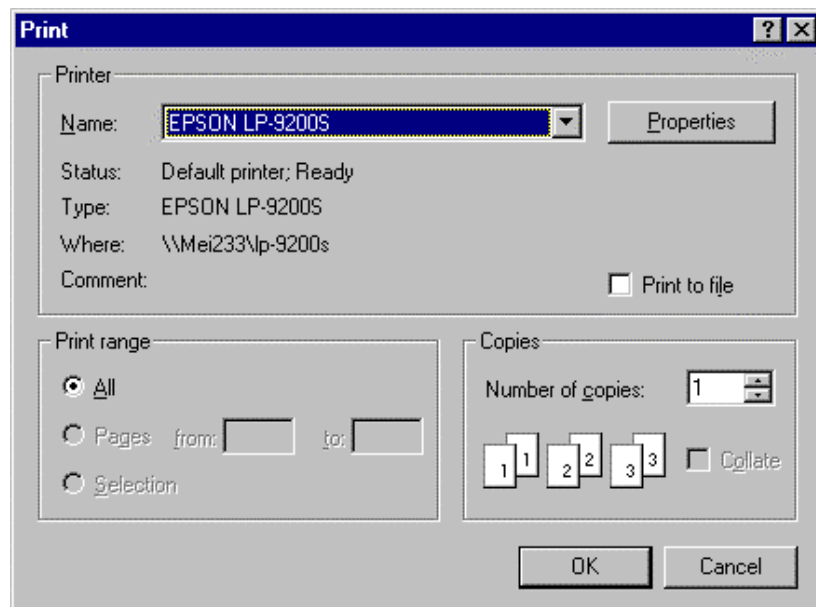
This function is supported in C1 or later version of this software.

(1) Print out

The **[Print]** button is displayed on the HAND Parameter screen or the Parameters screen that is displayed via the parameter setting menu.



If the **[Print]** button is clicked, the Print screen is displayed.



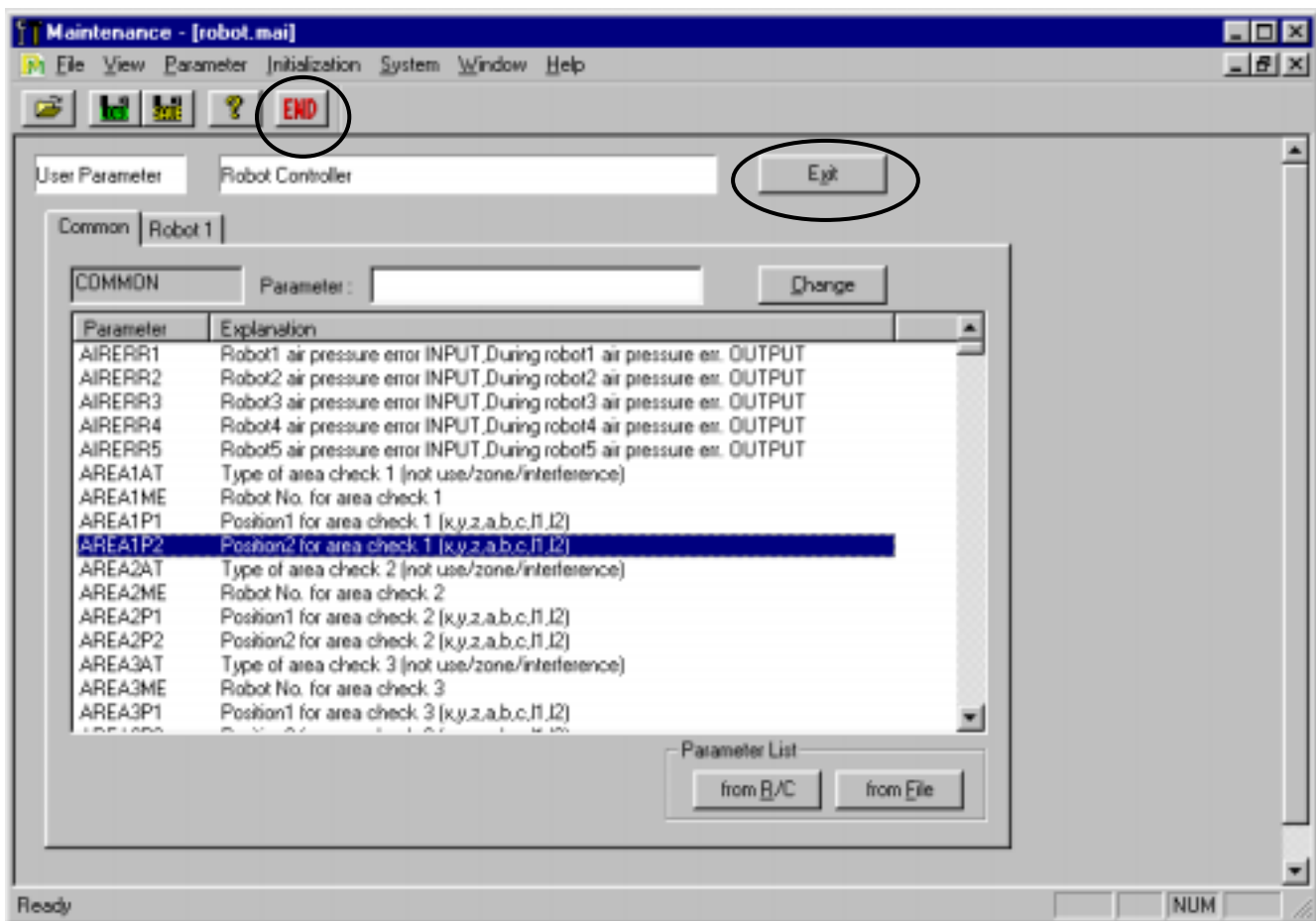
Confirm the printer to print, and then click the **[OK]** button. The parameter information that is currently being displayed is output to the designated printer.

(2) Print to file

If a printer is not connected to the PC, the loaded information can be output to a file. Check **[Print to file]** on the Print screen above, and then click the **[OK]** button. At this time, the file is saved in the text format so that it can be read by a general text editor (MemoPad, WordPad, etc.)

5. Explanation of each tool

5.3.10. Exiting the maintenance tool



Click on the [Exit] button on the screen or click on the "End" button on the tool bar.

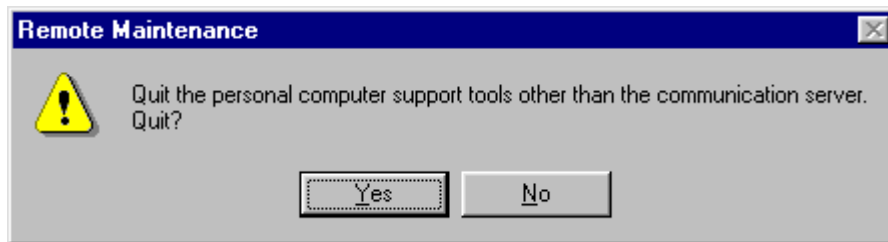
5.4. Remote maintenance

The data of a robot at a remote location can be monitored and serviced over a telephone line.

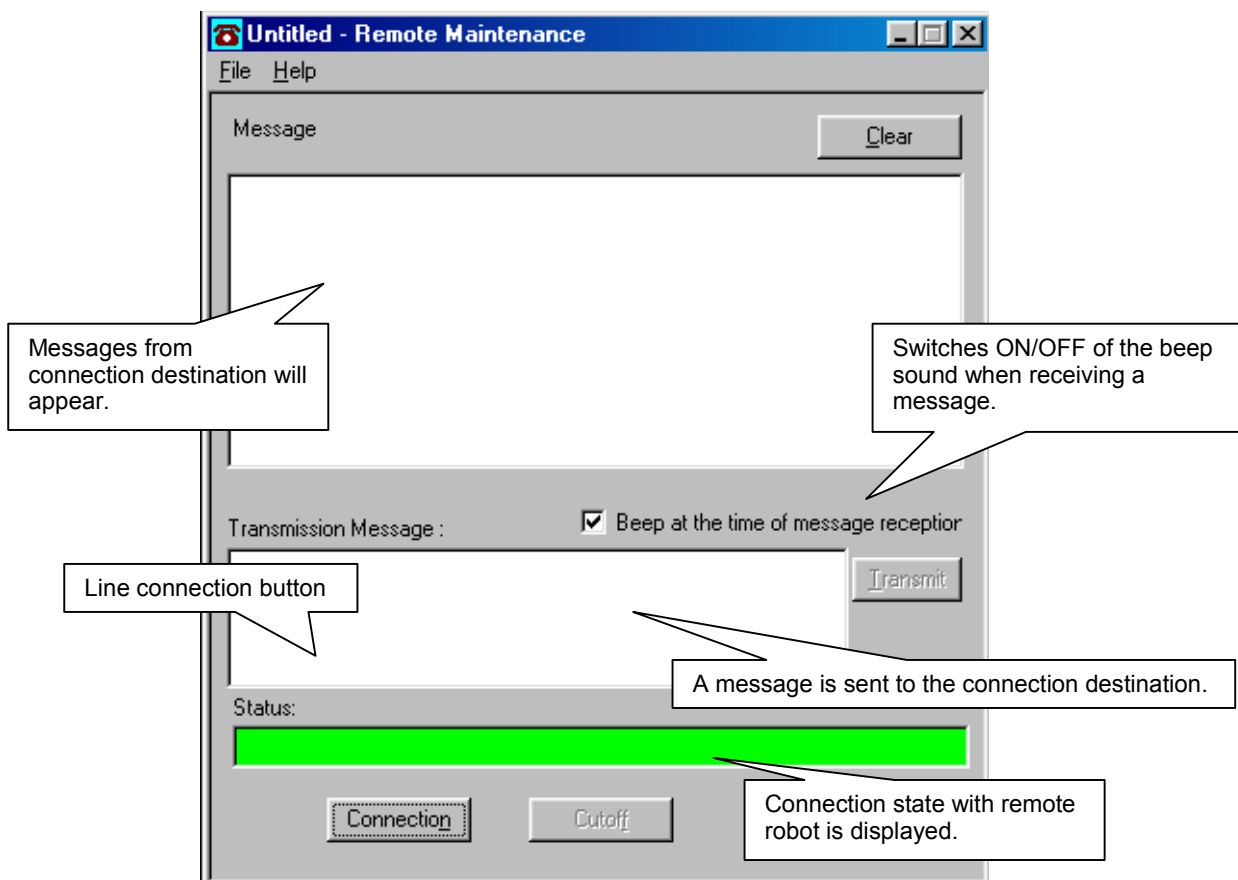
Prepare a personal computer with modem and telephone line at the robot side and at the remote location. Connect each personal computer with the following method.

(1) Connecting with the remote robot

Click on the "Remote maintenance" button on the main screen. The remote maintenance tool will start, and the following type of dialog will appear.



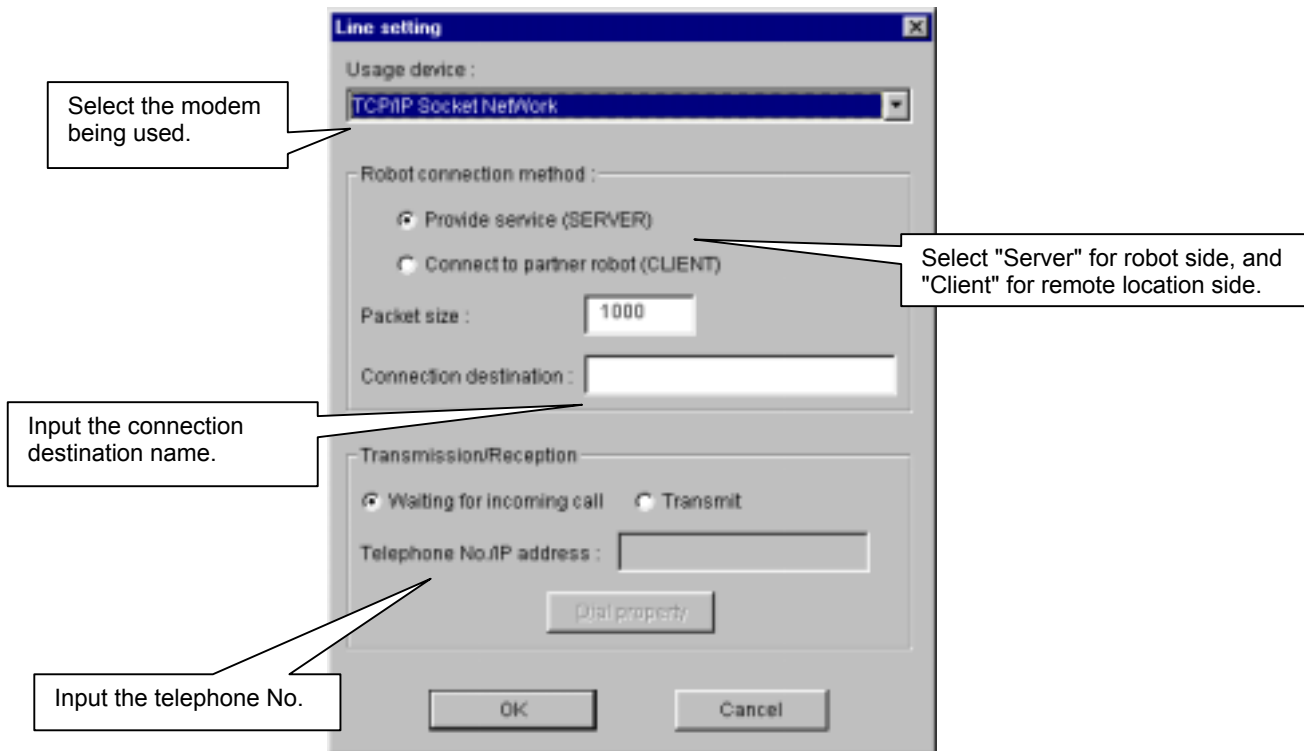
If the program editing tool, monitoring tool or maintenance tool is running, exit the tool, and then click on the "OK" button. The following main screen will appear.



5. Explanation of each tool

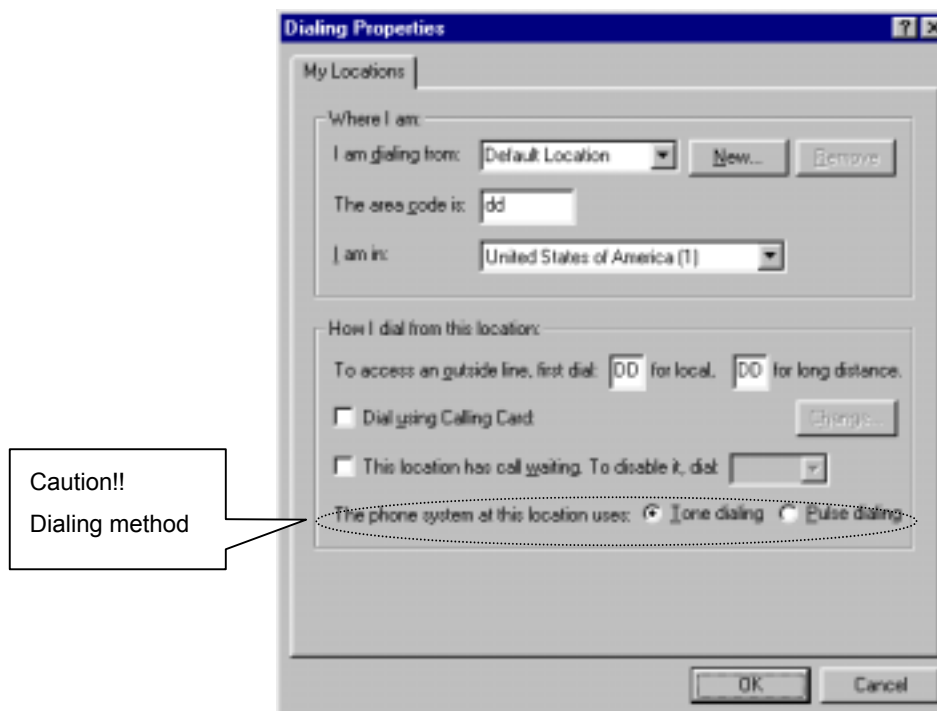
Click on the "Line connection" button.

The following line setting screen will appear. Input the telephone No., and type of modem being used, and then click on the "Start" button. The remote connection will start.



(**Caution:** If the personal computer being used is connected to an intranet, connection with LAN instead of a telephone line is possible.)

The following screen will appear when the "Dial property" button is clicked on. The details regarding the telephone being used can be set. Take special care to the dialing method (tone (push line) or pulse (dial line)).



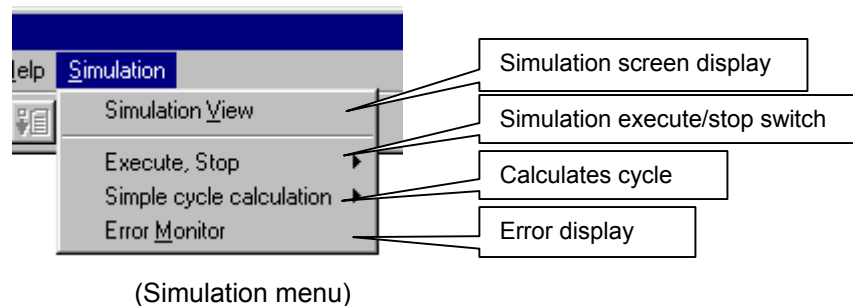
5.5. Simulation (Only for standard installation)

The simulation methods are explained in this section.

Caution : The simulation function is compatible only with the "Standard personal computer support software version".

Note that the simulation function also cannot be used when the MOVEMASTER commands are selected with the "Standard personal computer support software version".

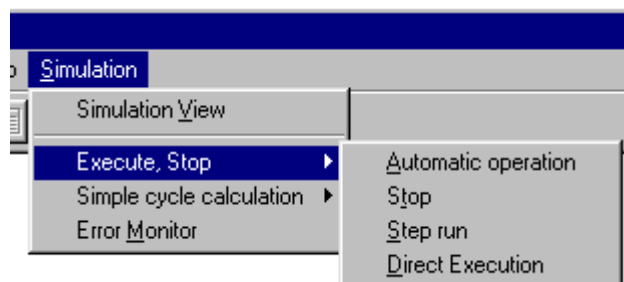
The robot program's operation can be simulated using the program editing tool. The simulation function can be used with the [Simulation] menu.



5.5.1. Simulating operation

The program being edited (only when always open) can be simulated. The types of simulation are the same as the robot controller, and include [Automatic operation], [Step execution] and [Direct execution].

To simulate the operation, select one of the following from the [Simulation] menu [Execute/stop].



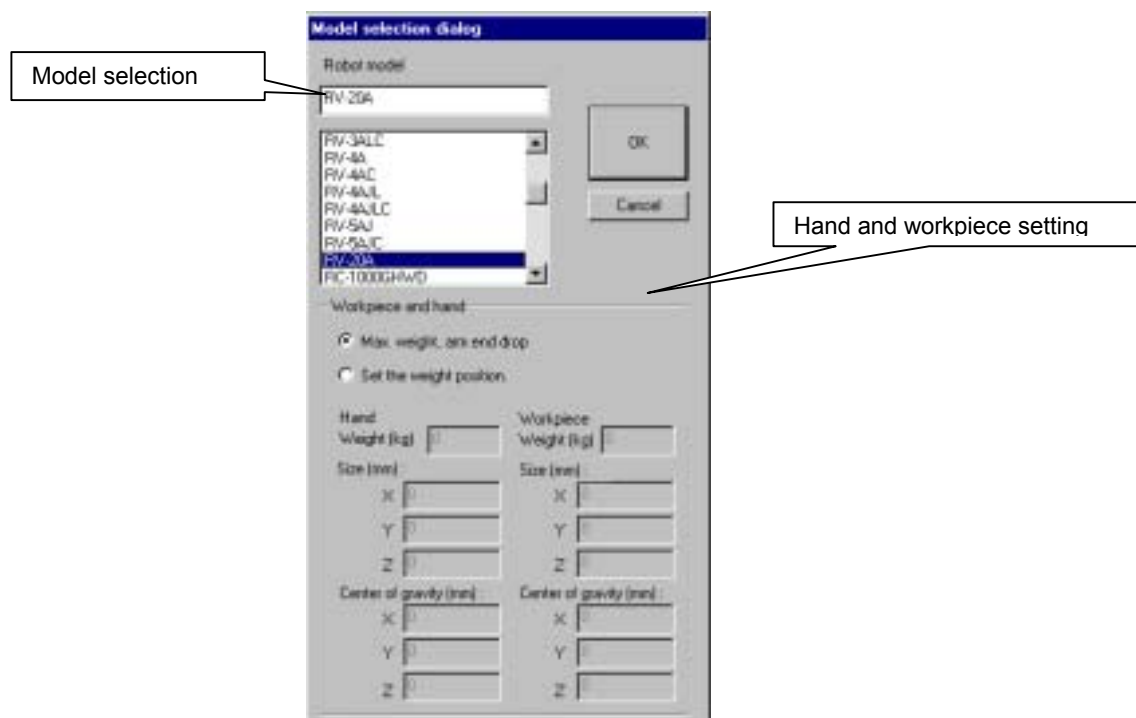
[Automatic operation]

[Step execution]

[Direct execution]

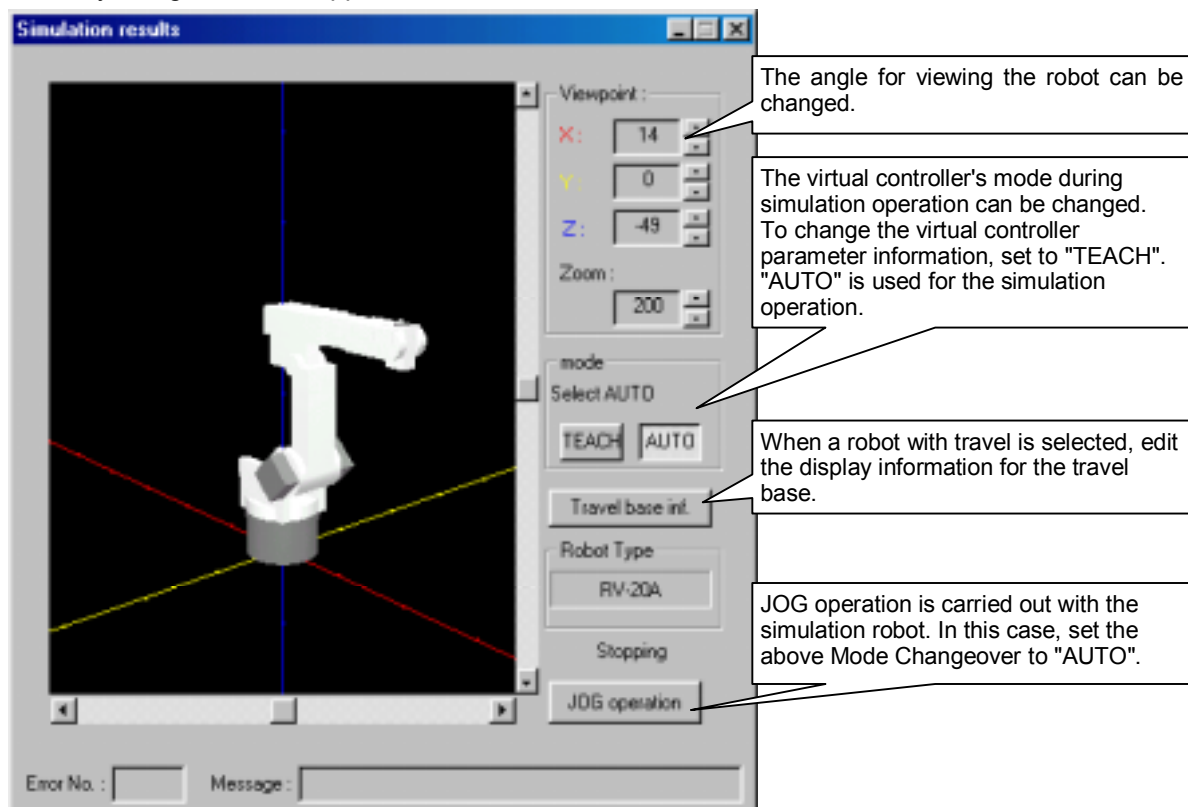
The following "Model selection" dialog will appear, so select the model and then click on the "OK" button.

5. Explanation of each tool



(Model selection dialog)

The following "Simulation results" view panel will appear, and the state of the simulation of the program currently being edited will appear.



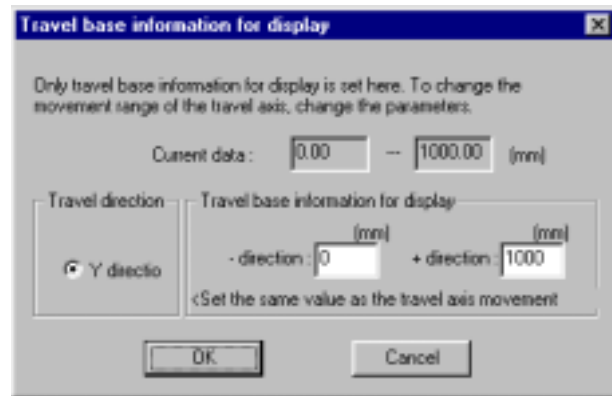
(Simulation panel)

Caution: Communication with the robot controller is not possible while the simulation panel is opened. To communicate with the robot controller, close the simulation panel ([Simulation] menu → Simulation view).

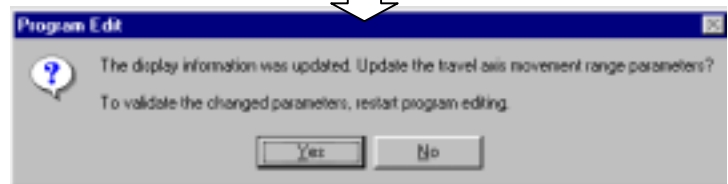
[Travel Base Display Information]

When a robot with travel is selected, edit the display information for the travel base.

The display information is changed here.



To change the robot's movement range, select "Yes (Y)". The parameter values will change. Note that program editing must be restarted to validate these parameters.



Automatic operation

The same operation as the controller's automatic operation takes place.

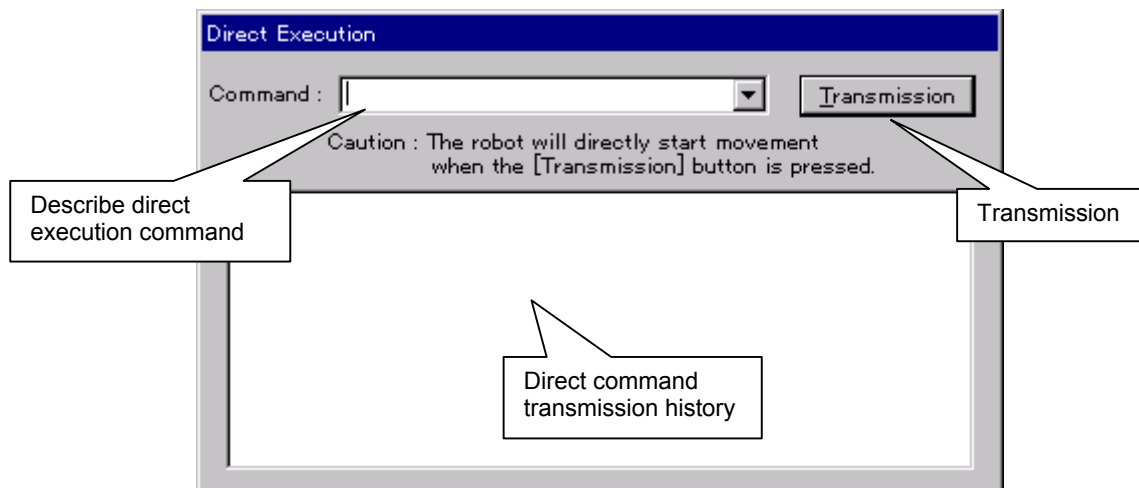
To stop, select [Stop] from the [Simulation] menu → [Execute/stop].

Direct execution

The robot can be directly operated by inputting command statements without line Nos.

When [Simulation] menu → [Execute/stop] → Direct execution

is selected, the following type of dialog for direct execution will appear.



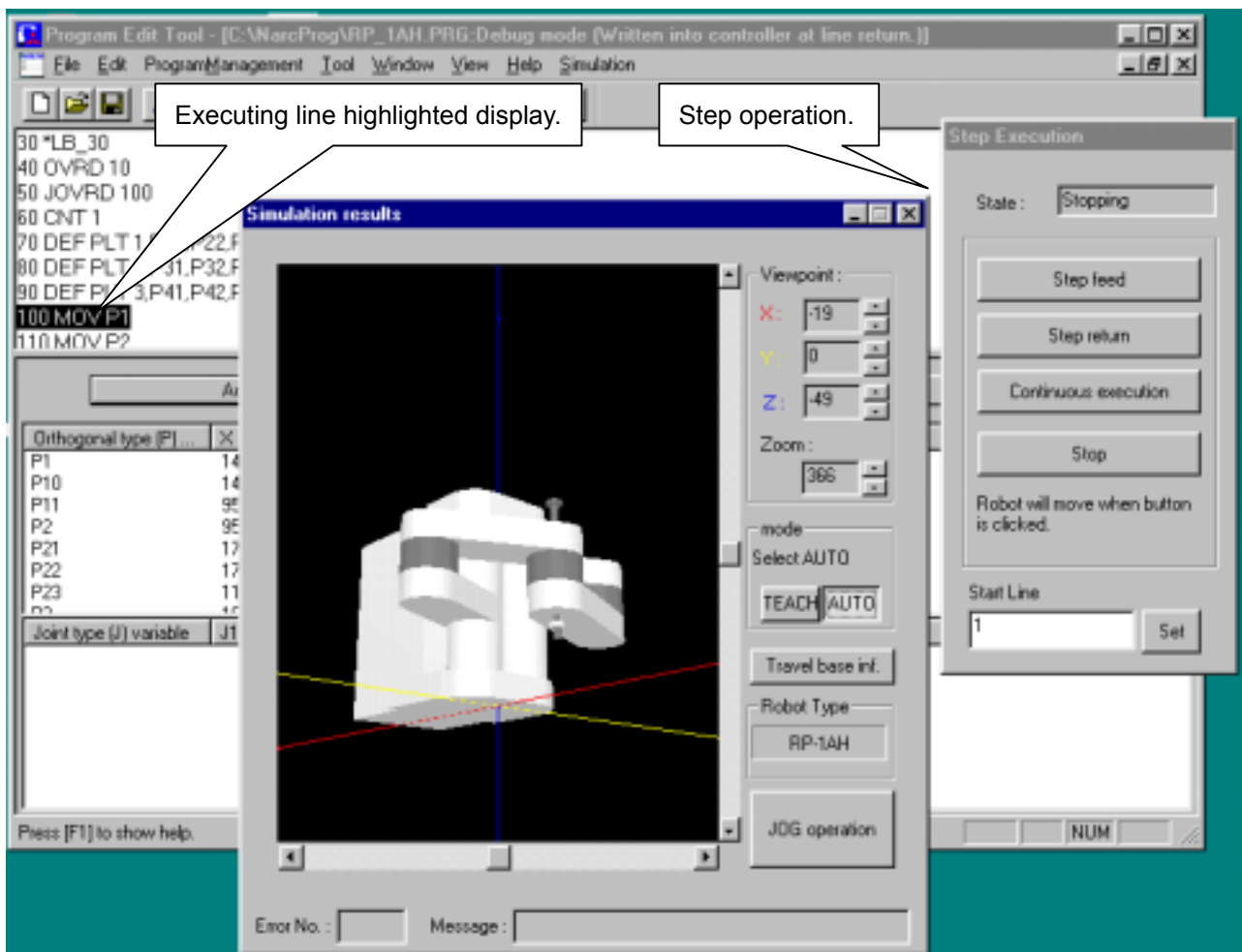
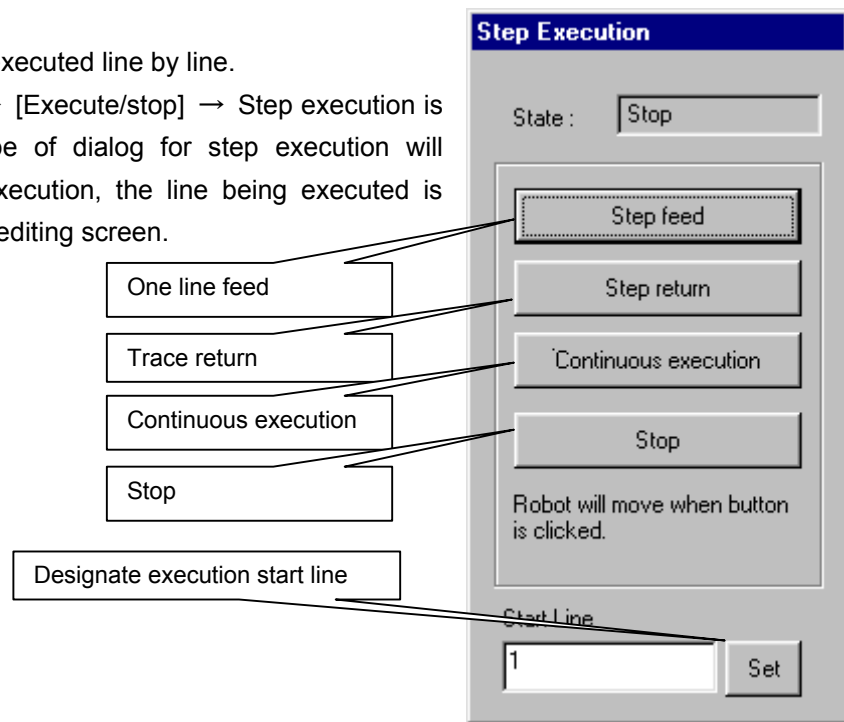
(Direct execution dialog)

5. Explanation of each tool

Step execution

The robot program can be executed line by line.

When [Simulation] menu → [Execute/stop] → Step execution is selected, the following type of dialog for step execution will appear. During the step execution, the line being executed is highlighted on the program editing screen.



(Step execution dialog and highlighted editing screen)

5.5.2 Tact time calculation

*** Quit all other applications before calculating the tact time.***

The tact time can be calculated with two methods using the personal computer support software.

- (1) Calculate the tact time from the created robot program.
- (2) Designate the robot movement's start point and end points, and calculate the movement tact time.

Each calculation method is described below.

Tact time calculation

The robot's movement tact time can be calculated with the personal computer support software using the simulation function.

Note that the calculated tact time will differ according to the performance of the personal computer used and the load state, and will not completely match the actual robot movement time (tact time).

Use this function as a guide for considering the tact time.

The results of the tact time calculated with the personal computer support software will have an error of approx. $\pm 3\%$ under the following conditions compared to the actual robot movement time (tact time).

Tact time calculation conditions

- (1) Do not start any application other than the personal computer support software's "Program Edit" function.

The tact time calculation by the personal computer support software has sections processed in the background of the personal computer. For example, if an application such as a Word document that carries out automatic save is started and running, the personal computer support software's background process will take longer and the correct tact time may not be calculated. (The tact time will increase.)

Quit all other applications before calculating the tact time with the personal computer support software.

- (2) There must be no general-purpose signal input waiting in the program.

Invalidate any signal input waiting, etc., before calculating the tact time.

When considering the tact time of this type of program, add the approximate input waiting time to the calculation results.

5. Explanation of each tool

Precautions for using personal computer support software

When calculating the tact time using this software's simulation function, the tact time may not be correctly calculated depending on the specifications of the personal computer being used or the robot's drawing posture.

Use under the following conditions to calculate the tact time.

Quit all applications other than the personal computer support software when calculating the tact time.

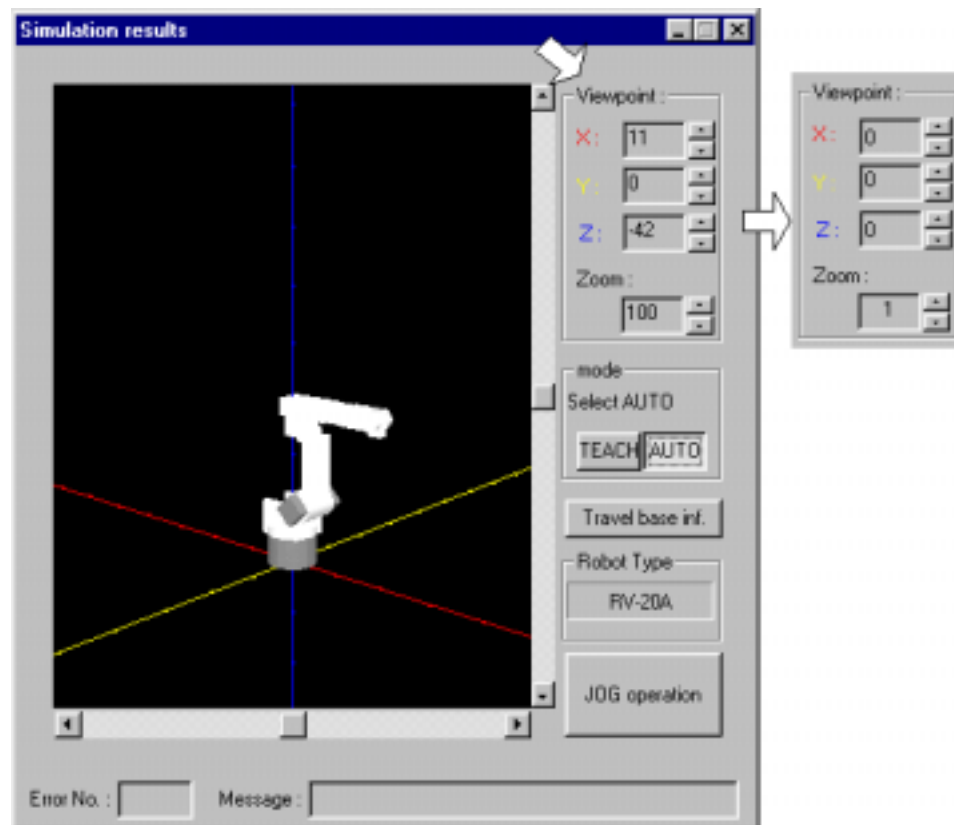
1. Recommended personal computer specifications

CPU over 233MHz
Memory over 128MByte

2. When using recommended specification personal computer

The following type of "Simulation Results" will appear when the tact time is calculated.

Set this window's focus as X axis = 0, Y axis = 0, Z axis = 0, Zoom = 1 as shown below.



This function does not need to be changed when using Windows NT.

3. When using non-recommended specification personal computer

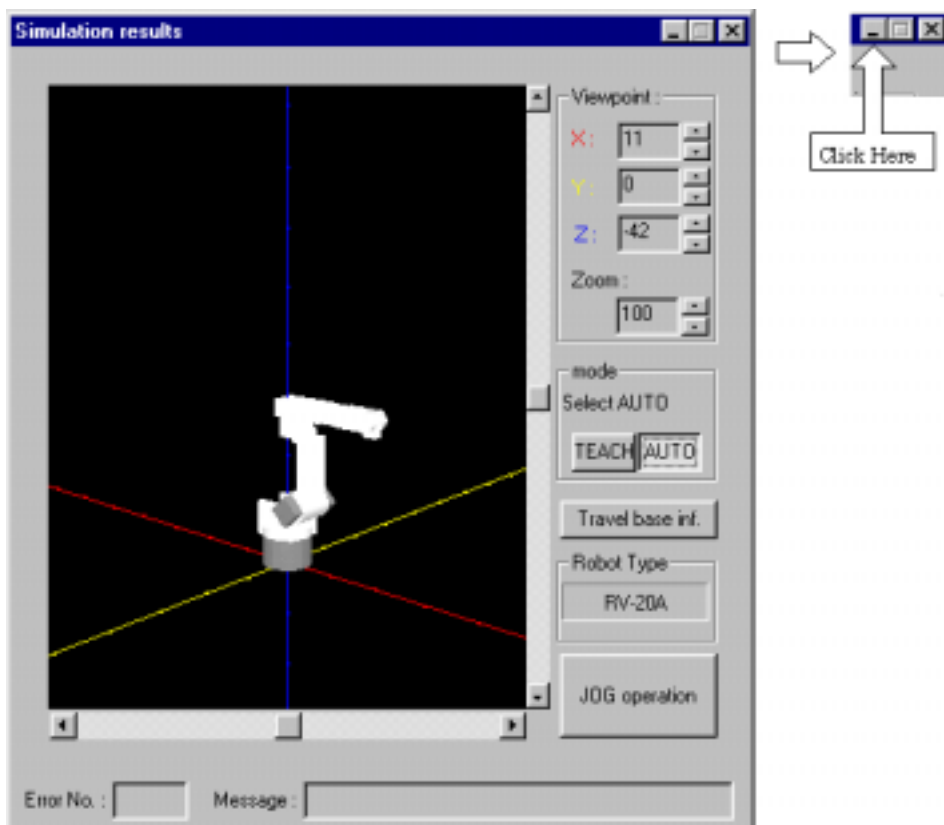
Iconize the "Simulation Results" window. The operation methods are shown below.

The drawing posture does not need to be changed when the window is iconized and the tact time is calculated.

[Operation methods]

- (1) Open the robot program for which the tact time is to be calculated.
- (2) Open "Simulation View" under the menu "Simulation", and select the robot model.
At this time, always set the conditions to "**High Accuracy**".
- (3) The "Simulation Results" window will open. Iconize this window.

The icon can be created by clicking the button on the upper right of the window.



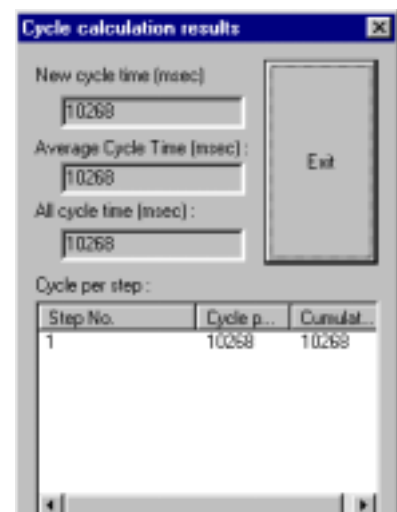
- (4) Select the execution section for which the tact time is to be calculated, and calculate the tact time by selecting "Simulation" - "Simple Tact Calculation" - "Tact Calculation". Refer to section "5.5.2 Calculating the Tact Time" for details on calculating the tact time.

- (5) Do not return the "Simulation Results" window to the original size while calculating the tact time.

When the tact time has been calculated, the "Tact Calculation Results" window as shown on the right will open.

Note: In some cases, this "Tact Calculation Results" window may be hidden behind a different window. In this case, press the [Alt] key on the keyboard, or click on the "**Tact Calculation Results**" icon displayed on the task bar.

This completes the calculation of the tact time.



5. Explanation of each tool

(1) Calculating the tact time from the robot program

The tact time of a designated range of the robot program can be calculated. Note that the following limits will apply.

Note: Observe the following points when calculating the tact time from the robot program.

1. The tact time for a program that falls into an infinite loop cannot be correctly calculated.
Confirm that the robot program does not fall into an infinite loop with a FOR statement or GOTO statement.
2. The tact time cannot be calculated for a program having signal input waiting, etc. Invalidate any signal input waiting, etc., before calculating the tact time.
3. With the PC support software prior to version C1, a program containing an END statement cannot calculate the tact time.

For example, the tact time cannot be calculated for the following types of programs.

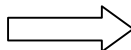
(The sections of the program highlighted in black are set so that the tact time can be calculated.)

```
10 ...  
: :  
100 MOV P01  
110 MOV P02  
120 MOV P03  
130 GOTO 10  
140 END
```

When the GOTO
statement's jump
destination is not
selected

```
10 MOV P00  
: :  
100 GOSUB *SUB1  
110 END  
120 :  
200 *SUB1  
210 MOV P01  
220 RETURN
```

When the subroutine's
jump destination is not
selected

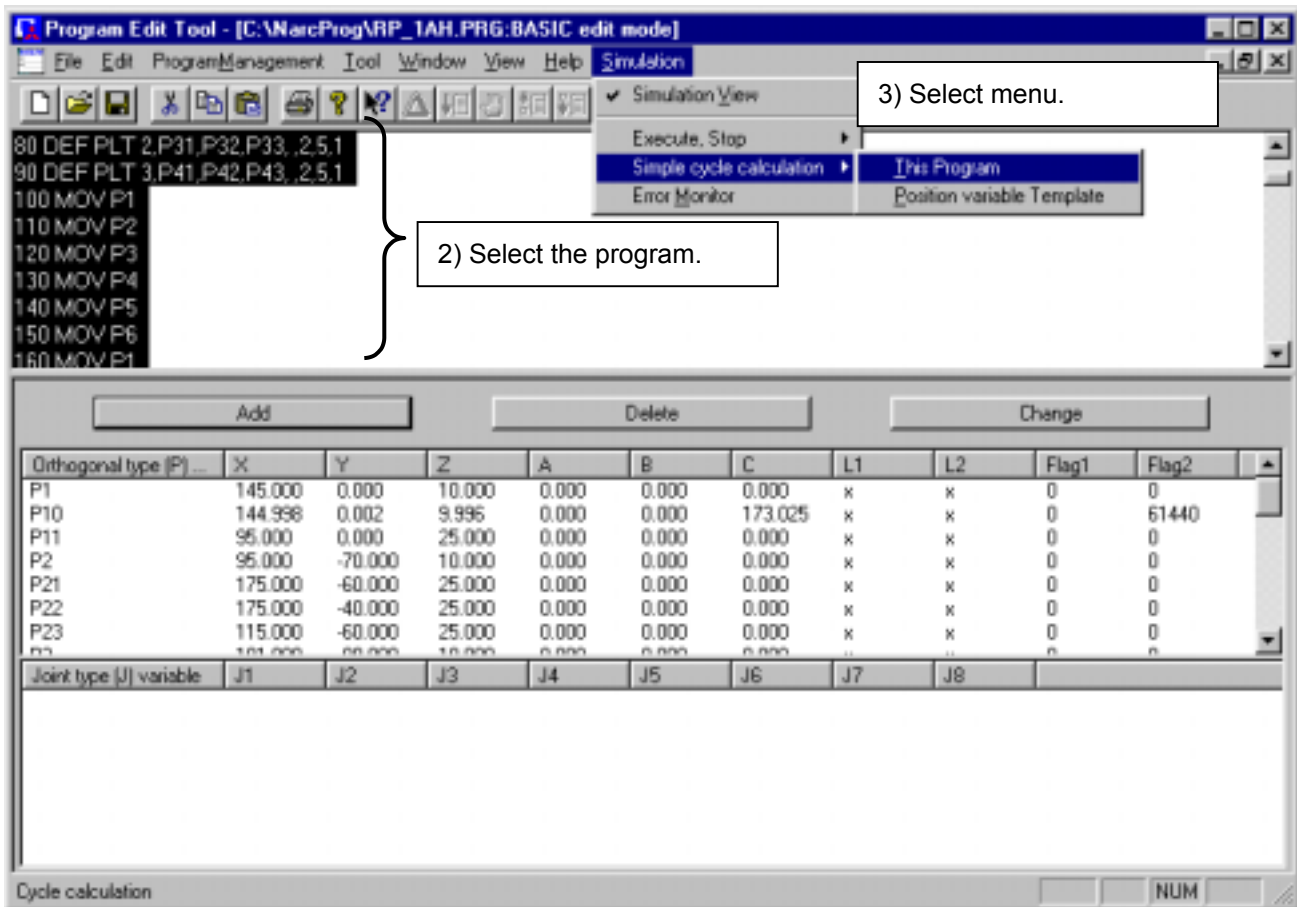


```
10 GOTO 120  
20 '  
30 *SUB1  
40 MOV P01  
50 RETURN  
60 '  
120 *START  
130 MOV P02  
140 MOV P03  
150 GOSUB *SUB1  
160 END
```


[Operation methods]

- 1) Open the robot program.
- 2) Select the execution section for which the tact time is to be calculated.

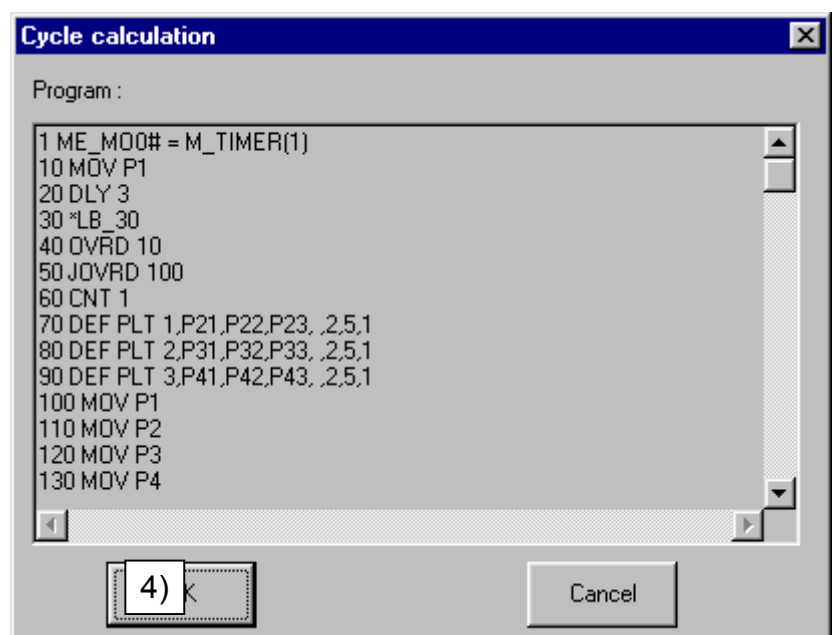
The section can be selected by dragging with the mouse.



- 3) Select the menu "Simulation" - "Simple Tact Calculation" - "Tact Calculation".

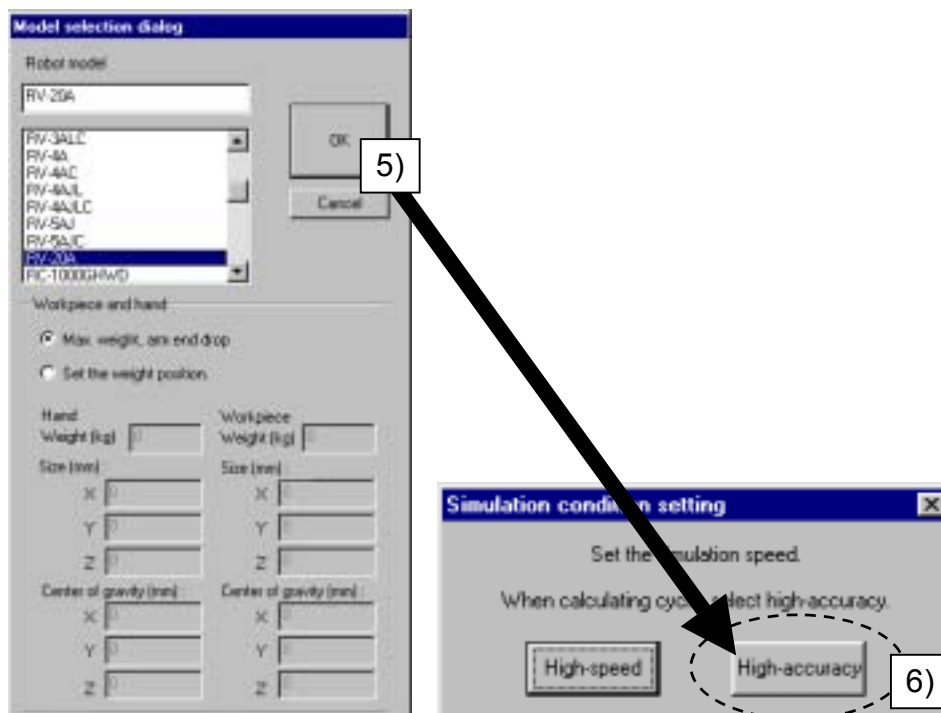
- 4) The Tact Calculation Range Confirmation window will appear. After confirming the range, click on [Execute].

Confirm that the selected program is correct.

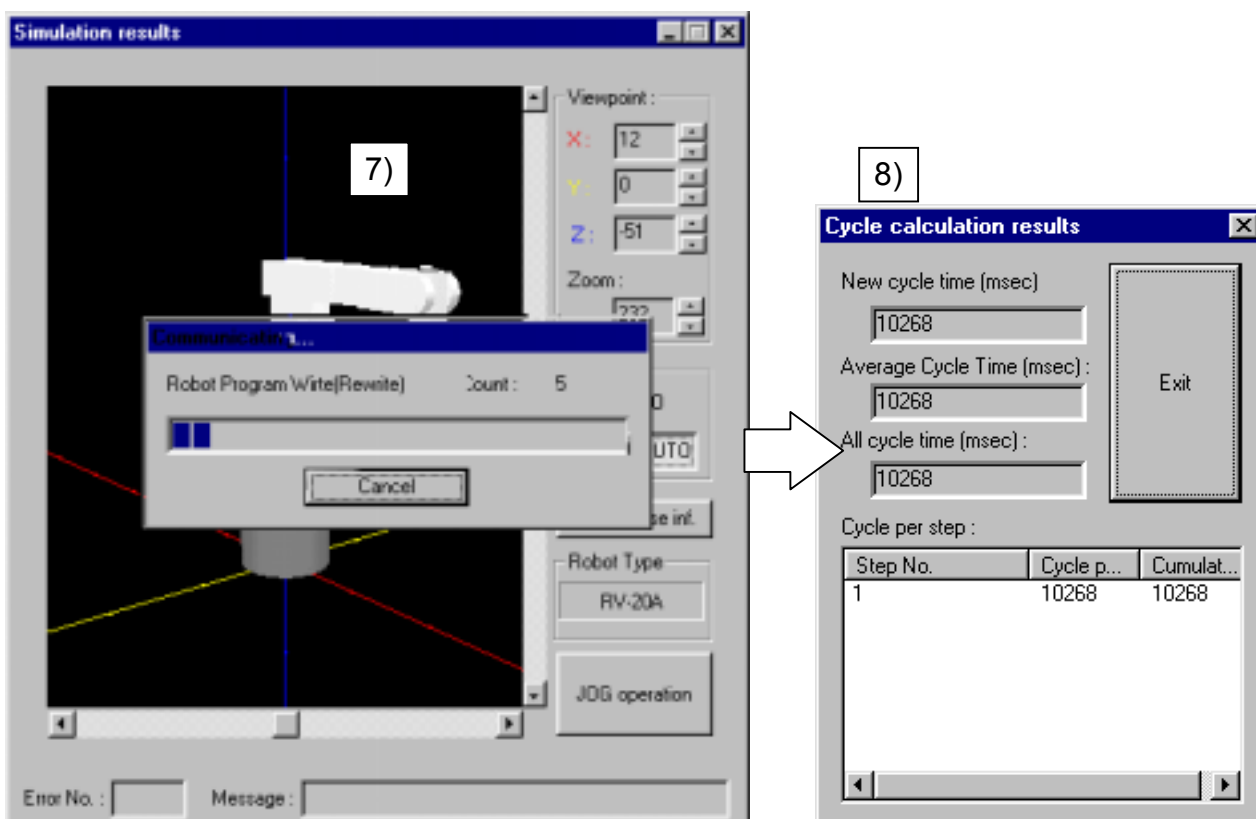


5. Explanation of each tool

- 5) If the Simulation Viewer is not opened, the robot model selection window will appear. Select the robot model, and then click on [OK].
- 6) When the Simulation Condition Setting window appears, click on [High Precision].

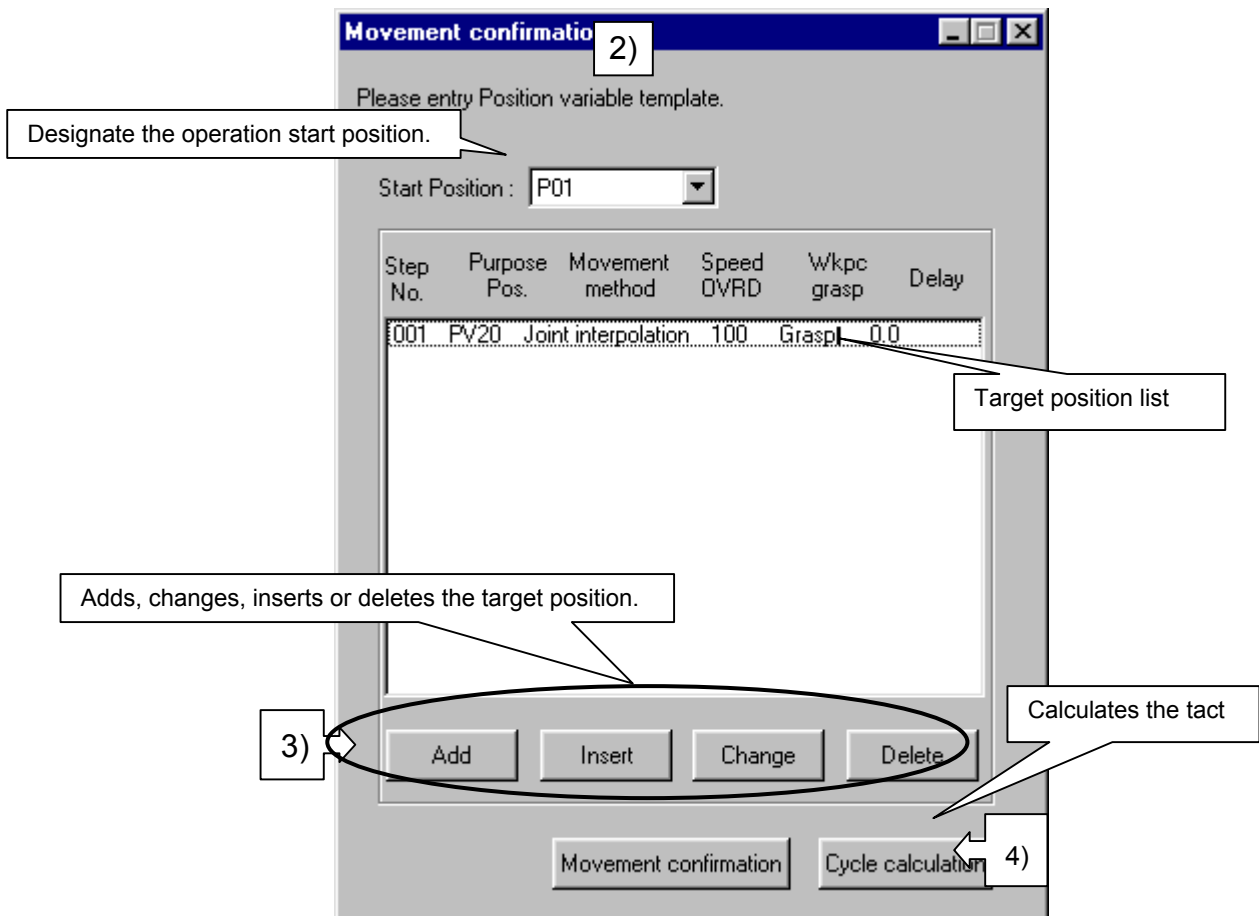
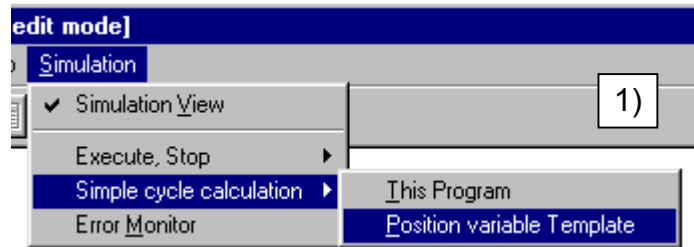


- 7) The robot movement will appear on the Simulation Viewer, and the tact calculation process will start.
- 8) When the tact calculation is completed, a window showing the tact calculation results will appear.



(2) Even without describing a robot program, the robot operation cycle time can be calculated by designating the robot movement transit point.

- 1). To calculate the cycle, select [Simulation] menu → [Simple cycle calculation]. The following general operation setting dialog will appear



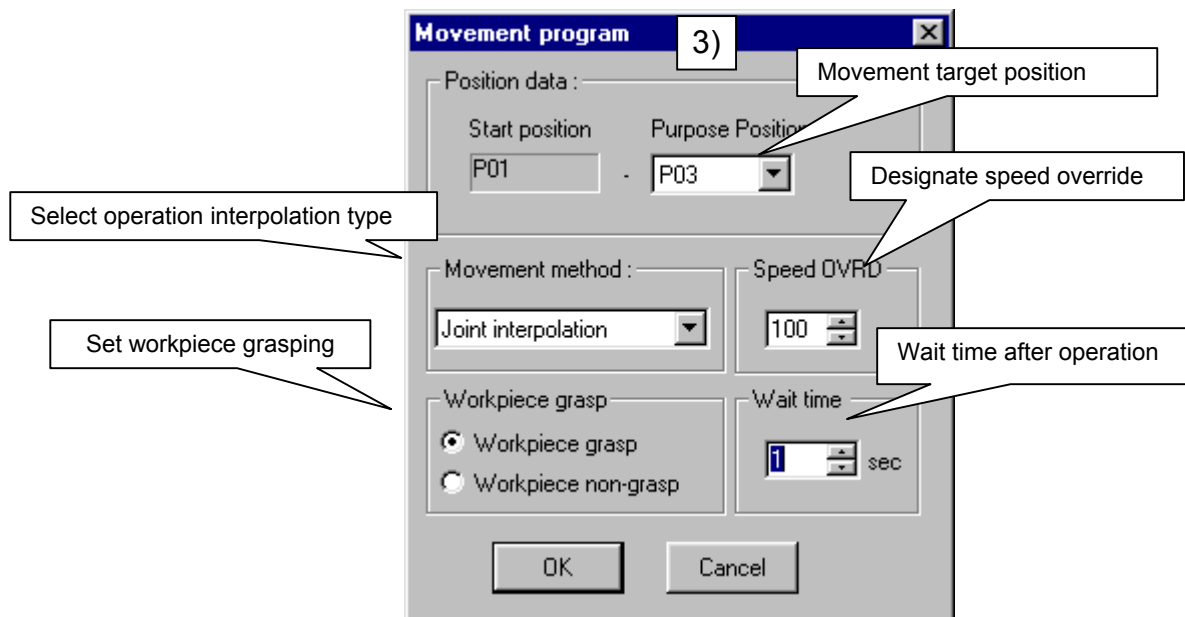
(General operation dialog)

- 2) Calculate the tact with the following procedure.

Select the robot operation start point from the [Start position]. The position variables registered in the position variable template (displayed with [Tool] menu → [Position variable template]) can be selected.

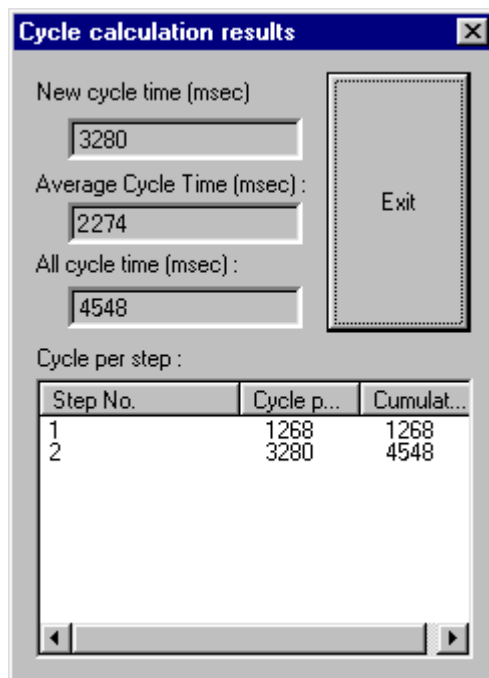
5. Explanation of each tool

- 3) When the [Add] button is clicked on, the target position addition dialog shown below will appear. Set the target point with this dialog. The position variables registered in the position variable template (displayed with [Tool] menu → [Position variable template]) can be selected.



(Transit point setting dialog)

- 4) Select the added transit point from the list, and then click on the [Cycle calculate] button.
5) The simulation view will open, the robot operation state will appear, and the cycle time will appear when the operation is completed.



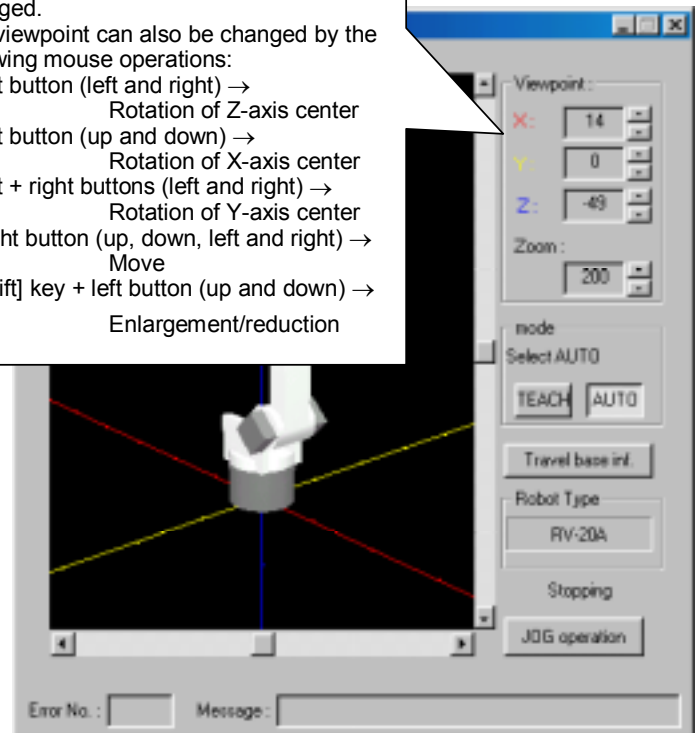
5.5.3. Editing the position variables with the simulation robot

The position variables can be edited by moving the simulation robot, and reading in the position variables from the simulation robot's posture. Use the following procedure.

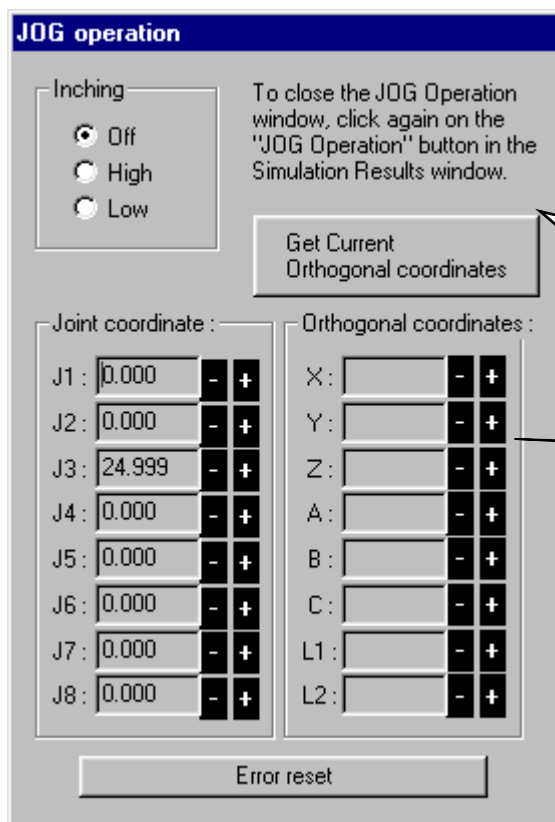
Start up the simulation screen by selecting [Simulation view] from the [Simulation] menu.

The angle for viewing the robot can be changed.
The viewpoint can also be changed by the following mouse operations:

- Left button (left and right) → Rotation of Z-axis center
- Left button (up and down) → Rotation of X-axis center
- Left + right buttons (left and right) → Rotation of Y-axis center
- Right button (up, down, left and right) → Move
- [Shift] key + left button (up and down) → Enlargement/reduction



(Simulation panel)



(JOG operation dialog)

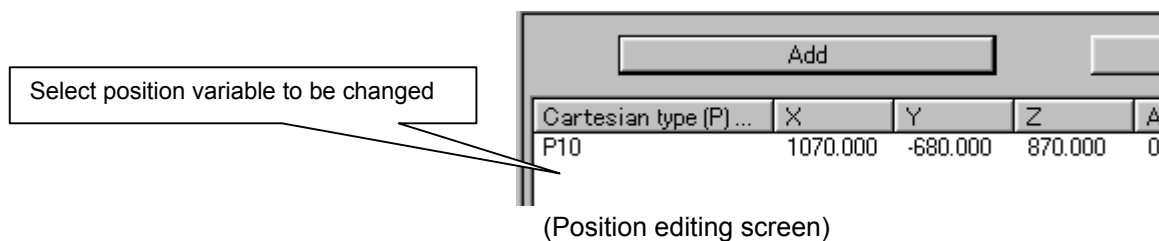
When the [JOG operation] button is clicked on, the following JOG operation dialog will appear. The simulation view robot can be operated with this dialog. Operate the simulation robot and set the robot to the required posture.

With JOG operation, only the joint coordinates can be monitored.
For the orthogonal coordinates, click on the "Retrieve Current Position" button after the robot movement stops.

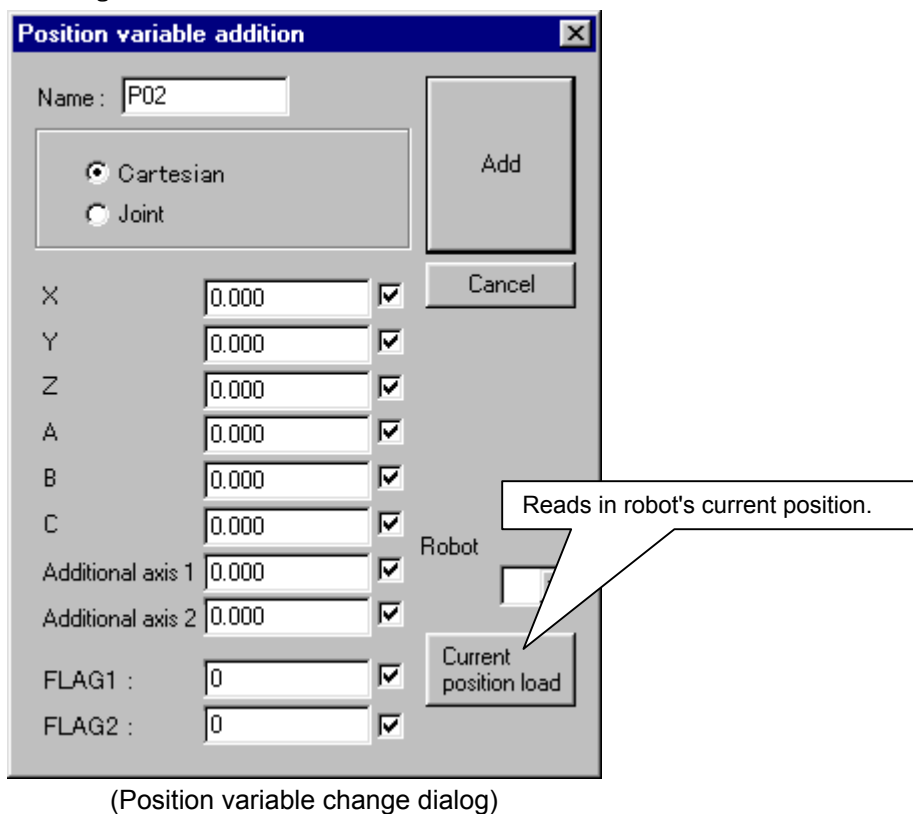
Click on the axis button for the direction to move to.

5. Explanation of each tool

Select the position variable to be edited from the program position variable editing screen being edited, and then click on the [Change] button. To newly add a position variable, click on the [Add] button.



The current position of the simulation robot will be read in when the [Current position read] button on the position variable change and add dialog is clicked on.



5.6. Program conversion

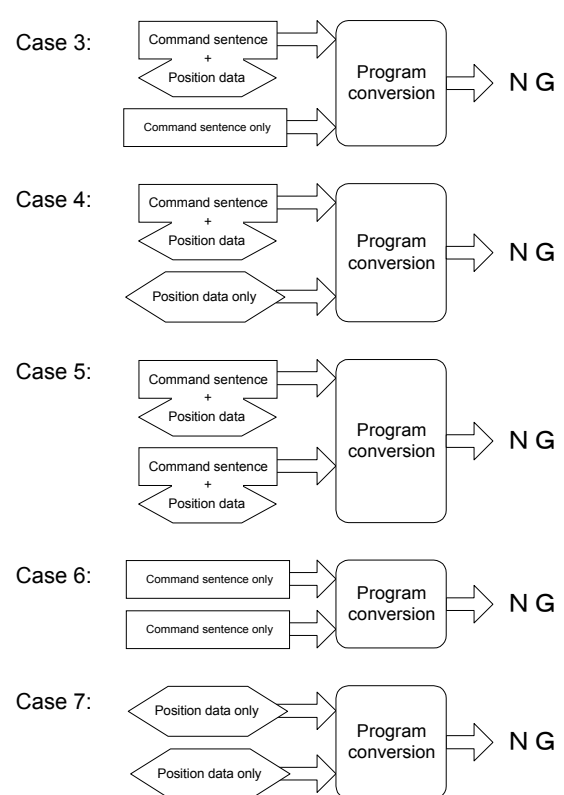
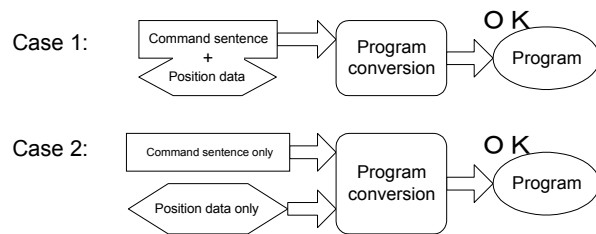
The personal computer support softwares for E/EN/M1 and M2 series (hereafter "old PC support S/W") and the support softwares for the new personal computer (hereafter "new S/W") differ in the configuration of position data. Therefore, the program prepared by using the old PC support S/W cannot be used as it is in the new S/W. The program conversion converts the position data as well as the instructions (DJ, MP, and PD instructions of the Move Master command) related to the position data so that the robot programs created and saved by the old PC support S/W can be used with this new S/W.

5.6.1 Precautions

Make sure to read through the items of caution given below before carrying out program conversion.

- Note 1:** The program conversion converts the position data as well as the instructions (DJ, MP, and PD instructions of the Move Master command) related to the position data. It is not possible to convert automatically from MELFA-BASIC III data to MELFA-BASIC IV data. Be sure to make grammatical check using the new S/W before using the program saved by the old PC support S/W.
- Note 2:** The program for M1 prepared by using the old PC support S/W has the base rotated 90°. When using the program for M1, convert M1 program into EN program using the old PC support S/W before carrying out the program conversion.
- Note 3:** In the case of the old PC support S/W, the data can be saved separately such as command sentence only, position data only or command sentence and position data. In this program conversion, it is possible to convert the file of command sentence only and the file of position data only to one program. The other combinations can not be made. See next page for reference.

5. Explanation of each tool

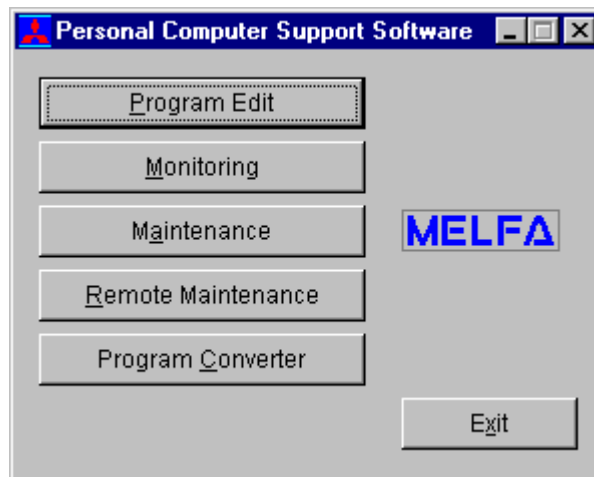


5.6.2 Start-up

There are two methods to start up.

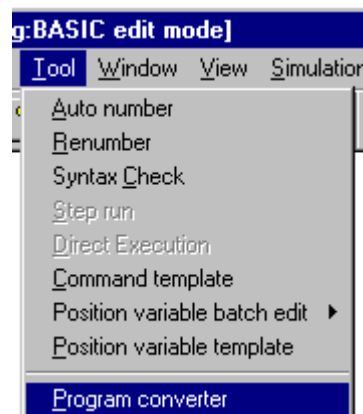
(1) Start up from MENU

Click on the [Program Converter] button using the MENU of the MELFA Personal Computer support software.



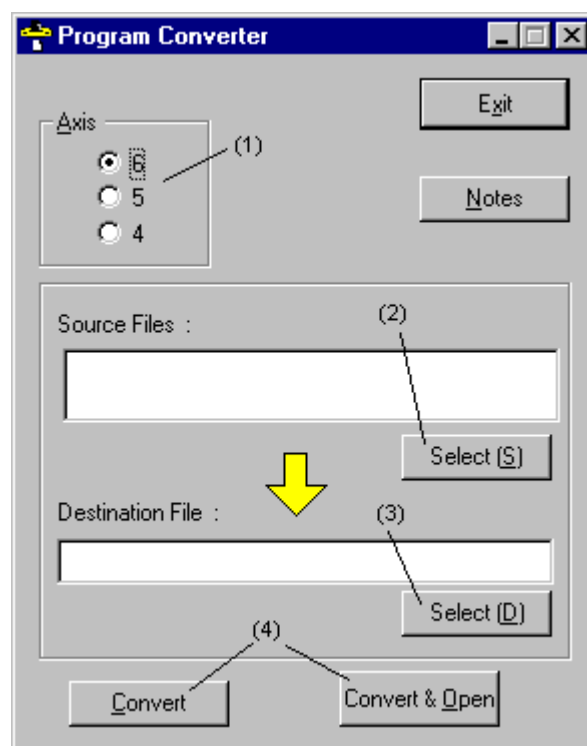
(2) Start up from PROGRAM EDIT

Click on the [Program converter] in the [Tool] menu of the 'Program Edit Tool - [Robot.prg: BASIC edit mode]' window.



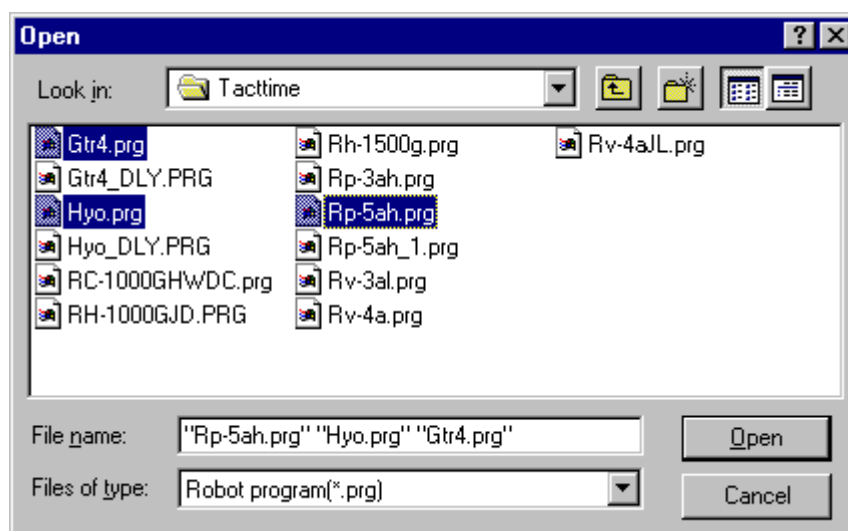
5.6.3 Operation methods

- (1) Select the number of axes of the robot. Make sure to select the correct number of axes, since the position data structure of the old PC support S/W differs according to the number of robot axes.



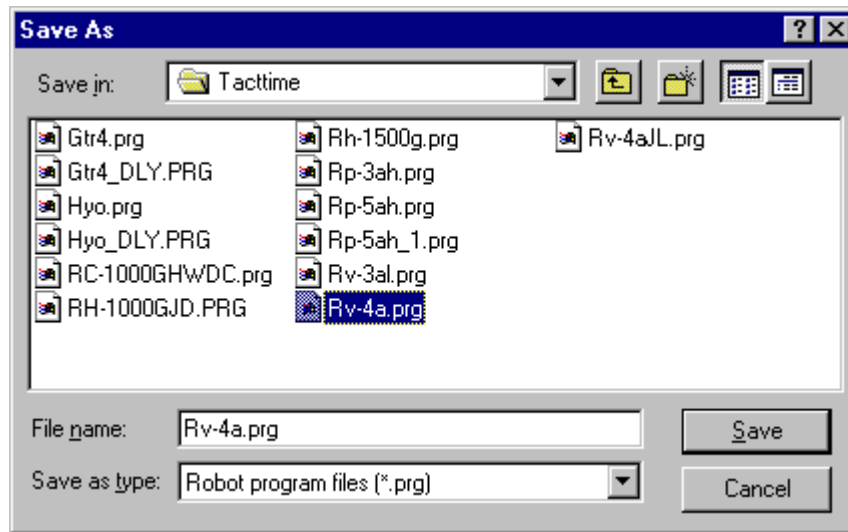
- (2) Designate the source file.

The [OPEN] window for file selection appears when the [Select] button of the [Source Files:] is clicked.



Click the file while pressing the [Ctrl] key for selecting several files. After selecting the file for conversion, click on the [Open] button. Then the selected files appear in the [Source Files:] box.

- (3) Designate the file to write down the converted program. Click on the [Select (D)] button of the [Destination File:] to display the [Save As] window for file selection.



Name the file in which the converted program is written before clicking on the [Save] button.

- (4) After designating the conversion source file and the conversion destination file, click on the [Convert] or [Convert & Open] button.
- When [Convert] is clicked, the designated file is converted and written down in the conversion destination file.
- When [Convert & Open] is clicked, the file is converted before being opened through program edit. Select according to the object.

5. Explanation of each tool

6. Setup of the communication middleware “MelfaRXM.ocx”

6.1 Overview

MelfaRXM.ocx is an ActiveX control that communicates with the robot controller.

MELFARXM.ocx is enclosed by the “Personal Computer Support Software” standard version. However, when using only the function of “Personal Computer Support Software”, it is not necessary to install MELFARXM.ocx.

By using MelfaRXM.ocx, the Windows application linked to MELFA ROBOT can be created easily on your equipment.

An instruction manual and notes of MelfaRXM.ocx are enclosed in CD-ROM of the standard version Personal Computer Support Software.

When MelfaRXM.ocx and Personal Computer Support Software live together in one personal computer, we ask you to surely read the instruction manual --“3.2 When making it coexist with personal computer support software” -- of the communication middleware “MelfaRXM.ocx”.

6.2 Description in CD-ROM

The CD-ROM(standard) contains the following items:

Refer to the instnction manual in CD-ROM for the usage of MelfaRXM.ocx.

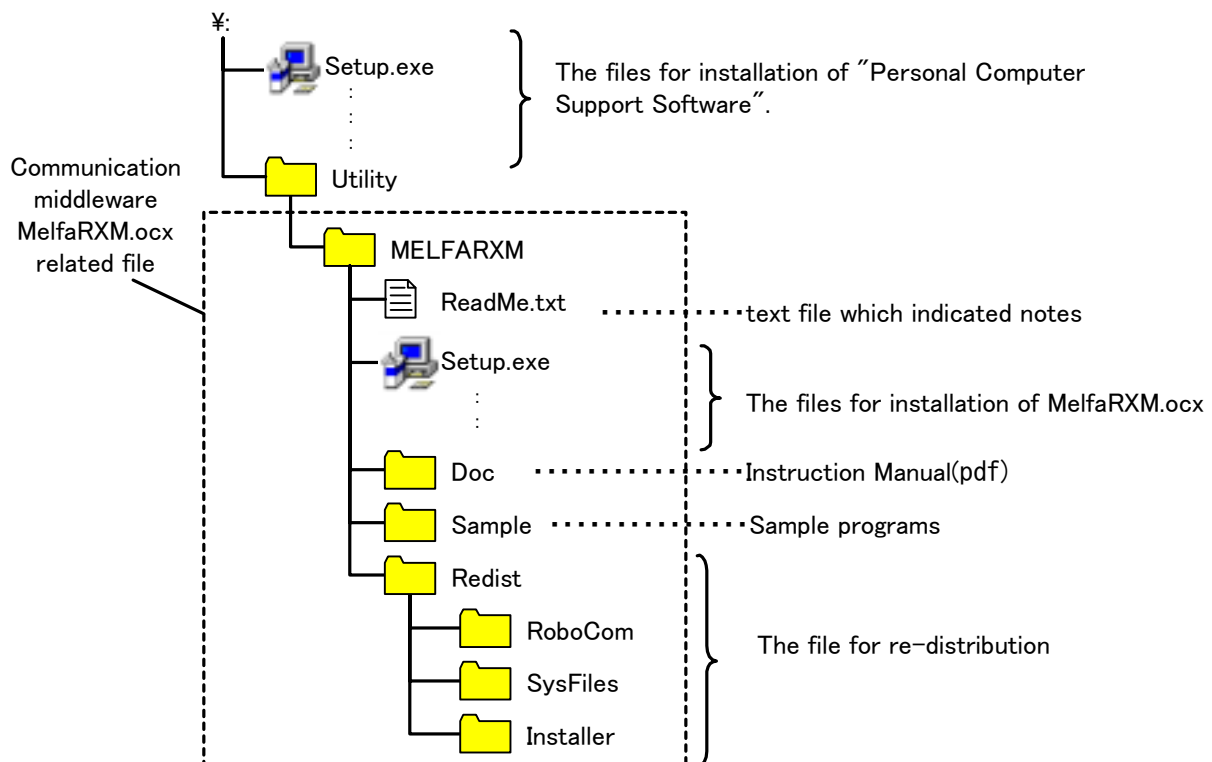


Figure 6- 1 Description in CD-ROM

6.3 Guide for reading the manual

The manual is in the CR-ROM as the Adobe PDF file.

D:/Utility/MELFARXM/Doc/MelfaRXME.pdf

* Example for the CD-ROM drive is “D:”.

(1) Prepare for reading

① Prepare of the personal computer

The personal computer with which is installed Microsoft Windows 95 or more and it has the CD-ROM drive is required.

② Prepare of the software for reading

For reading the manual, Adobe Acrobat Reader Ver.5.0 or more is required.

If Adobe Acrobat Reader isn't installed, please download from following Adobe Systems Incorporated URL(As of December, 2002).

URL: <http://www.adobe.com>

(2) How to read

① Case of starting from Explorer

Start Explorer and select the file. Then Acrobat Reader starts and the manual is displayed.

② Case of starting directly from Acrobat Reader

Start Acrobat Reader and select the file. Then the manual is displayed.

Microsoft and Windows are either registered trademarks or trademarks of Microsoft Corporation in the United States and/or other countries.

Acrobat Reader Copyright 1987-1999 Adobe Systems Incorporated. All rights reserved.

Adobe, the Adobe logo, Acrobat, and the Acrobat logo are trademarks of Adobe Systems Incorporated.

Reference to registered trademarks and trademarks are omitted in this manual.

6.4 Installation

Perform installation according to the following procedure:

- (1) Insert the program CD-ROM into the CD-ROM drive of your personal computer. Setup of “Personal computer support software” automatically starts. Please click “cancel” .
- (2) Select [Run] from the [Start] button.

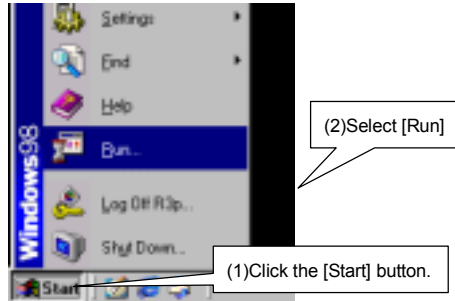


Figure 6-1 : Selecting [Run]

- (3) Check the drive name of the CD-ROM drive. Enter the following and click the [OK] button. :
 "drive name":/Utility/MelfaRXM/Setup.exe
 (If the CD-ROM drive is “D”, enter “D”:/Utility/MelfaRXM/Setup.exe”)

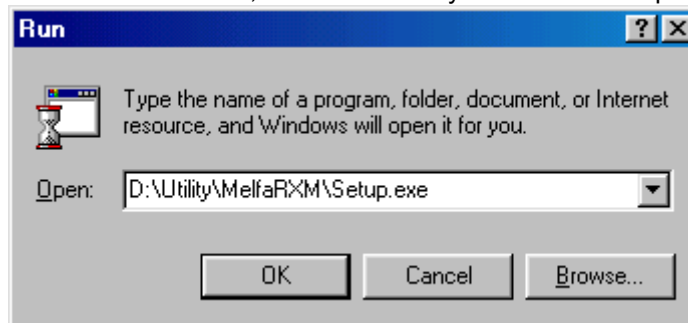
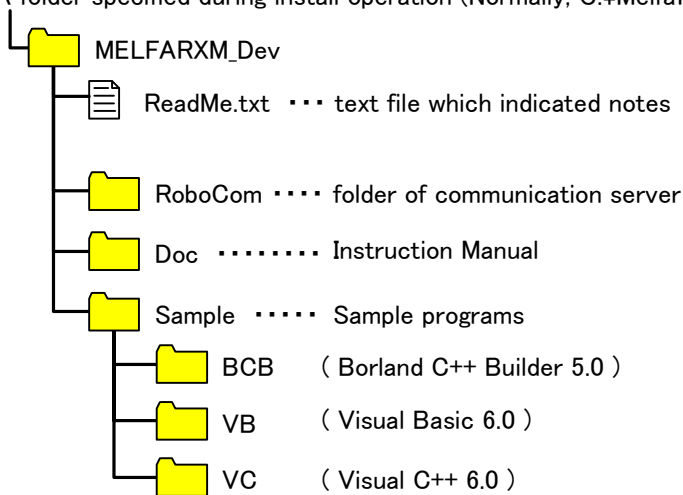


Figure 6-2 : [Run] Screen

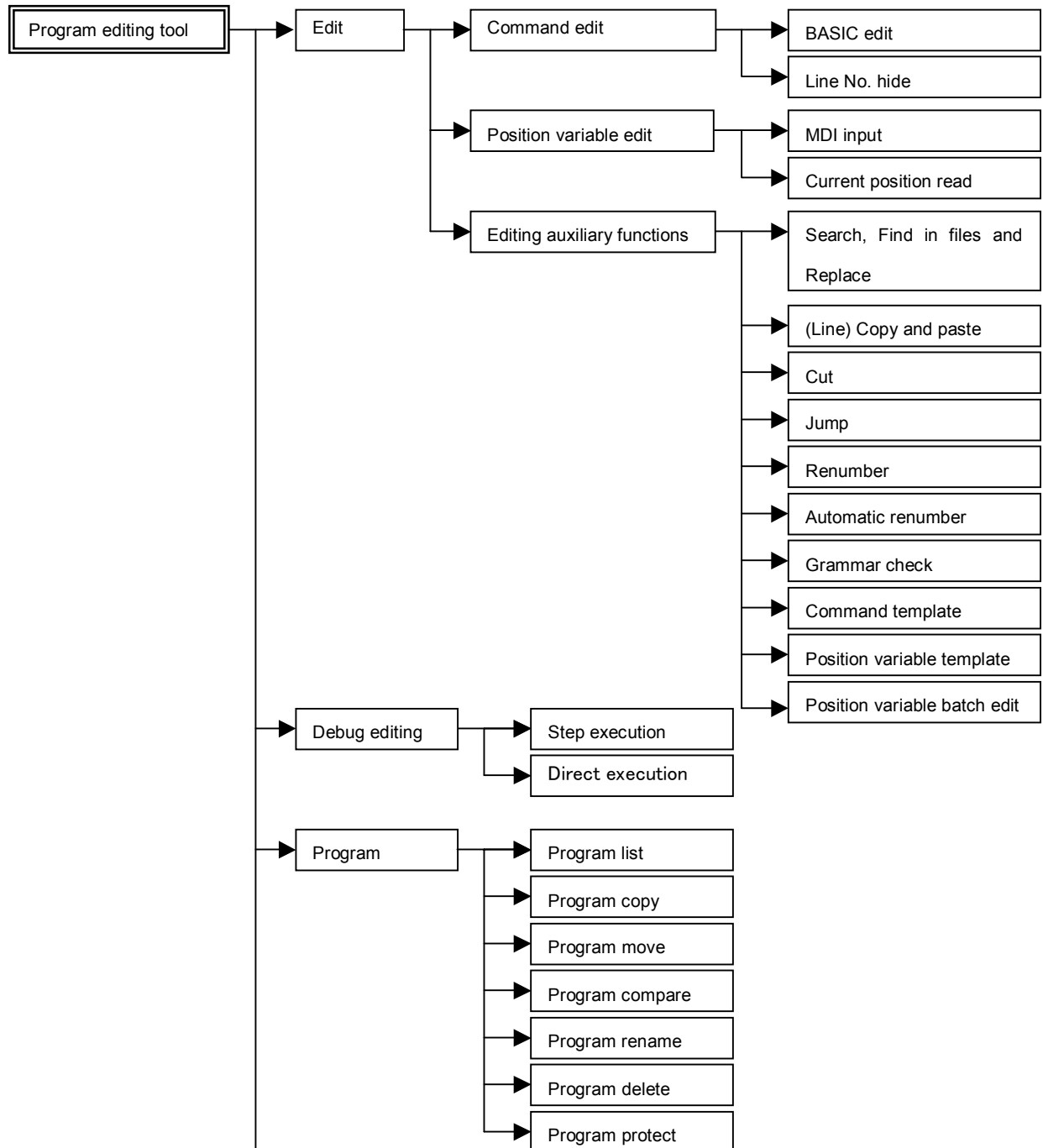
- (4) Installer starts and the Setup screen appears. Install according to the instructions that appear on the screen. The files listed in Table 1 are installed at the destinations as indicated.

6. Setup of the communion middleware “MelfaRXM.ocx”

Table 1 : Files to be Installed

No.	Description	Install destination
1	MelfaRXM.ocx	/Windows/System folder ex) 95/98/Me : /Windows/System NT/2000 : /Winnt/System32 Xp : /Windows/System32 MelfaRXM.ocx and EzSocketRC.dll are registered in the registry.
2	EZSocketRC.dll (communication DLL)	
3	NarcServerApiM.dll (communication DLL)	
4	RoboCom.exe (communication DLL)	A folder specified during install operation (Normally, C:\MelfaRXM is used.)  <pre> graph TD MELFARXM_Dev[MELFARXM_Dev] --> ReadMe_txt[ReadMe.txt] MELFARXM_Dev --> RoboCom[RoboCom] MELFARXM_Dev --> Doc[Doc] MELFARXM_Dev --> Sample[Sample] RoboCom --> RoboCom_server[folder of communication server] Doc --> Instruction_Manual[Instruction Manual] Sample --> Sample_programs[sample programs] Sample --> BCB[BCB] Sample --> VB[VB] Sample --> VC[VC] BCB --- BCB_desc["(Borland C++ Builder 5.0)"] VB --- VB_desc["(Visual Basic 6.0)"] VC --- VC_desc["(Visual C++ 6.0)"] </pre>
5	Instruction Manual	
6	Sample programs	

A. Appendix (Function tree)

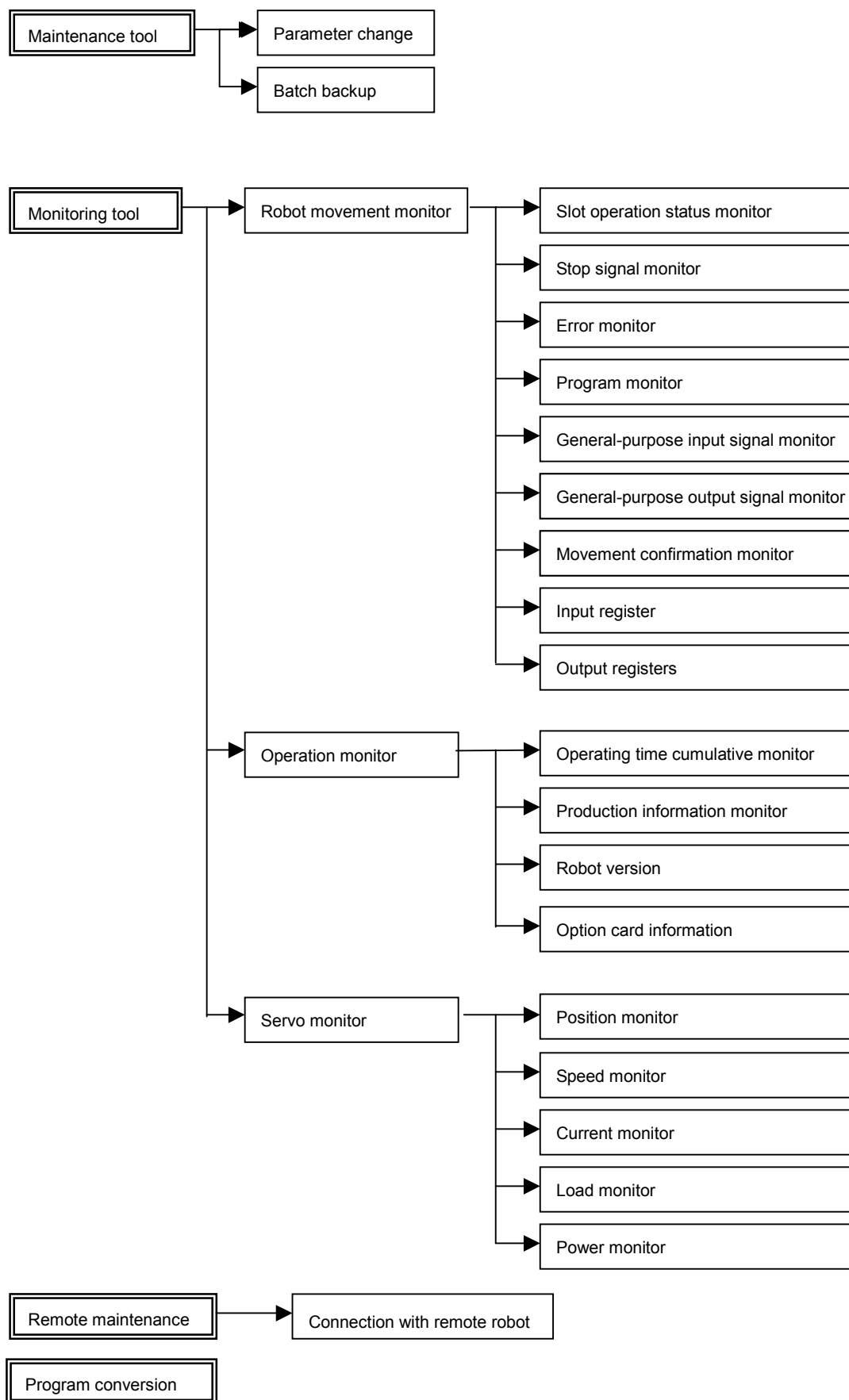


The simulation function is compatible only with the "Standard personal computer support software version".

The function cannot be used with the "Personal computer support software mini". Note that the simulation function also cannot be used when the MOVEMASTER commands are selected with the "Standard personal computer support software version".

Note that the simulation function also cannot be used when the MOVEMASTER commands are selected with the "Standard personal computer support software version".

6. Appendix (Function tree)





HEAD OFFICE : MITSUBISHI DENKI BLDG MARUNOUCHI TOKYO 100-8310 TELEX : J24532 CABLE MELCO TOKYO
NAGOYA WORKS : 1-14, YADA-MINAMI 5, HIGASHI-KU, NAGOYA, JAPAN